BIOLOGY GRADE 10: ANIMAL GASEOUS EXCHANGE AND RESPIRATION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.3 (12 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 3.0: Physiology of Life Processes
Sub-Strand	Sub-Strand 3.3: Animal Gaseous Exchange and Respiration
Total Duration	12 lessons × 40 minutes = 480 minutes total
Sub-Strand Content	– General characteristics of respiratory surfaces in animals – Structure and adaptations of respiratory structures in animals (skin, gills, tracheae, lungs) – Mechanisms of gaseous exchange in humans (breathing mechanics, alveolar exchange) – Aerobic respiration (equation, stages, energy yield) – Anaerobic respiration (equation, products in animals and yeast, lactic acid in muscles) – Respiratory quotient (RQ) for carbohydrates, fats, and proteins – Importance of gaseous exchange and respiration in animals – Factors affecting the rate of respiration (temperature, oxygen concentration, substrate availability, surface area)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe the general characteristics of respiratory surfaces in animals, b) explain the structure and adaptations of respiratory structures in different animals, c) describe the mechanisms of gaseous exchange in humans, d) differentiate between aerobic and anaerobic respiration and write their balanced equations, e) calculate the respiratory quotient (RQ) for different food substrates, f) explain the importance of gaseous exchange and respiration in animals, g) analyse the factors that affect the rate of respiration, h) appreciate the importance of efficient gaseous exchange and respiration for the survival of animals.
Core Competencies	– Communication and Collaboration: Learners discuss and present findings from breathing-rate and pulse-rate practicals, collaboratively construct and revise gaseous exchange models, and debate the implications of aerobic vs. anaerobic respiration during exercise in Kenyan athletes such as marathon runners from Iten. – Critical Thinking and Problem Solving: Learners analyse data on respiratory quotients for different food substrates (e.g., maize starch vs. nyama choma fat), identify patterns, and draw evidence-based conclusions about energy metabolism. – Creativity and Imagination: Learners design and iteratively improve annotated diagrams and physical models of the human respiratory system and alveolar gas-exchange surfaces. – Digital Literacy: Learners use the ARES offline platform to watch simulations and videos on oxygen transport through the circulatory system and interpret virtual spirometer data. – Self-Efficacy: Learners develop confidence in conducting and interpreting quantitative practicals (measuring breathing rate,

	pulse rate, and calculating RQ values) and connecting results to their own lived experience of physical activity.
Core Values	– Responsibility: Learners handle laboratory apparatus (e.g., limewater, thermometers, stopwatches) with care and follow safety protocols during respiration experiments, reflecting responsibility for self and peers. – Respect: Learners appreciate diverse body responses observed during the breathing-and-pulse practical, recognising that physiological variation is normal and avoiding ridicule of peers' results. – Unity: Learners work collaboratively in mixed-ability groups during model-building and practical activities, valuing each member's contribution to the collective understanding of gaseous exchange. – Social Justice: Learners discuss how access to clean air and adequate nutrition (e.g., sufficient carbohydrates from ugali and githeri) affects respiratory health, linking biology to equity in Kenyan communities.
Science & Engineering Practices	– SEP 1 – Asking Questions and Defining Problems: Learners generate testable questions from the anchoring phenomenon (e.g., "Why does breathing rate increase during exercise?") and post them on the Driving Question Board at the start of the unit. – SEP 2 – Developing and Using Models: Learners iteratively build and revise annotated diagrams and 3-D models of the human respiratory system, alveolus, and circulatory loop, adding detail as new content is encountered across all 12 lessons. – SEP 3 – Planning and Carrying Out Investigations: Learners design and conduct controlled experiments measuring breathing rate and pulse rate at rest and after exercise, and investigate factors affecting yeast respiration rate. – SEP 4 – Analysing and Interpreting Data: Learners tabulate and graph breathing-rate and pulse-rate data, calculate RQ values from experimental gas volumes, and interpret spirometer traces. – SEP 6 – Constructing Explanations: Learners write evidence-based explanations of why oxygen debt occurs during anaerobic respiration in athletes, linking cellular chemistry to observable symptoms (muscle cramps, heavy breathing after a race). – SEP 8 – Obtaining, Evaluating, and Communicating Information: Learners read structured texts on circulatory and respiratory system functions, evaluate the reliability of sources, and communicate findings in written reports and class presentations.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners connect efficient gaseous exchange and respiration to personal health choices, including the importance of physical activity and avoiding smoking — a growing public-health concern in urban Kenya (Nairobi, Mombasa). – Environmental Conservation: Learners examine how air pollution from matatu emissions and industrial activity in Kenyan cities impairs respiratory surfaces, and discuss community responsibility for clean air. – Safety and Security: Learners follow proper laboratory safety procedures when using limewater (an irritant) and conducting breathing-rate experiments, and learn about dangers of oxygen-deficient environments (e.g., enclosed grain stores in rural Kenya). – Life Skills Education: Learners develop decision-making and self-management skills by reflecting on how lifestyle factors (diet of carbohydrates from ugali vs. fats from nyama choma, exercise, altitude in the Rift Valley) affect their own rate of respiration and energy availability. – Financial Literacy: Learners discuss how understanding fermentation (anaerobic respiration in yeast) underpins food-preservation and brewing industries in Kenya, connecting biology to entrepreneurship opportunities.
Career Connections	– Medicine and Clinical Practice: Understanding gaseous exchange mechanisms and respiratory disorders (asthma, pneumonia) is foundational for doctors, clinical officers, and nurses serving in Kenyan hospitals such as Kenyatta National Hospital. – Sports Science and Athletic Coaching: Knowledge of aerobic vs. anaerobic respiration and oxygen debt is directly applied by coaches training Kenya's world-class distance runners in Iten and Eldoret. – Environmental Health and Public Health: Public-health officers and environmental scientists monitor air quality and respiratory health impacts in Kenyan cities, requiring a deep understanding of how pollutants impair gaseous exchange. – Biochemistry and Nutritional Science: Calculating respiratory quotients and understanding substrate oxidation is essential for

	nutritionists and food scientists advising on balanced diets for Kenyans. – Biotechnology and Food Technology: Anaerobic respiration (fermentation) is exploited in Kenya's food and beverage industry (e.g., fermented porridge, commercial yeast production), requiring knowledge of microbial respiration. – Veterinary Science: Veterinarians working with livestock (cattle, goats) and aquatic animals at Lake Victoria fish farms must understand comparative gaseous exchange across animal groups.
Focus for Lessons	This unit explores how animals — from earthworms in Kenyan red soil to tilapia in Lake Victoria and human athletes racing on Eldoret's roads — obtain oxygen and release carbon dioxide at both the organ and cellular level. Using the anchoring phenomenon of measurable changes in breathing rate and pulse rate during exercise, learners move from observable whole-body responses to molecular explanations of aerobic and anaerobic ATP production, building an increasingly detailed and accurate model across all 12 lessons. The pedagogical approach is phenomenon-driven and inquiry-based: learners first predict and observe, then construct explanations grounded in evidence, continuously updating a class Driving Question Board and their individual gaseous-exchange models.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does my body work so much harder to get oxygen into my cells during exercise — and what happens inside those cells when there is not enough? KICD KEY INQUIRY QUESTIONS: 1. How do animals exchange gases with their environment? 2. How do animals obtain energy from food?
Anchoring Phenomenon	ANCHORING PHENOMENON — "Your Body Changes When You Move" At the start of the unit, every learner measures their own resting breathing rate (breaths per minute) and resting pulse rate (beats per minute) using a stopwatch and fingers placed on the radial artery at the wrist. Values are recorded in a class data table on the board. Learners then perform 2 minutes of vigorous step-ups on their chair and immediately re-measure both rates. Without exception, both rates have risen sharply — a learner who breathed 14 times per minute at rest may now breathe 28 times, and a pulse of 72 bpm may have jumped to over 120 bpm. This is puzzling on multiple levels: Why does the body need more oxygen so urgently that it doubles its breathing effort? Why does the heart also beat faster at the same time — what is the connection between the lungs and the heart? Why do some learners feel a burning sensation in their leg muscles after the exercise, even when they were breathing hard the whole time? Where exactly does the oxygen go once it enters the lungs, and what happens to it inside the muscle cells? These questions — all arising from a 2-minute activity every learner has just personally experienced — anchor the entire unit and are progressively answered as learners investigate respiratory surfaces, breathing mechanics, cellular respiration chemistry, and the role of the circulatory system.
Storyline Thread	Lesson 1 — PREDICT & ANCHOR: Learners measure resting breathing rate and pulse rate (practical), record class data, perform step-up exercise, re-measure, and record changes. Learners write individual predictions explaining WHY both rates increased. Class Driving Question Board (DQB) is created: all student questions are posted. Learners draw a first "naïve model" showing how they think oxygen gets from air to muscles. Reading on circulatory system functions assigned on ARES platform. Lesson 2 — OBSERVE: Characteristics of Respiratory Surfaces. Learners examine prepared slides / diagrams of skin (earthworm), gill lamellae (fish), tracheal tubes (insect), and alveoli (mammal). They complete a comparison table (surface area, moisture, vascularity, thickness). Limewater demonstration performed. Learners revisit DQB: which questions can now be partially answered?

	Lesson 3 — EXPLAIN: Structure and Adaptations of Respiratory Structures. Teacher-guided sensemaking of how each structural feature (large surface area, thin walls, moist surface, rich blood supply) solves the problem of efficient gas exchange. Learners annotate diagrams and add a "comparative respiratory surfaces" layer to their growing model. Kenyan context: tilapia gill adaptation discussed using Lake Victoria fishery examples. Lesson 4 — OBSERVE & EXPLAIN: Mechanisms of Gaseous Exchange in Humans — External Respiration. Learners watch an ARES offline video/animation of inhalation and exhalation (diaphragm and intercostal muscle movement, pressure changes). Learners then model pressure changes using a bell-jar diaphragm model or improvised plastic-bottle version. Key vocabulary: tidal volume, inspiration, expiration, pleural cavity. Lesson 5 — EXPLAIN: Mechanisms of Gaseous Exchange in Humans — Alveolar & Cellular Exchange. Learners trace the journey of one oxygen molecule: alveolus → capillary → haemoglobin → heart → muscle cell. Reading on circulatory system functions (ARES platform) is discussed. Learners update their individual model to show blood as the transport link between lungs and muscles, directly connecting to the anchoring phenomenon. Lesson 6 — OBSERVE & EXPLAIN: Aerobic Respiration. Learners examine the word equation and balanced chemical equation for aerobic respiration. Worked example: calculating ATP yield conceptually. Learners annotate a mitochondrion diagram showing glucose + oxygen → carbon dioxide + water + ATP. Kenyan context: energy released from ugali (starch/glucose) fuelling a Rift Valley farmer's day of work. Learners update DQB — the "normal" oxygen demand during exercise is now explained. Lesson 7 — OBSERVE & EXPLAIN: Anaerobic Respiration. "Yeast Balloon" practical (supporting phenomenon 4) conducted. Learners compare aerobic vs. anaerobic equations side-by-side (animals: lactic acid pathway; yeast: ethanol + CO_2 pathway). "Burning legs" video clip (supporting phenomenon 1) revisited. Learners explain in writing: why lactic acid builds up in muscles during intense exercise and what "oxygen debt" means. DQB updated — the burning-leg mystery is solved. Lesson 8 — EXPLAIN: Respiratory Quotient (RQ). Learners are introduced to the concept and formula: RQ = CO_2 produced ÷ O_2 consumed. Teacher works through RQ calculations for carbohydrates (RQ = 1.0), fats (RQ ≈ 0.7), and proteins (RQ ≈ 0.9) using Kenyan food examples (maize starch from ugali, fat from nyama choma, protein from beans/githeri). Learners practise 3–4 calculation problems. Discussion: what does an athlete's RQ during a marathon tell us about their fuel source? Lesson 9 — PRACTICAL & DATA ANALYSIS: Factors Affecting Rate of Respiration. Learners design and conduct a controlled investigation into the effect of temperature on yeast respiration rate (measuring CO_2 bubble count or balloon size at 10 °C, 25 °C, 37 °C, 50 °C). Data are tabulated and graphed. Learners identify the optimum temperature and explain enzyme denaturation at high temperatures. Connection to DQB: why does body temperature rise during exercise? Lesson 10 — EXPLAIN: Factors Affecting Rate of Respiration (continued) & Importance of Gaseous Exchange. Class discussion on remaining factors: substrate concentration, oxygen availability, surface area. Learners then construct a concept map on the importance of respiration (energy for movement, growth, active transport, heat regulation, reproduction). Kenyan context: importance of aerobic capacity for agricultural labour, sports performance, and survival at high altitude in the Rift Valley. Lesson 11 — MODEL BUILDING & DQB CONSOLIDATION: Learners produce a final, fully annotated cumulative model that integrates: (a) comparative respiratory surfaces, (b) human breathing mechanics, (c) alveolar and cellular gas exchange, (d)

	aerobic and anaerobic respiration equations, and (e) factors affecting respiration rate. DQB is reviewed: all original questions are answered in writing. Peer review of models using a teacher-provided checklist. Lesson 12 — SUMMATIVE REVIEW & TRANSFER: Learners return to the original anchoring phenomenon data (resting vs. post-exercise breathing and pulse rates) and write a comprehensive scientific explanation using all unit vocabulary and concepts. Exit task: a novel scenario (a Kenyan swimmer crossing the Winam Gulf of Lake Victoria) asks learners to predict and explain respiratory and metabolic responses. Unit closes with learner self-assessment and DQB celebration of questions answered.
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — "The Burning Legs of a Iten Runner": A short video clip shows a 5 000 m runner from Iten crossing the finish line, doubling over, breathing heavily for minutes afterward even though the race is over. Learners are asked: why does breathing not stop the moment the race ends? This supports the lesson on anaerobic respiration and oxygen debt. – SUPPORTING PHENOMENON 2 — "Fish Out of Water": Learners observe (live or via video) a tilapia removed from Lake Victoria water, noting its rapid gill-cover movements and eventual suffocation. This supports comparative study of gill structure and the requirement for a moist, thin, well-vascularised respiratory surface. – SUPPORTING PHENOMENON 3 — "Limewater Turns Milky": Learners breathe gently through a straw into limewater and observe it turn milky white (calcium carbonate precipitate), while a control tube of fresh air does not change. This tangible, visible evidence that exhaled air contains elevated CO_2 supports the lesson on breathing mechanics and gas composition. – SUPPORTING PHENOMENON 4 — "Yeast Balloons": Learners set up two flasks — one with sugar solution + yeast at 37 °C and one at 4 °C (ice water) — each sealed with a balloon on top. Over 20 minutes the warm flask's balloon inflates while the cold one does not, making CO_2 production from anaerobic fermentation visible and introducing the effect of temperature on respiration rate.

LESSON 1 (80 minutes): Your Body Changes When You Move — Anchoring the Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To anchor the unit phenomenon by having learners personally experience and measure the changes in breathing rate and pulse rate during exercise, generate predictions, and open the Driving Question Board.
Knowledge	Learners will know that breathing rate and pulse rate increase measurably during physical exercise, and that both rates return toward resting values during recovery.
Skills	Learners will be able to measure resting and post-exercise breathing rate and pulse rate using a stopwatch and radial pulse technique, record data accurately in a class data table, and construct a naïve model diagram showing their current understanding of how oxygen travels from air to muscle cells.
Attitudes	Learners will appreciate that their own body is a source of scientific data, and will show curiosity and intellectual humility by acknowledging what they do not yet understand and posting genuine questions on the Driving Question Board.
Key Inquiry Question	How do animals exchange gases with their environment? How do animals obtain energy from food?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — 'Your Body Changes When You Move' — by making every learner a data point. The personal, embodied experience of doubled breathing rate and a racing pulse creates authentic puzzlement that will drive inquiry across all subsequent lessons. The naïve models drawn today will be revised progressively as learners investigate respiratory surfaces, breathing mechanics, cellular respiration chemistry, and the circulatory system. The Driving Question Board opened today remains live for the entire unit.
Safety Notes	Before the step-up exercise, ask learners to declare any known heart conditions, asthma, or recent illness — these learners should act as timekeepers and data recorders rather than performing the exercise. Ensure chairs are stable on a flat floor surface; learners should hold the back of the chair for balance. Stop the exercise immediately if any learner feels dizzy, chest pain, or severe breathlessness beyond normal exertion. Keep water available.

B. LESSON OVERVIEW

Learners open the unit by becoming experimental subjects. Working in pairs, they measure their own resting breathing rate and resting pulse rate (radial artery method), record values in a shared class data table on the board, then perform 2 minutes of vigorous step-ups on their chair. They immediately re-measure both rates and record post-exercise values. The sharp, universal rise in both measures — visible across the entire class dataset — creates genuine puzzlement: Why does oxygen demand double so quickly? Why does the heart speed up at the same moment the lungs speed up? Why do some learners feel a burning sensation in their leg muscles even though they were breathing hard? Learners write individual written predictions, share and discuss in small groups, then post their unanswered questions on the class Driving Question Board (DQB). Each learner draws a first 'naïve model' — a labelled diagram showing, in their own current understanding, how oxygen moves from the air outside their body into the muscle cells of their legs. The lesson closes with a brief orientation to the ARES platform reading on circulatory system functions, assigned as preparation for Lesson 2.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Laplace transform to solve an equation] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** B and T Cell Response (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are seated. Without any prior instruction, each learner silently counts their own resting breathing rate for 60 seconds (a partner watches the clock) and then locates their radial pulse at the wrist and counts beats for 60 seconds. Each learner records both values on a sticky note. The teacher then draws a two-column table on the board (RESTING BREATHING RATE | RESTING PULSE RATE) and every learner posts their sticky note. Learners observe the spread of class data — most breathing rates cluster between 12–20 breaths/min, most pulse rates between 60–90 bpm. Learners are then asked to write, individually and silently, a prediction in their notebooks: 'If you perform 2 minutes of vigorous step-ups right now, what do you predict will happen to your breathing rate and pulse rate? Write a number prediction AND a reason — WHY do you think that will happen?' Learners are not yet told whether they are correct; predictions are sealed in anticipation.	Begin by saying: 'Before I tell you anything about today's lesson, I need everyone to be completely silent for 60 seconds. Place one hand flat on your chest and count every time your chest rises. Your partner watches the clock on the board. Ready — go.' [Allow 60 seconds of silence.] 'Now stop. Write that number down — that is your resting breathing rate.' Repeat the protocol for pulse rate, guiding learners to place two fingers (index and middle) on the inside of the wrist below the thumb: 'Press gently until you feel a pulse. Count beats for 60 seconds — your partner times you.' After data is posted on the board, give 10 seconds of wait time while learners scan the class data, then cold-call 3 learners: 'Achieng — what pattern do you notice in the breathing rate column?' / 'Kamau — does everyone have the same resting pulse? What does that tell us?' / 'Fatuma — why do you think there is variation across the class?' Then say: 'Now, without looking at anyone else's work, open your notebook and write your prediction. I want a number AND a reason. You have 3 minutes — go.' Circulate and read predictions silently; do not confirm or deny. After 3 minutes: 'Underline your reason. We will come back to it.'	Strategy: INDIVIDUAL WRITTEN PREDICTION (Science & Engineering Practice: Asking Questions and Defining Problems). By committing a prediction to paper before observation, learners activate and expose their prior mental models. The act of writing a 'reason' forces learners to articulate a causal mechanism — even if naïve — which creates a cognitive anchor that will be productively disturbed in the next phase. The class data table makes variation visible, prompting natural curiosity about biological norms and individual differences.	Observable indicator: Every learner has written two numbers (predicted breathing rate and pulse rate post-exercise) AND at least one sentence of causal reasoning. Teacher scans notebooks during circulation — flag learners who have written only numbers without a reason and prompt: 'Tell me why you think that will happen.' Listen for common naïve ideas (e.g., 'lungs get tired,' 'heart pumps harder because legs need more blood') — these will be revisited explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** [Laplace transform to solve an equation] Source: Khan	Learners stand beside their chairs. On the teacher's signal, they perform continuous step-ups (step up with right foot, then left foot onto the chair seat, then step down right, then left) for exactly 2 minutes at a vigorous pace — the	Before exercise: 'Safety first — if anyone has asthma, a heart condition, or feels unwell, please come and sit with me at the front. You will be our official timekeeper and data recorder — that is an equally important scientific role.'	Strategy: WHOLE-CLASS DATA TABLE ANALYSIS (Science & Engineering Practice: Analysing and Interpreting Data). By aggregating every learner's data into one visible table, the class can see that the increase in both rates	Observable indicator: Every learner's notebook contains three written observations (breathing rate change with numbers, pulse rate change with numbers, physical sensations). Check that learners are comparing numbers,

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** B and T Cell Response (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	teacher counts aloud every 15 seconds to maintain urgency. Immediately when the teacher calls 'STOP,' learners sit, locate their radial pulse, and count for 60 seconds while a partner times them. A second partner simultaneously counts the learner's breathing rate for 60 seconds. Both post-exercise values are written on a second sticky note (a different colour if available) and added to the class table next to their resting values. Learners then silently compare: their own resting vs. post-exercise values, and the pattern across the whole class. Key observable data: breathing rate typically doubles (e.g., 14 → 28 breaths/min); pulse rate typically increases by 50–80% (e.g., 72 → 120+ bpm). Some learners report a burning sensation in thigh muscles. Learners write three observations in their notebook: (1) what happened to breathing rate, (2) what happened to pulse rate, (3) any other physical sensations noticed.	Check chairs are stable. Then: 'Everyone else, stand beside your chair. When I say GO, step up onto the seat with your right foot, bring up your left, step down right, step down left — keep going continuously. I will call out the time every 15 seconds. Make it vigorous — this is science, not a gentle stroll.' [Run 2 minutes, calling '15 seconds!', '30 seconds!', '45 seconds!', '1 minute!', etc.] Call 'STOP!' sharply. 'Sit down immediately. Find your pulse — NOW. Partner, start timing. Count for 60 seconds.' [Allow 60 seconds.] 'Write that number. Now count breathing rate — partner times again.' After data is posted, give 15 seconds of wait time for learners to scan the updated table. Then cold-call 4 learners: 'Omondi — what happened to your breathing rate? By how much did it change?' / 'Wanjiru — was your prediction close?' / 'Otieno — did everyone's rates go up by the same amount? What do you notice?' / 'Halima — you mentioned a burning feeling in your legs — where exactly, and when did it start?' Write the word BURNING on the board and circle it. Say: 'That burning sensation is one of the most important clues in this entire unit. Hold onto that observation.'	is universal — not individual — which establishes it as a biological pattern rather than a personal quirk. The teacher deliberately highlights the outlier of muscle burning as a 'discrepant event' within the phenomenon, seeding the later investigation of anaerobic respiration and lactic acid without naming those concepts yet.	not just writing 'it went up.' Cold-call responses reveal whether learners are reading the class data table accurately. Flag any learner who reports no change in pulse rate — check measurement technique (finger placement, counting method) before drawing conclusions.
Explain Phase 📹 **VIDEO:** Laplace transform to solve an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners return to their written predictions from Phase 1 and compare them to their actual observations. In groups of four, they discuss: (1) Was your predicted number close to reality? (2) Was your REASON correct, partially correct, or wrong? (3) Can your group agree on one sentence that explains WHY breathing rate	Say: 'Open your notebooks to your prediction from the start of the lesson. Read your reason again.' [10 seconds wait time.] 'In your group of four, I want one person to read their reason aloud. Then discuss: was the reason right, partially right, or missing something important? You have 4 minutes.' Circulate and listen — do	Strategy: NAÏVE MODEL CONSTRUCTION (Science & Engineering Practice: Developing and Using Models). The naïve model serves a dual purpose: it makes implicit knowledge explicit and visible, and it creates a baseline artefact that learners will physically revise and annotate in every subsequent lesson. By	Observable indicator: Every learner has a labelled diagram with at least three structures identified (e.g., nose/mouth, lungs, heart, blood, muscles) and at least one question mark. Teacher circulates and photographs 4–5 models on a phone to display in later lessons as evidence of growing understanding. Listen during group

📖 **HTML:** [B and T Cell Response (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	and pulse rate both increase during exercise? Groups share out, and the teacher records all proposed explanations on the board without judgment. Learners then individually draw their NAÏVE MODEL — a labelled diagram in their notebooks showing, in their own current understanding, the pathway oxygen takes from the air outside their body, into and through the body, all the way to the muscle cells of their legs during exercise. They label every structure they know. They draw arrows showing movement. Where they are unsure, they draw a question mark. After drawing, learners write one sentence: 'The biggest thing I do NOT understand yet is ___.' This sentence becomes their personal DQB anchor question.	not correct yet. After 4 minutes, cold-call one spokesperson per group (rotate so the same learners are not always called): 'Group by the window — Mwangi, what did your group agree on?' Record key phrases on the board. Common responses will include: 'muscles need more energy,' 'oxygen is needed for energy,' 'lungs need to work harder,' 'heart pumps faster.' Affirm all as partial truths: 'Every one of these ideas has something important in it. Over the next lessons, we are going to find out exactly how they connect.' Then: 'Now I want you to draw. No textbook, no notes — just what YOU think right now. Draw the path oxygen takes from the air to the muscles in your legs. Label everything you know. Use question marks where you are unsure. You have 7 minutes.' Play quiet background music or allow silence. After drawing: 'At the bottom of your model, complete this sentence: The biggest thing I do not understand yet is ___. Be honest — question marks are more valuable to a scientist than false certainty.' Cold-call 3 learners to share their sentence aloud.	explicitly asking learners to mark question marks, the task normalises scientific uncertainty and positions gaps in knowledge as productive starting points rather than failures. The teacher's neutral recording of group explanations (without correction) preserves learner agency and avoids premature closure.	discussion for learners who conflate breathing with cellular respiration (e.g., 'the muscles breathe') — note these learners for targeted questioning in Lesson 2. The 'biggest thing I do not understand yet' sentence is a direct window into misconceptions.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Laplace transform to solve an equation] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner writes their 'biggest unanswered question' from their naïve model on a sticky note (or a torn slip of paper if sticky notes are unavailable). The teacher collects all questions and reads them aloud to the class, then sticks them on a dedicated section of the classroom wall labelled 'OUR DRIVING QUESTION BOARD — Sub-Strand 3.3.' The class, guided by the teacher, sorts the questions into clusters: questions about BREATHING (the mechanics of	Distribute sticky notes or paper slips. Say: 'Write your one most burning question about what we just observed. Not a question you already know the answer to — a real question, something that genuinely puzzles you. One question per slip. You have 2 minutes.' Collect all slips. Read each aloud expressively: 'Where does the oxygen actually go after it enters the lungs?' / 'Why did my legs burn if I was breathing in oxygen?' / 'Is the heart connected	Strategy: DRIVING QUESTION BOARD CONSTRUCTION (Science & Engineering Practice: Asking Questions and Defining Problems). The DQB transforms individual curiosity into a shared community knowledge-building artefact. By physically clustering questions, learners practice the scientific skill of categorisation and begin to see the unit's conceptual architecture before instruction begins. The DQB also creates accountability — learners will	Observable indicator: Every learner has contributed at least one genuine question to the DQB. Teacher checks that questions are causal or mechanistic in nature (beginning with 'Why,' 'How,' 'What causes') rather than factual recall questions (beginning with 'What is the name of'). If a learner's question is purely recall-based (e.g., 'What are the parts of the lungs?'), the teacher redirects privately: 'That is a good start — can you turn it into a WHY or HOW

📖 **HTML:** B and T Cell Response (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	how air enters and leaves), questions about OXYGEN TRANSPORT (how oxygen moves through the body), questions about ENERGY IN CELLS (what happens inside muscles), and questions about EXERCISE RESPONSE (why both heart and lungs respond together). The unit's overarching driving question — 'Why does my body work so much harder to get oxygen into my cells during exercise — and what happens inside those cells when there is not enough?' — is printed or written in large letters at the top of the DQB. Learners copy the four cluster headings and 2–3 representative questions from each cluster into their notebooks as a unit roadmap.	to the lungs?' / 'What makes a breath happen — is it automatic?' As you read, invite brief class responses: 'Hands up — who else had a question like this one?' After reading all, say: 'Let's sort these together.' Involve learners physically in grouping: 'Auma, where would you put this question — breathing, transport, energy, or exercise response? Tell us why.' Pin questions under cluster headings. Then point to the unit driving question at the top: 'Every single question you have posted is hiding inside this one big question. By the end of this unit, you will be able to answer all of them — but only if you stay curious and keep connecting your new knowledge back to what you felt in your own body today.' Instruct learners: 'Copy the four cluster headings and two questions from each cluster. This is your learning roadmap for the next several lessons.'	return to it in every lesson to mark questions as 'answered' and to add new, more sophisticated questions as their understanding deepens.	question? Why are those parts shaped the way they are?' The four cluster headings in learner notebooks serve as a formative check at the end of the lesson.
Model Building Phase 📹 **VIDEO:** Laplace transform to solve an equation Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** B and T Cell Response (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners pair up and share their naïve models with a partner — not to be corrected, but to notice: 'What did your partner draw that you forgot to include? What did you draw that they did not?' Each learner adds one new element to their model in a different colour pen/pencil, labelled 'Added after partner discussion — Lesson 1.' This small revision practice establishes the habit of cumulative model revision that will continue throughout the unit. The teacher then briefly orients learners to the ARES platform reading assigned for homework/independent study: a short text on the functions of the circulatory system — how the heart, blood, and blood vessels	Say: 'Find a partner you have not worked with yet today. Spend 3 minutes sharing your naïve models — take turns. No correcting each other. Just notice: what did they include that you missed?' [Allow 3 minutes.] 'Now pick up a different colour pen. Add one thing to your model that you learned from your partner. Write next to it: Added after partner discussion — Lesson 1. Date it.' Circulate and observe — comment positively on specific details: 'Njoroge, I notice you added arrows showing the direction of blood flow — excellent scientific habit.' Then project or hold up the ARES reading title. Say: 'Tonight on ARES, you will read about the circulatory system.	Strategy: CUMULATIVE MODEL REVISION with COLOUR-CODED LAYERS (Science & Engineering Practice: Developing and Using Models). Establishing the colour-coded revision habit in Lesson 1 — even before learners have learned any new content — normalises the idea that scientific models are always incomplete and always improvable. The different ink colours will create a visual timeline of growing understanding across the unit. The targeted ARES reading assignment bridges the lesson to Lesson 2 by directing attention to the circulatory system without delivering the explanation — learners must find the evidence themselves.	Observable indicator: Every learner's naïve model now has at least one element added in a second colour, dated and labelled 'Lesson 1.' Exit reflection sentences contain specific numbers from the class data table (e.g., 'I observed my breathing rate went from 16 to 30 breaths per minute, and I am now wondering where exactly the oxygen goes after it leaves the lungs'). Teacher collects 5–6 exit slips at random — use these to plan the opening of Lesson 2. Any learner whose exit reflection contains no numbers or whose wonder question is absent should be followed up at the start of Lesson 2.

	work together to transport materials. Learners are told: 'As you read, look for evidence that helps you answer the cluster question: How does oxygen get transported from the lungs to the muscles? Annotate your reading — circle anything that connects to what you felt in your body today.' The lesson closes with a 60-second 'exit reflection': learners write one sentence — 'Today I observed ___, and I am now wondering ___.'	Here is your reading mission: find at least TWO sentences in the reading that help explain why your heart beat faster during exercise. Write them in your notebook with a star next to them. We will start Lesson 2 by sharing what you found.' Close with the exit reflection: 'Last 2 minutes — on the bottom of your notebook page, complete this: Today I observed ___, and I am now wondering ___. Be specific — write actual numbers from today's data.' Collect or visually scan exit reflections as learners leave.		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did the class data table show a universal increase in both rates for all learners? If any learners showed no change, was it a measurement error or a technique issue — and how will you address pulse measurement technique in Lesson 2? (2) Which cluster on the Driving Question Board attracted the most questions — breathing mechanics, oxygen transport, cellular energy, or exercise response? This tells you where learner curiosity is strongest and where prior knowledge is weakest. Plan to honour the most-populated cluster early in the unit sequence. (3) What were the most common naïve model errors? (Common ones: oxygen going directly from lungs to muscles without blood; the heart being drawn separately from the lungs with no connecting pathway; no representation of cells at all.) Document these as targets for explicit model revision in Lessons 2–3. (4) Did the burning-in-the-legs observation surface naturally from learners, or did only one or two mention it? If it did not surface widely, plan a brief 2-minute re-opening at the start of Lesson 2: 'Raise your hand if you felt any discomfort in your leg muscles after the step-ups. Let us talk about that for 2 minutes — because that sensation is a clue to something happening inside your cells that is completely different from normal breathing.' (5) How will you ensure the DQB remains a live, evolving document — not just a Day 1 display? Plan to begin each subsequent lesson by pointing to the DQB and asking: 'Which of our questions can we now answer, even partially, based on what we learned today?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners measured their own resting breathing rate and pulse rate, performed 2 minutes of step-up exercise, then measured both rates again immediately after. The class data table showed that breathing rate approximately doubled (e.g., 14 → 28 breaths/min) and pulse rate rose sharply (e.g., 72 → 120+ bpm) in every learner. Some learners also reported a burning sensation in their thigh muscles during or after the exercise.
What did I learn?	Learners confirmed through their own body data that both breathing rate and pulse rate increase measurably and universally during vigorous physical exercise. They observed that the two systems — breathing and heartbeat — appear to respond together at the same time. They also noted that the burning sensation in leg muscles is a separate, puzzling observation that their current knowledge cannot yet explain.
How does this explain the phenomenon?	Learners generated their own first-draft explanations in small groups: muscles need more energy during exercise, oxygen is somehow needed for that energy, and the lungs and heart must be working together to supply it. These explanations are partial and incomplete — they do not yet account for how oxygen moves from the lungs to the muscles, what happens to oxygen inside

	cells, or why muscles burn even when oxygen is being breathed in. These gaps are now posted on the Driving Question Board and will be systematically answered across the unit.

LESSON 2 (80 minutes): Characteristics of Respiratory Surfaces — What Makes a Good Gas Exchange Surface?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to identify and compare the structural characteristics of respiratory surfaces across different animals and relate these features to efficient gas exchange.
Knowledge	Learners will be able to: (a) identify the key structural features of respiratory surfaces (large surface area, thin walls, moist surface, rich blood/fluid supply); (b) compare respiratory surfaces in earthworms (skin), fish (gill lamellae), insects (tracheal tubes), and mammals (alveoli); (c) explain how each structural feature enhances the rate of diffusion of gases.
Skills	Learners will be able to: (a) observe and interpret diagrams and prepared slides of respiratory surfaces; (b) complete a structured comparison table using evidence gathered from observations; (c) carry out and interpret a limewater demonstration as evidence that exhaled air contains carbon dioxide.
Attitudes	Learners will: (a) appreciate the precision of biological structures in meeting the functional demands of organisms; (b) demonstrate curiosity and careful observation during slide examination and demonstration; (c) show responsibility when handling laboratory materials including limewater.
Key Inquiry Question	How do animals exchange gases with their environment?
Purpose in Storyline	In Lesson 1, learners anchored their investigation in the personal phenomenon — their own breathing and pulse rates doubling after step-ups. They began asking: where does the oxygen actually go, and how does it get into the body in the first place? Lesson 2 zooms in on the entry point — the respiratory surface itself. By examining four different respiratory surfaces (skin, gills, tracheal tubes, alveoli), learners gather the structural evidence needed to answer why the mammalian lung, with its millions of alveoli, can supply enough oxygen to fuel a sprinting human being. The limewater demonstration provides direct chemical evidence that gas exchange is actually happening — CO_2 is a real product of the process — linking structure directly back to the phenomenon and priming Lesson 3's investigation of breathing mechanics.
Safety Notes	Limewater (calcium hydroxide solution) is an irritant. Learners must not pipette by mouth. Provide safety goggles and warn learners to avoid contact with eyes or skin. If limewater contacts skin, rinse immediately with water. Prepared slides are fragile — demonstrate correct handling. Dispose of limewater safely by diluting with water before pouring down the sink.

B. LESSON OVERVIEW

Learners open the lesson by revisiting their personal data from the anchoring phenomenon and the Driving Question Board (DQB) populated in Lesson 1. They make written predictions about what a respiratory surface must look like in order to allow gases to move quickly in and out. They then rotate through four observation stations — earthworm skin diagrams, fish gill lamellae diagrams/slides, insect tracheal tube diagrams, and mammalian alveoli diagrams/slides — completing a structured comparison table for each. A whole-class limewater demonstration provides chemical evidence that exhaled air contains CO_2, confirming gas exchange is actively occurring. In the Explain phase, learners use their comparison tables to construct a generalised list of features shared by all efficient respiratory surfaces, explicitly connecting surface area, thickness, moisture, and vascularity back to their own doubling breathing rates. The lesson closes with a DQB update and a cumulative revision of learners' class model of what happens when the body works harder.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Surface of Revolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners re-read their own recorded data from Lesson 1 (resting vs. post-exercise breathing rate and pulse). They are shown a single prompt on the board: 'If your breathing rate doubled during step-ups, your lungs had to move twice as much oxygen into your blood in the same amount of time. Draw or list three features you think the surface where this gas exchange happens must have.' Learners write and sketch individually in their science notebooks for 3 minutes, then share predictions with a shoulder partner for 2 minutes. Selected pairs share with the class. Predictions are recorded on a sticky-note cluster on the DQB under the heading 'Our Predictions About Respiratory Surfaces.' No predictions are corrected at this stage — all are honoured as starting points.	Re-display the class data table from Lesson 1 on the board. Say: 'Look at your own numbers — your resting breathing rate and your rate after step-ups. That surface inside your chest had to double its work in seconds. Take 3 minutes — write or sketch in your notebook: what must that surface look like to do that job?' Allow 3 minutes of silent individual thinking (WAIT TIME enforced — do not circulate and prompt during this period). After partner share, cold-call 4–5 learners using the class register: 'Wanjiku, what did you and your partner predict? Say one feature.' After each response, ask: 'Can anyone add to that — a different feature?' Record predictions verbatim on sticky notes and place on DQB. Say: 'We are not deciding if these are right yet. We are building our best prediction before we look at evidence — just like a scientist would.'	Elicit-and-Record Prediction (Predict-Observe-Explain framework). By committing predictions to the DQB in writing before observation, learners create a cognitive anchor that makes the subsequent evidence meaningful. Misconceptions made visible here (e.g., 'the surface must be hard and thick to be strong') become productive targets for revision during the Explain phase.	Teacher observes notebook sketches and sticky-note predictions for the presence or absence of surface-area reasoning (e.g., 'folded,' 'many small parts,' 'big area') and thickness reasoning ('thin walls'). If fewer than half of learners mention either feature, the teacher notes this as a concept gap to address explicitly during the Explain phase.
Observe Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Surface of Revolution (Flexbook) Source: CK-12 🔍 Search ARES	Learners rotate through four stations (approximately 8 minutes each, teacher signals rotation). Each station has: a large labelled diagram (A3 print or projected image), a guiding observation card with three focus questions, and — where available — a prepared slide and hand lens or light microscope. Station A: Earthworm skin (cutaneous exchange) — cross-section diagram showing mucus layer, capillary network, thin epidermis. Station B: Fish gill lamellae — diagram of gill arch with filaments and lamellae,	Before rotations begin, model the observation process at Station D for 2 minutes: 'I am going to show you how to read this station. I look at the diagram first — I notice the wall is extremely thin, about one cell. I write: wall thickness — one cell thick. I ask: is there a blood supply? Yes — dense capillaries. I write that. Now you will do the same at every station.' Circulate during rotations, spending approximately 2 minutes per station cluster. At Station B (gills), prompt: 'Look at the shape of the lamellae — why might this shape	Structured Observation with a Comparison Table (Evidence-to-Pattern strategy). The comparison table forces learners to process each surface against the same four criteria, making cross-animal patterns visible. The four-criteria framework (surface area, thickness, moisture, vascularity) is the analytical lens that transforms raw observation into generalisable biological principles. The limewater demonstration adds a chemical evidence layer — learners see that CO_2 is a real, detectable output of the process, connecting structural	Teacher checks comparison tables for two key indicators: (1) Learners correctly identify alveoli as having the largest relative surface area and thinnest walls. (2) Learners note moisture in every entry. If either is missing from more than one-third of tables during circulation, teacher pauses the class for a 90-second whole-group clarification before the next rotation. Observable indicator for limewater: learner verbal responses correctly identify 'CO_2' (not just 'gas' or 'breath') as the cause of the milky change — this

for similar readings	showing countercurrent flow. Station C: Insect tracheal system — diagram of spiracles, tracheal tubes, and tracheoles reaching body cells directly. Station D: Mammalian alveolus — diagram and slide showing alveolar wall (one cell thick), dense capillary network, surfactant moisture layer. At each station, learners complete one row of a pre-printed comparison table in their notebooks: columns are 'Respiratory surface,' 'Surface area (small/medium/large/very large),' 'Wall/surface thickness,' 'Moist? (Yes/No — how maintained?),' 'Blood/fluid supply,' 'Animal and habitat.' A whole-class limewater demonstration runs between Station C and D rotations: the teacher blows gently through a straw into limewater and learners observe the solution turn milky, confirming CO_2 in exhaled air.	matter for surface area? Think for 10 seconds before you write.' (10-second WAIT TIME.) Cold-call one learner per station visit: 'Otieno, what did you write for blood supply at this station — and why does that matter?' For the limewater demonstration, say: 'Watch carefully. I am blowing air from my lungs — the same air that was just in my alveoli — into this limewater. What do you predict will happen? You have 10 seconds.' (10-second WAIT TIME, then take 2 predictions by cold-call.) Perform demonstration. Ask: 'What changed? What does this tell us about what is coming out of our respiratory surface?' After all rotations, give 3 minutes for learners to complete any gaps in their comparison tables.	observation to biochemical reality and back to the phenomenon (why does breathing rate rise? — because CO_2 must be removed).	distinguishes surface-level observation from conceptual understanding.
Explain Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Surface of Revolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners look across their completed comparison tables and individually write one sentence answering: 'What features do ALL four respiratory surfaces share — and why does each feature matter?' They then share in groups of four to build a consensus list. Each group writes their agreed list on a strip of manila paper and posts it on the board. The class then co-constructs a single master list under the heading 'Features of an Efficient Respiratory Surface.' Learners copy the master list and annotate each feature with one sentence explaining how it helps gas exchange by diffusion. Finally, each learner writes a short paragraph connecting the alveolar features back to the anchoring phenomenon: 'Explain why the	After groups post their strips, lead a gallery-walk comparison: 'Walk to the strips posted by other groups. Put a tick next to any feature your group also identified. Put a question mark next to anything you do not understand.' (Allow 3 minutes gallery walk.) Return to seats. Say: 'Let us build one master list. What feature appeared on the most strips?' Cold-call 5 learners in sequence to contribute one feature each. For each feature, ask the probing question: 'How does this feature help a gas molecule move from the air into the blood — or from the blood into the air? Think for 15 seconds before you answer.' (15-second WAIT TIME enforced before accepting response.) When the master list is complete, say:	Consensus List + Annotated Master Framework (Generalisation from Evidence strategy). By deriving the feature list from their own observational evidence rather than receiving it from the textbook, learners own the conceptual framework. The mandatory connection paragraph re-anchors the abstract features to the personal, lived phenomenon — preventing the knowledge from becoming inert facts disconnected from the driving question. Connecting diffusion to the alveolar structure prepares learners for the quantitative treatment of Fick's Law in a later lesson.	Connection paragraphs are the primary formative artefact. Teacher looks for three elements: (1) mention of at least two structural features (surface area and wall thickness are minimum); (2) use of the word 'diffusion' or equivalent concept; (3) explicit link to the learner's own measured breathing rate increase. Paragraphs missing element (3) indicate learners have acquired knowledge but not yet connected it to the phenomenon — teacher flags these learners for targeted questioning in Lesson 3's opening review.

	structural features of alveoli allow your breathing rate to double during exercise and still supply enough oxygen to your muscle cells.'	'Now write the connection paragraph in your notebook. Use your own step-up data from Lesson 1. You have 5 minutes.' Circulate and read over shoulders — if a learner's paragraph does not mention diffusion or surface area, write a margin prompt: 'How does surface area affect how many molecules can cross at once?' Do not correct — prompt only.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Area of a Surface of Revolution (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	The class DQB (physical board or displayed digitally) is revisited. Learners look at the questions posted in Lesson 1 and the prediction sticky notes posted at the start of this lesson. Working in pairs, learners identify: (a) one question on the DQB that they can now partially answer using today's evidence — they write a brief answer on a green sticky note and attach it to the relevant question; (b) one new question that today's observations have raised — written on a yellow sticky note and added to the board. The teacher then reads aloud two or three of the partial answers and two or three of the new questions, narrating the class's growing understanding: 'We started today not knowing what a respiratory surface looks like. Now we know four features. But notice — we have new questions too. That is what real science looks like.'	Point to the DQB and say: 'In Lesson 1 we asked: where does the oxygen actually go once it enters the body? Look at what we observed today. Can any group now give a partial answer to that question?' Cold-call one group to read their green sticky note aloud. Ask the class: 'Do you agree? What evidence from today supports this partial answer?' Take 2–3 responses. Then say: 'Now — what new questions do today's observations raise for you? What do you still not understand?' Allow 2 minutes of silent individual writing before pairs share and choose one question for the yellow sticky note. After sticky notes are posted, read two new questions aloud. Say: 'These are exactly the questions we will investigate over the next lessons — about how breathing actually works, and what happens inside the muscle cell when oxygen arrives. Hold onto your curiosity.'	Driving Question Board Update (Iterative Sense-Making strategy). The DQB functions as a public, cumulative record of the class's epistemic progress. Colour-coded responses (green = partial answer, yellow = new question) make the growth of understanding visible and honour the generative nature of inquiry — new questions are a sign of deeper thinking, not failure. This phase also provides metacognitive practice: learners must evaluate what they now know versus what they still need to investigate.	Teacher reads all green sticky notes posted. Observable indicator: partial answers must reference at least one structural feature and connect it to gas exchange (e.g., 'The alveoli have a large surface area so oxygen can diffuse into the blood quickly'). Vague answers ('the oxygen goes into the lungs') indicate the learner has not yet connected structure to function — teacher notes these pairs for follow-up. New yellow questions that reference 'what happens inside the cell' or 'how does oxygen get from the alveolus to the muscle' indicate learners are tracking the storyline correctly toward cellular respiration.
Model Building Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan	Learners return to the simple whole-class model begun in Lesson 1 (a sketch of a person with arrows showing air in and out, and rising numbers for breathing rate and pulse). Each learner now adds a 'zoom-in' panel to their own	Say: 'Our model from Lesson 1 showed air going in and out and numbers going up. But it was a black box — we did not show what happens at the surface. Today we can fix that. Add a zoom-in panel to your model showing the	Cumulative Model Revision with 'I Used to Think / Now I Think' Frame (Conceptual Change Modelling strategy). The zoom-in panel technique teaches learners that scientific models operate at multiple scales simultaneously —	Teacher reviews notebook model revisions for two indicators: (1) The zoom-in alveolus diagram includes at least three of the four key features with labels. (2) The 'I used to think / Now I think' sentence shows genuine

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Area of a Surface of Revolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	notebook copy of the model: a labelled diagram of an alveolus showing its four key features (large surface area via curvature, one-cell-thick wall, moist inner surface, dense capillary network) with small arrows showing O_2 moving in and CO_2 moving out by diffusion. Below the diagram, learners write: 'I used to think the respiratory surface was __________. Now I think it is __________ because __________.' The class model on the board is updated by a volunteer who adds the zoom-in panel with teacher guidance. Learners note which parts of the driving question the model can now help answer and which parts remain unanswered (breathing mechanics, cellular respiration, heart role).	alveolus with all four features.' Allow 5 minutes for individual model revision. Circulate and prompt with: 'Where is the blood in your diagram? How thick is the wall — show that with your line.' After 5 minutes, invite one volunteer to draw the zoom-in on the class board model. Ask the class: 'Does this model match your evidence from the stations? Is anything missing?' Cold-call 3 learners to evaluate the class model. Close by asking: 'Our model now explains how gas gets across the respiratory surface. But it still does not explain what happens inside the muscle cell once the oxygen arrives — or why we feel that burning in our legs. What do we need to find out next?' Take 2–3 responses without correcting — let learner curiosity drive the bridge to Lesson 3.	the whole-body level (breathing rate) and the cellular level (alveolar structure) must be connected in one coherent model. The 'I used to think / Now I think' frame makes conceptual change explicit and metacognitive, consolidating the lesson's learning while building scientific identity. The unresolved questions about the muscle cell deliberately create productive tension that motivates Lesson 3.	conceptual shift (not merely 'I used to think nothing, now I think alveoli exist'). Models that show only two features or no directional arrows for gas movement indicate the learner needs additional structural reinforcement — teacher targets these learners with a brief one-to-one conversation at the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) During the station rotations, which respiratory surface caused the most confusion or generated the most questions — was it the insect tracheal system (no blood supply, gas delivered directly to cells)? If so, how will you address the conceptual leap this requires at the start of Lesson 3? (2) Did learners' connection paragraphs in the Explain phase successfully link alveolar structure back to their own step-up data, or did the majority write about structure in isolation from the phenomenon? If the phenomenon connection was weak, plan a 5-minute re-anchor at the start of Lesson 3 using two or three learner-written paragraphs (anonymised) as discussion models. (3) Review the new questions posted on the DQB. Count how many reference 'what happens inside the cell' or 'cellular respiration' — these learners are tracking ahead of the storyline and may benefit from extension prompts in Lesson 4. (4) Was the limewater demonstration sufficient to establish CO_2 as a real, chemical product of gas exchange, or did learners describe the milky change without connecting it to CO_2? If the latter, consider adding a second demonstration at the start of Lesson 3 using bicarbonate indicator to make the CO_2 identification more explicit. (5) Note the names of any learners whose model revisions were missing two or more alveolar features — prepare a differentiated diagram scaffold (partially labelled outline) for these learners to use in Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined diagrams and slides of four respiratory surfaces — earthworm skin, fish gill lamellae, insect tracheal tubes, and mammalian alveoli — and completed a comparison table recording surface area, wall thickness, moisture, and blood/fluid supply for each. They also observed limewater turning milky when exhaled air was blown through it.

What did I learn?	All efficient respiratory surfaces share four structural features: large surface area, thin walls (one to a few cells thick), a moist surface (to keep gases in solution for diffusion), and a rich blood or fluid supply (to maintain a concentration gradient). Mammalian alveoli have the most extensive surface area and the thinnest walls, making them highly adapted for the rapid, high-volume gas exchange needed during exercise. Carbon dioxide is a real, chemically detectable product that exits through the respiratory surface.
How does this explain the phenomenon?	The structural features of the alveolus directly explain why the breathing rate can double during exercise and still supply enough oxygen: the enormous combined surface area of millions of alveoli maximises the number of gas molecules that can diffuse simultaneously, while the one-cell-thick moist wall minimises the distance gases must travel. This structural efficiency is the reason a human sprinting near Uhuru Park in Nairobi can sustain high muscle activity — the respiratory surface is built for exactly this demand.

LESSON 3 (80 minutes): Structure and Adaptations of Respiratory Surfaces	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explain how the structural features of respiratory surfaces are adaptations that solve the problem of efficient gas exchange in animals.
Knowledge	Learners identify and explain the four key adaptations of respiratory surfaces (large surface area, thin walls, moist surface, rich blood supply) and describe how these features are present in mammalian alveoli, fish gills (tilapia), and insect tracheae.
Skills	Learners annotate biological diagrams of respiratory surfaces, construct comparison tables, and revise their cumulative explanatory model to include a 'comparative respiratory surfaces' layer.
Attitudes	Learners appreciate how structural form is precisely matched to biological function, and value the Lake Victoria tilapia fishery as a real-world context connecting biology to Kenyan livelihoods.
Key Inquiry Question	How do animals exchange gases with their environment?
Purpose in Storyline	In Lessons 1 and 2, learners gathered evidence that breathing rate and pulse rate both rise sharply during exercise, and began asking WHERE oxygen goes after entering the lungs and HOW it crosses into the blood. Lesson 3 answers that structural question by revealing that every efficient gas-exchange surface shares four key adaptations — and that understanding those adaptations is the bridge to explaining why the alveolus is fast enough to supply working muscle cells. This sensemaking lesson directly advances the driving question by explaining the anatomical 'gateway' through which oxygen must pass before it can reach the muscle cells that were burning during step-ups.
Safety Notes	No hazardous materials are used in this lesson. If preserved fish specimens or fresh tilapia gills are brought in for observation, learners should wash hands thoroughly before and after handling. Remind learners not to touch their faces during specimen observation. Dispose of any biological material as directed by the school's waste-management protocol.

B. LESSON OVERVIEW

This is an EXPLAIN lesson — the third in the evidence-gathering sequence. Learners move from the observable evidence of Lessons 1 and 2 (rising breathing and pulse rates) into structured sensemaking about WHY the body can exchange gases quickly enough to meet the demand. The teacher guides learners through a 'Structure → Adaptation → Function' framework applied to three respiratory surfaces: human alveoli, tilapia gills (Lake Victoria fishery context), and insect tracheae. Learners annotate diagrams, build a comparison table, and add a new 'comparative respiratory surfaces' layer to their cumulative explanatory model. By the end of the lesson, learners can explain how each structural feature solves a specific gas-exchange problem, directly linking alveolar anatomy back to the doubled breathing rate they personally measured at the start of the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Elements and](#)	Learners are shown a large projected image of two surfaces side by side: (A) a smooth, flat	Display the two surface images on the projector without labelling them. Say: 'Do not look anything	Prediction Journaling with Public Record: Learners externalise prior knowledge by committing	Observable indicator: Every learner has at least three written prediction reasons in their exercise

atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plants' Adaptations for Life on Land (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	sheet of plastic (like a plastic bag) and (B) a crumpled, wet sponge. Without any explanation, learners are asked: 'If oxygen had to pass from one side to the other as quickly as possible, which surface would work better — and why? Write three reasons in your exercise book before we discuss.' Learners write individually for 3 minutes, then share predictions with a shoulder partner. Selected pairs share with the class. The teacher records key prediction words on the board (e.g., 'more surface', 'thinner', 'wet', 'faster') without evaluating them — these will be revisited at the end of the lesson. Learners are then asked to hold their prediction and connect it back to the phenomenon: 'Think about the step-ups from Lesson 1. Your alveoli had to pass enough oxygen into your blood to double what your muscles needed. Do you think a flat plastic surface could have done that job? Why or why not?' Learners record their prediction in their books.	up. This is your prediction — your best scientific thinking right now.' Allow 3 minutes of individual silent writing. Use a visible countdown timer. After writing, say: 'Turn to your shoulder partner. You have 90 seconds — share your three reasons.' [WAIT TIME: 10 seconds after giving the instruction before starting the timer.] Cold-call 3 pairs (not volunteers — use the class register to select): ask each pair to share ONE reason. Record their exact words on the board under the heading 'Our Predictions'. Do NOT correct or affirm. Say: 'We are going to test every one of these ideas today. Keep your predictions — we will come back to them.' Transition: 'Now let us connect this to our own bodies from Lesson 1. [WAIT TIME: 12 seconds] Think about this — your breathing rate doubled during step-ups. That means your alveoli had to work twice as hard to get oxygen across into your blood. What would an alveolus need to look like to do that job so fast?' Cold-call 2 learners. Record answers. Do not resolve — preserve cognitive tension.	predictions to writing before instruction. Recording predictions publicly on the board creates accountability and a reference point for conceptual change. The connection to the step-up phenomenon ensures predictions are grounded in personal evidence rather than rote recall.	book before discussion begins. Teacher circulates during the 3-minute writing period and scans books — look for learners who write only one word (e.g., 'thin') without explanation; prompt them quietly: 'Why does being thin help? Write the reason.' Listen to pair discussions to identify the most common misconception (likely: 'surface area doesn't matter as long as it is thin') — note this to address explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plants' Adaptations for Life on Land (Flexbook)	Learners rotate through three structured observation stations (10 minutes each, teacher signals rotation). STATION 1 — Human Alveoli: Learners examine a printed cross-section diagram of alveoli with capillaries (distributed as an A4 handout). Using a ruler and the scale bar on the diagram, they measure and record the estimated wall thickness of one alveolus and the estimated diameter of one alveolus. They also count the number of capillaries visible touching a single	Before stations open, say: 'You are scientists collecting structural evidence. You are NOT reading about function yet — just observe and measure what you see. Your job is to notice features, not explain them.' [WAIT TIME: 10 seconds.] Assign station groups using a pre-prepared seating plan to ensure mixed ability. Post a laminated 'Station Task Card' at each station with clear numbered instructions so learners are self-directed. Circulate — spend approximately 3 minutes at each	Structured Observation with Measurement: By requiring learners to measure (wall thickness, number of lamellae, branching count) rather than simply look, the station activity transforms passive viewing into active evidence collection. Anchoring Station 2 in the Lake Victoria tilapia fishery gives Kenyan relevance and makes the biology feel locally meaningful. The three-station comparison primes learners to look for shared patterns — the four adaptations —	Observable indicator: Each learner's observation table has numerical entries (not just 'thin' or 'many') for at least two of the three stations. Teacher checks tables during the final 5-minute recording period — if a learner has only written adjectives without measurements or counts, prompt: 'Can you give me a number? Estimate and record it.' Listen for the word 'surface area' emerging spontaneously during station discussions — note which learners use it correctly (they are ready to

Source: CK-12 🔍 Search ARES for similar readings	alveolus in the diagram. They record all measurements in a structured observation table in their books. STATION 2 — Tilapia Gill (Lake Victoria context): If available, a fresh or preserved tilapia gill arch is placed in a tray for direct observation with hand lenses. If specimens are unavailable, a high-resolution printed photograph of a tilapia gill with gill filaments and lamellae visible is used. Learners observe and sketch the gill arch, labelling: gill arch, gill filaments, gill lamellae. They record: estimated number of filaments on one arch (count or estimate), whether the surface appears moist, and whether blood vessels are visible. A printed card at the station reads: 'Tilapia from Lake Victoria are one of Kenya's most important food fish. Fishers at Kisumu, Homa Bay, and Mbita land thousands of tonnes each year. A tilapia's gills must extract dissolved oxygen from water efficiently enough to keep the fish alive and active.' STATION 3 — Insect Tracheal System: Learners examine a printed diagram of an insect tracheal system showing spiracles, tracheae, and tracheoles reaching muscle cells. They trace with a pencil the path oxygen travels from the spiracle to a muscle cell. They record: how many times the tube branches before reaching a cell, and the estimated diameter of a tracheole compared to a trachea on the diagram. After rotating through all three stations, learners return to seats and complete their observation table (5 minutes).	station per rotation cycle. At Station 2 (tilapia gill), ask: 'What do you notice about the number of lamellae? What does that do to the total surface available?' [WAIT TIME: 12 seconds.] Do NOT answer — let learners record their thinking. At Station 1 (alveoli), prompt: 'Use your ruler. How thick is that wall compared to a strand of your hair? Record the comparison.' At Station 3 (tracheal system), prompt: 'Count the branches from spiracle to muscle. What pattern do you notice?' At rotation signal, say: 'Record your final observation — you have 30 seconds.' After all rotations, give 5 minutes for learners to complete their observation tables and write one 'I notice...' sentence per station in their books. Cold-call 4 learners (from different stations) to share one observation each. Record on board. Say: 'Hold these observations — in the next phase, we will explain what every one of them means.'	before those patterns are named in the Explain phase.	lead in the Explain phase) and which do not (they need the explicit instruction).
Explain Phase 📹 **VIDEO:**	The teacher introduces the 'Structure → Adaptation →	Write the 'Structure → Adaptation → Function' framework on the	Structure–Adaptation–Function Framework with Diagram	Observable indicator 1: Every learner's diagram has at least

Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plants' Adaptations for Life on Land (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Function' framework using a class anchor chart written on the board: STRUCTURE (what you can see/measure) → ADAPTATION (how it is shaped by natural selection) → FUNCTION (the problem it solves for the organism). Learners are guided through each of the four key adaptations in turn, using the evidence they just collected at the stations. For each adaptation, the teacher explains the concept, then learners immediately annotate their station diagrams. ADAPTATION 1 — Large Surface Area: Teacher explains that folding or subdividing a surface dramatically increases the area available for diffusion without increasing the body's overall size. Learners annotate their alveolus diagram: draw arrows to the folded inner surface; label 'large surface area'. Teacher connects to tilapia: 'The gill lamellae of a Nile tilapia from Lake Victoria look like the pages of a book — hundreds of thin plates stacked together. Each plate is a surface for gas exchange. Why does the fish need so many?' [WAIT TIME: 12 seconds.] Cold-call 2 learners. ADAPTATION 2 — Thin Walls (Short Diffusion Distance): Teacher explains Fick's Law in accessible terms: 'Oxygen diffuses faster when it has a shorter distance to travel. The wall between the air in the alveolus and the blood in the capillary is only one cell thick — about 0.2 micrometres. That is why your body can double its gas exchange rate in seconds when you start running.' Learners annotate: label 'one-cell-thick wall' on the alveolus diagram and 'thin lamella wall' on	board before learners return from stations. Say: 'Every structure you observed at the stations is an answer to a problem. Your job now is to figure out which problem each structure solves.' Introduce each adaptation with a direct question before explaining: ADAPTATION 1: 'Look at your alveolus diagram. How many alveoli do you think are in two human lungs? Record your guess.' [WAIT TIME: 10 seconds.] Reveal: 'About 700 million. If you spread them all flat, they would cover half a badminton court — roughly 70 square metres. Why does that matter for someone doing step-ups?' Cold-call 2 learners. ADAPTATION 2: 'What did you measure as the alveous wall thickness at Station 1? Compare it to the thickness of this paper.' Hold up the handout. 'This paper is about 100 micrometres thick. The alveolus wall is 0.2 micrometres — 500 times thinner. What does that do for the speed of diffusion?' [WAIT TIME: 12 seconds.] Cold-call 2 learners. ADAPTATION 3: 'At the tilapia station, did the gill surface look dry or wet? Why does that matter?' [WAIT TIME: 10 seconds.] Cold-call 1 learner. Connect explicitly: 'When you were doing step-ups and breathing fast, your mouth dried out — that is because water evaporates as you exhale. Your alveoli must stay moist. How does the body solve this problem? [WAIT TIME: 12 seconds.] We will find out in a later lesson — add it as a question on the DQB.' ADAPTATION 4: After explaining concentration gradient, say: 'Imagine trying to carry water out of a flooded room in one bucket versus a team of 100 people with	Annotation: This strategy (a form of guided note-making) makes the explanatory logic visible and transferable. By requiring learners to annotate the same diagrams they observed at stations, the lesson creates a direct perceptual link between evidence and explanation. The Comparison Table builds analogical reasoning — learners see that evolution arrived at the same four solutions across very different animals, which deepens understanding of the underlying diffusion principles. The Function Summary sentence stem scaffolds scientific writing while requiring learners to integrate three levels of biological explanation.	three of the four adaptations annotated with both a label AND a brief functional note (not just the word 'thin' but 'thin → short diffusion distance'). Teacher does a rapid gallery walk during the Comparison Table activity — scan for any learner who has left the 'Rich Blood Supply' column blank (this is the hardest concept; prompt with: 'What happens to the concentration gradient if oxygen is not removed from the blood side?'). Observable indicator 2: Function Summary sentences use the three-part structure (structure → feature → mechanism). Collect three books at random at the end of the lesson to check — if the mechanism clause is missing, this is the target for the next lesson's opening review.

	the tilapia diagram. ADAPTATION 3 — Moist Surface: Teacher explains that oxygen must dissolve in a film of water before it can diffuse across the membrane. Learners annotate: 'moist lining' on the alveolus; 'water bathing lamellae' on the tilapia diagram. Teacher asks: 'What would happen to gas exchange if the alveolus dried out? Think about what happened to your breathing during the step-ups — you may have noticed your mouth got drier. Is that a coincidence?' [WAIT TIME: 15 seconds.] Cold-call 3 learners. ADAPTATION 4 — Rich Blood Supply: Teacher explains that a dense capillary network maintains a steep concentration gradient by continuously removing oxygen from the exchange surface. Learners annotate: 'dense capillary network' and 'steep concentration gradient maintained' on the alveolus diagram. Learners then complete a Comparison Table (pre-printed or drawn in books) with columns: Organism | Respiratory Surface | Large SA | Thin Walls | Moist | Rich Blood Supply — filling in Human (alveoli), Tilapia (gill lamellae), Insect (tracheoles). Finally, learners write a 3-sentence 'Function Summary' for ONE adaptation of their choice, using the sentence stem: 'The [structure] is adapted by [feature] which [solves the problem of] because [mechanism].'	buckets. Which clears the room faster? That is what the capillary network does for oxygen.' For the Comparison Table, give learners 6 minutes to complete independently, then pair-share for 2 minutes. Cold-call 3 pairs to confirm one row each. For the Function Summary, model the sentence stem with one example: 'The alveolus is adapted by having thin walls, which solves the problem of slow diffusion, because a shorter distance means oxygen molecules cross faster.' Then say: 'Now write your own for a DIFFERENT adaptation.' [WAIT TIME: allow 4 minutes of silent writing.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Elements and atoms]()	Learners return to the class Driving Question Board (a large poster or section of the board reserved across all 12 lessons). The teacher reads aloud the driving question: 'Why does my body work so much harder to get	Read the driving question aloud slowly and say: 'Point to the part of this question that Lesson 3 has helped us answer.' [WAIT TIME: 10 seconds.] Cold-call 3 learners and ask them to read the specific phrase they pointed to. Guide the	Cumulative Model Revision (Progressive Model Building): The DQB and model-building strategy makes learning visible as an ongoing process rather than a series of disconnected lessons. By physically adding a new layer to an	Observable indicator: Each learner's cumulative model now has a labelled alveolus cross-section with at least three of the four adaptations named AND an arrow with a question mark pointing toward 'muscle cell'.

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plants' Adaptations for Life on Land (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	oxygen into my cells during exercise — and what happens inside those cells when there is not enough?' The teacher then guides learners through a three-part DQB update: (A) ANSWERED TODAY — Learners identify which part of the driving question Lesson 3 has helped answer. They write on a sticky note (or the teacher scribes on the board): 'We now know that oxygen crosses into the blood through alveoli because they have a large surface area, thin walls, a moist surface, and a rich blood supply.' This note is placed in the 'Evidence We Now Have' column of the DQB. (B) NEW QUESTIONS RAISED — Learners write one new question that today's lesson created. Likely questions: 'How does oxygen actually get from the blood INTO the muscle cell?' / 'What happens to the oxygen once it is inside the cell?' / 'Why did our mouths dry out — how does the body keep the alveolus moist?' / 'What is the difference between breathing and respiration?' These are added to the 'Still Wondering' column. (C) MODEL CONNECTION — Learners open their cumulative model (a diagram they have been building since Lesson 1) and add a new labelled layer: a simplified cross-section of an alveolus with the four adaptations labelled, connected by an arrow to the label 'Blood → Heart → Muscle Cell' (a placeholder for what will be explained in later lessons). The teacher points out: 'Look at your model. You can now explain the first gateway — the alveolus. The oxygen is now in the blood. The next question is: what happens	class to agree: 'We have answered HOW oxygen gets into the blood — through the adaptations of the alveolus.' Say: 'But have we answered what happens INSIDE the muscle cells? [WAIT TIME: 8 seconds.] No — that is still ahead of us. What question does that leave?' Cold-call 2 learners to propose new DQB questions. Add them publicly. For the model update, say: 'Open your cumulative model. You have 4 minutes to add the alveolus cross-section layer. Label all four adaptations. Draw an arrow leaving the alveolus labelled: Oxygen → blood. We do not yet know what happens next — draw a question mark after the arrow.' Circulate and verify that every learner's model now includes the alveolus layer. For any learner whose model is incomplete from earlier lessons, note their name and address in the next lesson's opening.	existing diagram, learners experience conceptual growth as spatial and visual accumulation. The 'question mark arrow' technique preserves productive uncertainty — learners know what they do not yet know — which primes curiosity for the cellular respiration lessons ahead. Connecting today's anatomical knowledge back to the phenomenon (step-ups, doubled breathing rate) ensures that structural knowledge is always understood as a functional explanation, not isolated fact.	Teacher photographs two or three models with a phone at the end of the lesson to track model sophistication across the sequence. Listen for the quality of new DQB questions — surface-level questions ('What is respiration?') suggest the learner has not yet connected today's anatomy to the cellular level; deeper questions ('How does oxygen get from the blood INTO the cell?') indicate readiness for the next lesson.

	next?'			
Model Building Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plants' Adaptations for Life on Land (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In the final phase, learners complete a 'Comparative Respiratory Surfaces' model revision activity. Each learner receives (or draws) a blank three-panel template in their book: Panel 1 = Human alveolus, Panel 2 = Tilapia gill (Lake Victoria), Panel 3 = Insect tracheole. For each panel, learners must draw a simplified labelled diagram showing the four adaptations, and write one sentence explaining how that surface is specifically suited to the organism's environment (air-breathing mammal; water-breathing fish in Lake Victoria; small air-breathing insect). Learners then write a 'Big Idea' sentence at the bottom of the three-panel model: 'All efficient respiratory surfaces share the same four adaptations because all animals face the same problem: getting oxygen to cells fast enough to release energy from food.' Finally, learners are asked to re-read their ORIGINAL prediction from the start of the lesson (the flat plastic sheet vs. the crumpled sponge). They write a 'Prediction Check': 'My prediction was [correct / partially correct / incorrect] because...' and explain the change in their thinking using at least one specific piece of evidence from today's lesson. This metacognitive close connects the lesson arc and reinforces that science is a process of refining ideas with evidence.	Say: 'We are now going to build the most important part of today's model — a comparison across three very different animals that shows they all solved the same problem the same way. You have 10 minutes.' [WAIT TIME: 10 seconds to let learners re-read the instructions.] Circulate actively — target learners who leave the 'environment sentence' blank and prompt: 'A tilapia lives in Lake Victoria. The water there has less dissolved oxygen than open ocean water — and the fish must keep swimming. What does that mean for its gill adaptations?' After 8 minutes of individual work, say: 'You have 2 minutes to write your Big Idea sentence. This is the most important sentence in your book today — write it carefully.' [WAIT TIME: allow 2 full minutes of silence.] Cold-call 3 learners to read their Big Idea sentence aloud. Listen for whether they include the phrase 'energy from food' — if not, say: 'Remember the step-ups. Why does the muscle need oxygen in the first place? Add that to your sentence.' Close the lesson by returning to the predictions board: 'Look at the predictions we wrote at the start. Which ones were right? Which were incomplete? [WAIT TIME: 12 seconds.] That is how science works — you make a prediction, gather evidence, and revise. You just did science.' Assign exit task: 'On a half-page, explain to a fisherman at Kisumu market why a tilapia with damaged gill lamellae would struggle to survive — use all four adaptations in your answer.'	Comparative Model Building with Metacognitive Prediction Check: The three-panel comparative model requires learners to apply the same explanatory framework across three biological contexts, which deepens generalisation and prevents the misconception that adaptations are organism-specific facts to be memorised rather than solutions to a universal problem. The Prediction Check is a metacognitive strategy — by explicitly comparing their original prediction to their current understanding and identifying the evidence that changed their mind, learners practise the reflective thinking that is central to scientific literacy. The Kisumu fisherman exit task grounds abstract biology in a Kenyan livelihood context and requires learners to translate scientific understanding into accessible explanatory language.	Observable indicator 1: The three-panel model is complete with labelled diagrams AND environment sentences for all three organisms. Teacher collects the models (or photographs them) as a formative artifact. Observable indicator 2: The Big Idea sentence includes reference to both gas exchange AND energy release from food — if 'energy' is absent, the learner has not yet connected the alveolus to the cellular respiration thread of the driving question. Observable indicator 3 (exit task): The Kisumu fisherman explanation uses at least two of the four adaptations correctly in context. This exit task is read by the teacher before Lesson 4 to identify which learners are ready to move to the cellular respiration thread and which need a brief review of the four adaptations at the start of the next lesson.

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions in your professional journal: 1. CONCEPTUAL CHANGE — Did learners' Prediction Check responses show genuine revision of their initial ideas, or did some learners simply copy the 'correct' answer without connecting it to their earlier prediction? If the latter, how could the prediction phase be made more personal or specific next time? 2. TILAPIA CONTEXT — Did the Lake Victoria tilapia context feel meaningful to your learners, or did it feel like an add-on? If learners in your class have direct experience with fishing (e.g., from Homa Bay, Kisumu, or Mbita), did you leverage that knowledge? Consider inviting a learner whose family fishes to describe the gills they have observed — this lived knowledge is more powerful than any diagram. 3. STATION MANAGEMENT — Were all three stations equally productive, or did one station become a bottleneck? The insect tracheal diagram is often the most abstract — consider whether a physical model (a branching straw structure) might make the tracheole system more concrete for future lessons. 4. FOUR ADAPTATIONS — Which of the four adaptations was hardest for learners to grasp? Typically 'rich blood supply maintaining a concentration gradient' requires more scaffolding because it involves understanding diffusion gradients dynamically rather than as a static feature. Note which learners still conflate 'surface area' and 'thin walls' — they will need explicit differentiation in Lesson 4. 5. DRIVING QUESTION CONNECTION — When learners updated the DQB, could they articulate clearly which part of the driving question had been answered and which remained open? If learners struggled to connect alveolar anatomy to the step-up phenomenon, plan a 5-minute 'phenomenon reconnect' at the start of Lesson 4 using the class data table from Lesson 1. 6. EXIT TASK DATA — Read the Kisumu fisherman exit tasks before Lesson 4. Tally: How many learners used all four adaptations? How many used only two? How many made an error about which adaptation is lost when lamellae are damaged? Use this data to plan the opening review of Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed diagrams and/or specimens of three respiratory surfaces — human alveoli, tilapia gill lamellae (Lake Victoria context), and insect tracheoles — at structured observation stations. They measured estimated wall thicknesses, counted lamellae and capillaries, and traced the branching path of tracheoles to muscle cells.
What did I learn?	Learners learned that all efficient respiratory surfaces share four key structural adaptations: (1) large surface area, (2) thin walls for a short diffusion distance, (3) a moist surface so oxygen can dissolve before crossing, and (4) a rich blood supply to maintain a steep concentration gradient. These adaptations explain why the alveolus can double its oxygen-delivery rate within seconds when exercise begins.
How does this explain the phenomenon?	Learners explained how the four adaptations of the alveolus solve the problem of getting oxygen into the blood fast enough to meet the doubled demand during step-ups. They also explained why a tilapia's gill lamellae must have the same adaptations in an aquatic environment, and began to model the pathway: oxygen → alveolus → blood → (next question) → muscle cell.

LESSON 4 (80 minutes): Mechanisms of Gaseous Exchange in Humans — External Respiration	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explain the mechanics of breathing (inhalation and exhalation) in terms of diaphragm and intercostal muscle movement, pressure changes, and air flow, and to connect these mechanics to the increased breathing rate observed during the anchoring phenomenon.
Knowledge	Learners will be able to: (a) describe the roles of the diaphragm and external/internal intercostal muscles during inhalation and exhalation; (b) explain how changes in thoracic volume create pressure gradients that drive air movement; (c) define tidal volume, inspiration, expiration, and pleural cavity; (d) relate the mechanics of external respiration to the increased breathing rate observed after exercise.
Skills	Learners will be able to: (a) observe and interpret an animation/video of the breathing mechanism; (b) construct and use an improvised bell-jar diaphragm model (plastic bottle version) to simulate pressure changes; (c) record and explain observations linking model behaviour to real lung mechanics; (d) annotate a diagram of the thoracic cavity to show pressure and volume changes.
Attitudes	Learners will: (a) appreciate that the body's breathing machinery is a precisely engineered system; (b) show curiosity about how a physical pressure change can move air in and out of the lungs without direct contact; (c) value careful, systematic observation as a science practice.
Key Inquiry Question	How do animals exchange gases with their environment?
Purpose in Storyline	In Lesson 3 learners identified the respiratory surfaces and structures of the human breathing system. Lesson 4 now answers the next layer of the driving question — not just WHERE exchange happens, but HOW air is actively pumped to and from those surfaces. By building and interrogating the bell-jar model, learners gather direct physical evidence that breathing is a pressure-driven mechanical process, explaining why the rate and depth of that process must increase so dramatically during the step-up exercise every learner experienced at the start of the unit.
Safety Notes	When cutting plastic bottles to construct the bell-jar model, the teacher should pre-cut bottles before the lesson or supervise cutting with scissors only — no razor blades in the hands of learners. Ensure cut edges are smoothed with tape to prevent cuts. Learners with asthma or respiratory conditions should not be asked to perform voluntary hyperventilation or breath-holding sub-tasks. Balloons are a choking hazard for small children but are safe for Grade 10 — however remind learners not to over-inflate. Dispose of used balloons responsibly.

B. LESSON OVERVIEW

Learners begin by revisiting their step-up exercise data from the anchoring phenomenon and noting that breathing rate doubled — but they have not yet explained the physical mechanism that makes each breath happen. The lesson opens with a focused PREDICT task: learners sketch what they think the diaphragm and chest wall do during a breath in. They then OBSERVE an ARES offline video/animation showing diaphragm contraction/relaxation, intercostal muscle action, thoracic volume changes, and the resulting pressure gradient. Immediately after the video, learner groups construct an improvised bell-jar diaphragm model from a 500 mL plastic bottle, two small balloons (lungs), a larger balloon stretched across the cut base (diaphragm), and a Y-shaped straw connector (bronchi). By pulling and pushing the base balloon, learners directly observe the inner balloons inflate and deflate, experiencing the pressure-volume relationship physically. An EXPLAIN phase uses the Claim–Evidence–Reasoning (CER) writing frame to connect model behaviour to real anatomy and to the phenomenon. The lesson closes with DQB updates and a cumulative model revision in which learners add pressure-change annotations to their thoracic cavity diagrams begun in Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** The lungs and pulmonary system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Parametric Forms and Calculus: Volume of Revolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their personal breathing-rate data recorded during the anchoring phenomenon (resting ~14 breaths/min; post-exercise ~28 breaths/min). The teacher asks: 'We know the rate doubled — but what physical movements inside your chest actually make ONE breath happen? In your notebook, draw a simple side-view outline of your chest. Mark where you think the diaphragm sits. Then draw arrows to show what you predict moves during a breath IN.' Learners work silently for 3 minutes, committing a prediction to paper before any instruction. Pairs then compare sketches for 1 minute, noting agreements and disagreements.	Display the class data table from the anchoring phenomenon on the board (or call a learner to read out their own numbers). Say: 'Wanjiku told us she went from 14 breaths per minute to 28. Baraka went from 16 to 30. The numbers are real — we measured them ourselves. Today we are going to find out exactly what physical machinery inside the chest is responsible for each one of those breaths.' Distribute the predict worksheet (or direct learners to a notebook page). Say: 'Sketch your chest outline now — you have 3 minutes of silent thinking time. There is no wrong answer here; I want to see YOUR current thinking before we look at any evidence.' Set a visible 3-minute timer. After the timer, say: 'Turn and share your sketch with your partner. Find ONE thing you agree on and ONE thing you disagree on.' After 1 minute of pair talk, cold-call 3 pairs: 'Pair at the window — what did you agree on?' Record key prediction terms (ribs move out, lungs expand, air rushes in) on the board under the heading PREDICTIONS — DO NOT ERASE.	Strategy: Predict-Before-Observe (PBO). By forcing learners to commit a physical sketch to paper before seeing the animation, PBO activates prior knowledge, surfaces misconceptions (e.g., 'the lungs pull air in by themselves'), and creates a cognitive anchor point. When the animation contradicts a prediction, the surprise drives deeper processing of the correct mechanism — learners are not passively receiving information but actively revising a mental model they already own.	Teacher circulates during the 3-minute silent sketch and looks for two specific indicators: (1) Does the learner place the diaphragm correctly (below the lungs, above the abdomen) or do they draw it at the throat? (2) Do learners show any understanding of volume change, or do they think the lungs actively suck air? Mentally note 2–3 learners with significant misconceptions to target during cold-calling in Phase 2. No marks are given — this is diagnostic only.
Observe Phase 📹 **VIDEO:** The lungs and pulmonary system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners watch the ARES offline animation (approximately 6–8 minutes) showing: (a) the position and shape of the diaphragm at rest, during inhalation, and during exhalation; (b) external intercostal muscles raising the rib cage outward and upward during inhalation; internal intercostal muscles pulling ribs inward and	Before pressing play, say: 'As you watch, you are a scientist gathering evidence. Your job is to fill in your observation table — one row per structure. I will pause the video after each structure so you have time to write.' Play until Timestamp 1 (diaphragm segment ends). PAUSE. Say: 'Diaphragm row — write now. You have 45	Strategy: Structured Note-Taking with Annotation (Observation Table + Prediction Revisit). The four-column observation table is not passive copying — it requires learners to translate visual motion (animation) into descriptive language and causal relationships. The prediction-revisit step immediately after the animation	Teacher collects (or photographs with phone) observation tables from two or three learners during the annotation pause at Timestamp 3. Look for: (1) Correct pairing of diaphragm CONTRACTION with INHALATION — if learners write 'diaphragm moves up during inhalation' this is a key

📖 **HTML:** [Parametric Forms and Calculus: Volume of Revolution (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	downward during exhalation; (c) a dynamic pressure-volume graph alongside the animation — as thoracic volume increases, intrathoracic pressure drops below atmospheric (≈ 101 kPa to ≈ 99 kPa), driving air inward; (d) a brief labelled sequence naming the pleural cavity and pleural fluid, explaining why the lungs follow the chest wall. Learners complete a structured observation table with four columns: Structure | Movement during Inhalation | Movement during Exhalation | Effect on Thoracic Volume. They pause and complete one row after each structure is shown (teacher pauses the animation at marked timestamps).	seconds.' Restart. PAUSE at Timestamp 2 (intercostal muscles). Say: 'Intercostal muscles row — write now.' Restart. PAUSE at Timestamp 3 (pressure-volume graph). Say: 'Look at your prediction sketches on the board. Put a tick next to any prediction the animation confirmed. Put a question mark next to anything that surprised you. You have 30 seconds of silent thinking time.' After the full animation, cold-call 3 learners (target those identified in Phase 1 with misconceptions): 'Amina — in your prediction, you drew the lungs pulling air in. What does the animation show actually creates that pulling effect?' WAIT 12 seconds. Probe: 'What word does the animation use for the space between the lung and the chest wall?' (pleural cavity). Record correct answers in a different colour on the PREDICTIONS board.	makes the learning visible: learners physically mark their own prior thinking as confirmed or revised. This is the NGSS Science and Engineering Practice of Analysing and Interpreting Data applied to secondary-source evidence.	misconception to address immediately. (2) Whether learners connect volume increase to pressure DECREASE (inverse relationship). If fewer than 60% of the class have the pressure direction correct, the teacher re-plays the pressure-volume graph segment and narrates it explicitly before moving to Phase 3.
Explain Phase 📹 **VIDEO:** [The lungs and pulmonary system] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Parametric Forms and Calculus: Volume of Revolution (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 construct the improvised bell-jar diaphragm model: a 500 mL plastic bottle with the base cut off serves as the thoracic cavity; two small balloons threaded onto a Y-shaped straw represent the lungs and bronchi; a large balloon stretched over the cut base represents the diaphragm. Learners pull the base balloon downward (simulating diaphragm contraction/flattening) and observe the inner balloons inflate; they push it upward and observe the inner balloons deflate. Groups record observations, then answer three scaffold questions: (Q1) When you pulled the base balloon DOWN, what happened to the volume inside the bottle? What happened to the inner balloons?	Distribute pre-assembled or part-assembled model kits (bottles pre-cut by teacher). Say: 'Before you touch the model — predict in one sentence what will happen to the inner balloons when you pull the base balloon down. Write it.' Give 30 seconds. Then say: 'Now test your prediction. Pull slowly — watch carefully. Record exactly what you see, not what you expected.' Circulate. If a group's inner balloons do not inflate, check for air leaks around the straw seal — help them re-seal with clay or tape. After 5 minutes of free exploration, pose the three scaffold questions on the board. Say: 'Q1 is purely observation — just describe what you saw. Q2 asks you to explain using a	Strategy: Claim–Evidence–Reasoning (CER) Framework. CER is an NGSS-aligned sensemaking structure (Constructing Explanations and Designing Solutions practice) that prevents learners from simply reporting observations without interpretation. By requiring a CLAIM (what is true), EVIDENCE (what was observed), and REASONING (why the evidence supports the claim, using scientific concepts), the framework pushes learners to integrate the three information sources of this lesson — their own body data, the animation, and the physical model — into a single coherent explanation. The explicit link back to the step-up phenomenon in Q3	Teacher reads CER paragraphs for three specific success criteria: (1) CLAIM correctly identifies pressure gradient as the driver of air movement (not 'the lungs pull air in'). (2) EVIDENCE cites at least one model observation AND one animation observation — not just one source. (3) REASONING uses the terms 'thoracic volume', 'pressure', and 'diaphragm' correctly and connects to the step-up exercise. Any CER missing criterion 3 is flagged for targeted feedback: teacher writes one question in the margin — e.g., 'Your model explanation is strong — now explain how this same mechanism worked faster during your step-ups.'

	(Q2) Using the word 'pressure', explain WHY the inner balloons inflated. (Q3) How does this model represent what happens in your chest when you breathe in after doing step-ups? Each learner then writes an individual CER paragraph: CLAIM — state what makes air move into the lungs; EVIDENCE — cite two observations from the model AND one observation from the animation; REASONING — explain using the concepts of thoracic volume, air pressure, and the diaphragm.	scientific word. Q3 is the most important — it connects your plastic bottle to your own body on the day of the step-ups. You have 8 minutes.' After group discussion, say: 'Each of you now writes your own CER paragraph — individual thinking, not a copy of your partner's words. You have 6 minutes of silent writing time.' Cold-call 2 learners to read their CER paragraphs aloud. After each, ask the class: 'Does their REASONING connect the model to the phenomenon — to the step-up exercise? Thumbs up or thumbs sideways.' Address gaps: 'What would we add to make the reasoning more complete?'	ensures the explanation is anchored to real experience, not abstracted away.	
Driving Question Board (DQB) Creation 📹 **VIDEO:** The lungs and pulmonary system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Parametric Forms and Calculus: Volume of Revolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class DQB (begun in Lesson 1) is displayed. Learners re-read the driving question: 'Why does my body work so much harder to get oxygen into my cells during exercise — and what happens inside those cells when there is not enough?' Each learner writes two sticky notes: one GREEN — a question from the DQB that today's lesson has now provided evidence to partially answer; one YELLOW — a new question today's lesson has raised. Learners post their notes on the DQB. The teacher reads 3–4 aloud and categorises them live: 'This green note says we can now explain WHY breathing rate increases — the diaphragm must contract more times per minute. That answers part of our driving question. This yellow note asks — but how does the oxygen actually get FROM the air sacs INTO the blood? That is the question Lesson 5 will tackle.'	Point to the specific section of the DQB where breathing mechanics questions were posted in earlier lessons. Say: 'We had a question posted in Lesson 1 — Kamau asked, what makes air move in and out? Look at your CER paragraph. Has today's lesson given Kamau an answer he could not have given on Day 1?' WAIT 10 seconds. Say: 'Write your green sticky note now — start it with the words: We now have evidence that… Write your yellow sticky note — start it with: We still don't know…' Give 3 minutes for writing. Direct learners to post notes. Read 3 green and 2 yellow notes aloud, narrating the storyline: 'Notice how our green notes are filling in the mechanism of breathing — but our yellow notes keep pointing forward to the blood, to the cells, to what happens when oxygen runs out. That gap is where our investigation is heading.'	Strategy: Driving Question Board Curation. The DQB is not decorative — it is a public, living record of the class's collective epistemic progress. By explicitly requiring learners to distinguish between questions that evidence has addressed (green) and new questions evidence has raised (yellow), the DQB develops the NGSS practice of Asking Questions and Defining Problems while also making metacognitive growth visible. The teacher's live categorisation models how scientists track the boundary between what is known and what remains to be investigated.	Teacher photographs the DQB after posting. Assess the green notes for accuracy — a green note claiming 'we now know how oxygen enters cells' is a misconception (that is Lesson 5–6 territory) and should be gently redirected: 'That is a great question — but do we actually have evidence for that yet, or is it still a yellow?' This also checks whether learners are conflating external respiration (today's lesson) with gas exchange at the alveolar surface (upcoming).
Model	Learners retrieve the labelled	Say: 'Take out your Lesson 3	Strategy: Cumulative Model	Exit-ticket check: before learners

Building Phase 📹 **VIDEO:** The lungs and pulmonary system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Parametric Forms and Calculus: Volume of Revolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	thoracic cavity diagram they began annotating in Lesson 3. Using a different coloured pen, they now add a second layer of annotations representing today's learning: (a) curved arrows on the diaphragm showing its direction of movement during inhalation and exhalation; (b) arrows on the rib cage showing outward/upward movement (inhalation) and inward/downward movement (exhalation); (c) pressure labels inside the thoracic cavity — 'pressure DECREASES → air flows IN' during inhalation and 'pressure INCREASES → air flows OUT' during exhalation; (d) a small inset box defining: tidal volume (the volume of air moved in one normal breath, approximately 0.5 L in an adult); pleural cavity (the fluid-filled space between the visceral and parietal pleura that transmits pressure changes). Learners compare their updated model with a partner and add any missing annotations, then write a two-sentence caption below the diagram that connects the model to the step-up phenomenon.	diagram — the one you labelled with structures. Today we add the MECHANISM layer. Use a different colour so future-you can see how your model has grown over time.' Project a partially complete reference diagram (missing the pressure annotations) on the board. Say: 'I have left the pressure labels blank on the board version — you are going to fill in YOURS first, then we will check together.' Give 5 minutes for individual annotation. Then ask: 'Which direction does the diaphragm move during INHALATION — up or down? Every person show me with your hand.' (All hands should move downward.) Cold-call one learner: 'Achieng — use the word pressure to explain why downward movement of the diaphragm causes air to flow in.' WAIT 12 seconds. Fill in the board version together. Close the lesson: 'Your model now shows both the structures from Lesson 3 AND the mechanism from today. In Lesson 5, we will zoom in on the alveolar surface and add the final layer — how oxygen actually crosses from air into blood. That will answer the next part of our driving question.'	Revision (Layered Annotation). Using a second pen colour to add a new layer of understanding to an existing diagram makes the process of scientific model-building tangible. Learners can visually see that models are not completed in one sitting — they are iteratively revised as new evidence is gathered. This directly mirrors the NGSS practice of Developing and Using Models and reinforces that the diagram is not an art task but an epistemic tool for tracking understanding.	close their notebooks, teacher says — 'Point to the pressure label on your inhalation diagram. Does it say pressure increases or pressure decreases? Hold up one finger for decreases, two for increases.' This instant whole-class visual check identifies any remaining pressure-direction confusion. Any learner holding up two fingers (pressure increases during inhalation) is given a targeted oral follow-up question before the next lesson: 'When the diaphragm flattens, what happens to the space inside your chest — does it get bigger or smaller? So what must happen to the pressure?'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) MECHANISM CONFUSION — Did a significant number of learners initially believe the lungs actively pull air in, rather than the diaphragm creating a pressure gradient? If so, consider adding a simple syringe-and-balloon demonstration at the start of Lesson 5 to reinforce the pressure-volume relationship before moving to alveolar gas exchange. (2) MODEL QUALITY — When learners held up fingers for the exit-ticket pressure check, what proportion showed confusion (two fingers)? If more than 25%, open Lesson 5 with a 3-minute revisit of the bell-jar model — pull the base balloon and ask learners to narrate the pressure change aloud before proceeding. (3) CER DEPTH — Did learners successfully link the model to the step-up phenomenon in their CER paragraphs, or did they describe the model in isolation? If the phenomenon connection was weak, use a pair-share at the start of Lesson 5 where learners read their CER back and add one sentence explicitly referencing their own step-up data. (4) DQB PROGRESSION — Are the yellow sticky notes pointing toward alveolar gas exchange and circulation (Lessons 5–6 territory), or are learners raising questions already answered? If learners are re-asking settled questions, this signals that prior lesson learning is not being retained — consider a brief 5-minute cumulative review protocol at the start of every lesson for the rest of the unit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners watched an ARES offline animation showing diaphragm and intercostal muscle movements, thoracic volume changes, and pressure gradients during inhalation and exhalation. Learners also constructed and manipulated an improvised bell-jar diaphragm model (plastic bottle, balloons, Y-straw) and directly observed inner balloons (lungs) inflate when the base balloon (diaphragm) was pulled downward.
What did I learn?	Learners learned that breathing is a mechanical, pressure-driven process: during inhalation, the diaphragm contracts and flattens while external intercostal muscles raise the rib cage, increasing thoracic volume and decreasing intrathoracic pressure below atmospheric pressure, causing air to flow in; during exhalation, these muscles relax, thoracic volume decreases, pressure rises, and air is expelled. Key vocabulary consolidated: tidal volume, inspiration, expiration, pleural cavity.
How does this explain the phenomenon?	Learners used the Claim–Evidence–Reasoning framework to explain that the doubled breathing rate observed after the step-up exercise is achieved by the diaphragm and intercostal muscles contracting more frequently and more forcefully, creating larger and more rapid pressure gradients so that more air — and therefore more oxygen — is moved to the lungs per minute. They connected the physical model behaviour to their own body data from the anchoring phenomenon.

LESSON 5 (40 minutes): Mechanisms of Gaseous Exchange in Humans — Alveolar & Cellular Exchange	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners trace the complete journey of an oxygen molecule from the alveolus into a muscle cell, and update their individual models to show blood as the transport link between the lungs and muscles — directly explaining why both breathing rate and heart rate rose during the step-up activity.
Knowledge	Learners explain how oxygen diffuses across the alveolar wall into blood capillaries, binds to haemoglobin in red blood cells, is transported via the circulatory system to muscle cells, and diffuses into those cells for use in cellular respiration; they also explain how carbon dioxide travels the reverse route back to the lungs.
Skills	Learners construct and annotate a sequential flow diagram tracing one oxygen molecule from alveolus → capillary → haemoglobin → heart → muscle cell, and revise their cumulative model to incorporate the circulatory system as the transport link.
Attitudes	Learners appreciate the elegant integration of the respiratory and circulatory systems and develop curiosity about how disruptions to either system (e.g., anaemia common in many Kenyan communities) affect oxygen delivery.
Key Inquiry Question	How do animals exchange gases with their environment? / How do animals obtain energy from food?
Purpose in Storyline	Lessons 1–4 established that breathing rate and heart rate both rise during exercise and that the lungs bring oxygen into the body through ventilation and diffusion at respiratory surfaces. Lesson 5 now closes the gap between the lungs and the muscle cells: learners discover that blood — specifically haemoglobin in red blood cells — is the delivery vehicle, and that the heart's faster beat during exercise is the body's mechanism for speeding up that delivery. This directly answers the long-standing puzzling observation: 'Why does the heart also beat faster at the same time?' and sets up Lesson 6 where learners investigate what muscle cells actually do with the oxygen once it arrives (cellular respiration).
Safety Notes	No laboratory hazards in this lesson. The ARES platform reading activity is conducted individually or in pairs on devices. Remind learners not to share devices without wiping screens first (hygiene). If any learner has a known blood or circulatory condition (e.g., sickle-cell anaemia) and the topic feels sensitive, handle with care and affirm that their experience is scientifically relevant and valued.

B. LESSON OVERVIEW

In this EXPLAIN lesson, learners use a guided reading on the ARES platform about circulatory system functions to trace the journey of a single oxygen molecule from the alveolus to a working muscle cell. The lesson opens by reconnecting to the step-up phenomenon — specifically the unanswered puzzle of why the heart rate also rose, not just breathing rate. Learners first predict the route oxygen takes after it enters the blood, then read for evidence, construct a sequential flow diagram ('The Oxygen Journey'), discuss in structured pairs, and finally update their cumulative individual models to show the circulatory system as the transport bridge between lungs and muscles. By the end, learners can articulate why both the lungs and the heart must work harder during exercise, advancing their answer to the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown the class data	Display the class pulse-rate data	Prior Knowledge Activation via	Teacher scans diagrams during

📹 **VIDEO:** [Red blood cells](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Enzymes in the Digestive System (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	table from Lesson 1 (resting vs. post-exercise pulse and breathing rates) displayed on the ARES platform or written on the board. They are asked: 'We know oxygen enters the blood at the alveolus. In your exercise book, draw a simple arrow diagram — starting from the word ALVEOLUS — to show where you think oxygen goes next until it reaches a muscle cell in your leg.' Learners work silently for 3 minutes, drawing their best-guess pathway. They are not corrected yet; predictions are held as a personal hypothesis to be tested against the reading. Learners write one sentence: 'I think the heart beats faster during exercise because ___.'	from Lesson 1 on the board and say: 'Look at Amina's data — her pulse jumped from 74 to 128 beats per minute after step-ups. We explained why her breathing rate doubled. But why did her HEART rate almost double too? Today we find out.' Pause 10 seconds. Issue the prediction task verbally and in writing on the board: 'Draw the oxygen journey from alveolus to muscle cell.' Circulate silently for 3 minutes — do NOT prompt or correct. After 3 minutes, cold-call 3 learners (select one who drew a direct lung-to-muscle arrow, one who included 'blood', one who mentioned 'heart') and ask each: 'Talk me through your diagram.' Do not affirm or correct — say: 'Interesting — let's see what the evidence tells us.'	Annotated Prediction Diagram — By committing a hypothesis to paper before reading, learners create a cognitive anchor. When the reading later confirms or corrects their prediction, the contrast drives deeper encoding than reading alone. Holding three different peer predictions publicly (lung-to-muscle direct; blood mentioned; heart mentioned) surfaces the range of prior knowledge and creates productive tension to resolve.	circulation: observable indicator = learner has drawn at least one arrow from 'alveolus' with a label attempt. Red flag: learner leaves page blank or draws oxygen going directly to muscle with no transport medium — note these learners for targeted questioning during the Observe phase. Collect one sentence predictions — check whether learners connect heart rate to transport speed or to some unrelated idea.
Observe Phase 📹 **VIDEO:** [Red blood cells](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Enzymes in the Digestive System (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners open the ARES platform reading: 'Circulatory System Functions and Gas Transport in Humans.' The reading covers: (a) structure of alveolar capillary network and the diffusion gradient for O_2 and CO_2; (b) how haemoglobin in red blood cells binds O_2 to form oxyhaemoglobin; (c) the route of oxygenated blood from lung capillaries → pulmonary veins → left side of heart → aorta → arteries → muscle capillaries; (d) diffusion of O_2 from capillary blood into muscle cells down a concentration gradient; (e) diffusion of CO_2 from muscle cells back into blood and transport to lungs. As they read, learners complete a structured 'Oxygen Journey' table in their exercise books with five columns: Location | What happens to O_2 here? | Direction of diffusion | What carries O_2? | Connection to exercise	Before learners open the reading, say: 'You are reading as a scientist checking your prediction — as you read, fill in the Oxygen Journey table. Every row must connect back to our step-up activity.' Write the five column headers on the board. Set a timer for 12 minutes. Circulate and use targeted questions with learners who are struggling: 'Find the paragraph that mentions haemoglobin — what does it say happens to oxygen there?' and 'Look at your prediction diagram — does the reading agree with your arrow?' At the 8-minute mark, cold-call one learner: 'Kamau, what does the text say carries oxygen in the blood?' — give 10 seconds wait time before accepting the answer. If the answer is incomplete, ask a follow-up: 'Is it the plasma or something inside the red blood cell?' Do not lecture; redirect learners back to	Structured Note-Taking with Phenomenon Connection Column — The five-column table is a disciplined evidence-extraction tool. The final column ('Connection to exercise phenomenon') is the critical sensemaking move: it prevents the reading from becoming abstract by requiring learners to explicitly link each biological location to the lived experience of doing step-ups. This mirrors the NGSS practice of 'Obtaining, Evaluating, and Communicating Information' anchored to a specific phenomenon.	Teacher checks the 'Direction of diffusion' column during circulation — observable indicator: learner correctly states O_2 diffuses from HIGH concentration (alveolus/blood) to LOW concentration (blood/cell). Common error: learners write 'oxygen moves towards the heart' without mentioning concentration gradient — prompt with: 'What makes oxygen move in that direction rather than the other way?' Also check whether the 'What carries O_2?' column says 'haemoglobin' or 'red blood cells' rather than just 'blood' (precision of language indicator).

	phenomenon. They annotate each row as they encounter evidence in the text.	the text.		
Explain Phase 📹 **VIDEO:** Red blood cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzymes in the Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners close their devices and work in pairs to construct a sequential flow diagram called 'The Oxygen Journey' — a horizontal chain of labelled boxes with arrows, starting at ALVEOLUS and ending at MUSCLE CELL MITOCHONDRIA. Each arrow must carry a label explaining the mechanism of movement (e.g., 'diffusion — high to low O_2 concentration'; 'pumped by left ventricle'; 'carried by haemoglobin'). Pairs then share with another pair (group of 4) and identify one point of disagreement or one question they still have. Groups nominate one remaining question to add to the Driving Question Board. Finally, the teacher leads a whole-class 'Molecule Narration': one learner narrates in first person — 'I am an oxygen molecule. I have just crossed the alveolar wall. Now I...' — while the class follows on their diagrams and corrects any errors in the narration.	Announce: 'Close your devices. Using ONLY your table notes and your memory, build the Oxygen Journey diagram with your partner — no devices.' Allow 7 minutes. Circulate and listen for pairs who stop the chain at 'the heart' without continuing to the muscle — prompt: 'What does the heart do with the oxygen — does it use it all, or does it send it somewhere?' After pairs share in groups of 4, ask groups: 'What is your one remaining question?' Write 3–4 questions on the board. Then launch the Molecule Narration — ask for a volunteer narrator, or cold-call a confident learner: 'Otieno, you are now an oxygen molecule. Start from the alveolus — go!' When the narrator makes an error or pauses, say to the class: 'Hands up if you want to correct or continue the story' — accept 2–3 contributions. End the narration when the molecule 'enters the muscle cell.' Ask: 'So why did Amina's heart beat at 128 bpm after step-ups? Turn and talk — 30 seconds.' Cold-call 2 pairs: give 10 seconds wait time each.	Molecule Narration (First-Person Perspective Storytelling) — Narrating as the oxygen molecule requires learners to sequence every mechanism in order and name the agent of each transport step. This strategy makes abstract diffusion gradients and circulatory routes concrete and sequential. It also reveals precisely where understanding breaks down (e.g., learners who say 'I enter the heart and stay there' have a misconception about the heart as a pump, not a destination). Paired flow diagram construction applies the NGSS practice of 'Developing and Using Models.'	Observable indicator during Molecule Narration: learner correctly identifies at least four sequential steps (alveolus → capillary blood / haemoglobin → heart pump → muscle capillary → muscle cell) without being prompted. Red flag: narrator skips the heart entirely or says oxygen 'travels through the lungs to muscles' — stop and redirect with: 'At which point does oxygenated blood leave the lungs — which vessel?' Check Turn-and-Talk responses: learners should explicitly link faster heart rate to faster delivery of oxygen to muscles (not just say 'the heart pumps harder').
Driving Question Board (DQB) Creation 📹 **VIDEO:** Red blood cells Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner writes on a sticky note (or an index card if sticky notes are unavailable): one CLAIM they can now make that answers part of the driving question, supported by one piece of EVIDENCE from today's reading or diagram. Example format: 'CLAIM: The heart beats faster during exercise because it needs to pump oxygenated blood to muscles more quickly. EVIDENCE:	Direct learners to the DQB and say: 'Look at the question we posted in Lesson 1 — WHY does the heart beat faster during exercise? Can we answer it now?' Give 10 seconds wait time. Accept 2 responses. Then say: 'Write your Claim-Evidence card — one claim, one piece of evidence. You have 3 minutes.' Circulate and prompt any learner who writes only a claim with: 'What from today's reading	Claim-Evidence Cards on the Driving Question Board — This is a public, cumulative knowledge-tracking structure. By requiring both a claim and evidence, the DQB card enforces scientific argumentation (NGSS practice: 'Engaging in Argument from Evidence'). Crossing out answered questions gives learners a visible sense of intellectual progress and sustains motivation across the 12-	Observable indicator: learner's card contains a testable claim (not a question) and cites a specific mechanism from the lesson (haemoglobin, diffusion gradient, heart pumping, concentration gradient) as evidence. Red flag: card says only 'oxygen goes to muscles' with no mechanism — these learners need Lesson 6 scaffolding. Collect cards at the end of class; review 5–6 before

📖 **HTML:** [Enzymes in the Digestive System (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Oxygen is carried by haemoglobin in red blood cells — more heart beats per minute = more deliveries per minute.' Learners post their cards on the class Driving Question Board under the column 'Why does the heart beat faster?' Previously posted questions from Lessons 1–4 are reviewed briefly — the teacher crosses out any questions that today's lesson has now fully answered.	supports that claim?' After cards are posted, read 2–3 aloud and model evaluating quality: 'This card says the heart beats faster — it has a claim. But does it explain WHY that helps oxygen reach muscles? Let's add that.' Cross out on the DQB the question 'What is the connection between the lungs and the heart?' and write next to it: 'ANSWERED — Lesson 5.' Point to the remaining unanswered question: 'What happens inside the muscle cell when oxygen arrives?' — say: 'That is what Lesson 6 will investigate.'	lesson unit. Connecting to the unanswered question previews Lesson 6 and sustains inquiry.	Lesson 6 to identify persistent misconceptions.
Model Building Phase 📹 **VIDEO:** [Red blood cells] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Enzymes in the Digestive System (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their individual cumulative model (started in Lesson 1 and updated in each subsequent lesson). Today's required addition: they must draw and label the circulatory link between lungs and muscles — showing (a) pulmonary capillaries around alveoli, (b) the heart as a pump, (c) systemic capillaries in muscle tissue, (d) arrows showing oxygenated blood flowing from lungs to muscles and deoxygenated blood returning to lungs, and (e) the label 'HAEMOGLOBIN carries O_2' on the blood vessels. Learners also add a colour code: red for oxygenated blood, blue for deoxygenated blood. In one sentence beneath the model, learners write: 'This explains why during step-ups, [complete with their own words].' Models are dated 'Lesson 5' to track progression.	Say: 'Take out your model. Today you are adding the missing link — the transport system that connects everything we have studied so far.' Display a simple teacher reference diagram on the board (hand-drawn is fine) showing the two loops: lung loop and body loop, with the heart at centre. Say: 'Your model must show both loops — use red for oxygen-rich blood and blue for oxygen-poor blood.' Give 5 minutes. Circulate and check for three non-negotiable elements: (1) heart labelled as pump, (2) haemoglobin label on at least one blood vessel, (3) the phenomenon sentence at the bottom. For learners who finish early, prompt extension: 'Add where CO_2 travels — opposite direction to oxygen. Label the concentration gradients.' Collect 4–5 models to photograph for a class model gallery on the ARES platform (with learner permission).	Cumulative Model Revision with Colour-Coded Transport Loops — Returning to and revising the same model across all 12 lessons makes learning visible as a growing, self-constructed explanation rather than a series of isolated facts. Colour-coding oxygenated vs. deoxygenated blood activates visual-spatial reasoning and reinforces the directional logic of gas transport. The mandatory phenomenon sentence ('This explains why during step-ups…') ensures the model remains anchored to lived experience rather than becoming a decontextualised diagram. This enacts the NGSS practice of 'Developing and Using Models' in its fullest iterative sense.	Observable indicator: model now shows a closed loop (lungs ↔ heart ↔ muscles ↔ heart ↔ lungs) with directional arrows and at least three labels (heart/pump, haemoglobin, diffusion). Red flag: learner draws oxygen going directly from lungs to muscle with no heart — this is a persistent pre-instructional model that needs explicit attention at the start of Lesson 6. The phenomenon sentence is the highest-value formative check: a complete sentence ('This explains why during step-ups, the heart beats faster to pump more oxygenated blood to the leg muscles faster') demonstrates integrative understanding.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) During the Molecule Narration, where exactly did learners' narration break down — was it at haemoglobin, the heart, or the concentration gradient into the muscle cell? That break point tells you the sharpest misconception to address at the start of Lesson 6. (2) How many Claim-Evidence cards on the DQB cited a specific mechanism (haemoglobin, diffusion gradient, heart pump) vs. vague statements ('oxygen goes to the blood')? If more than one-third of cards lacked a mechanism, plan a 5-minute retrieval review at the opening of Lesson 6. (3) Did the Oxygen Journey table's final column ('Connection to exercise phenomenon') reveal any learners who still see the circulatory system as separate from the breathing changes they observed? These learners may need a one-on-one prompt: 'If the heart stopped pumping faster, would breathing harder still get oxygen to your leg muscles?' (4) Were there learners who mentioned anaemia, sickle-cell disease, or malaria (all prevalent in Kenyan communities) in the context of haemoglobin — if so, note this as a rich context for Lesson 6 extension. (5) Review 4–5 photographed models before Lesson 6: do they show a closed loop, or do learners still draw oxygen as a one-way trip with no CO_2 return path? The return path (CO_2 from cell to lungs) is the conceptual bridge into cellular respiration in Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined class data showing that both breathing rate and heart rate rose sharply after 2 minutes of step-ups (e.g., pulse from 72 to over 120 bpm). They read evidence on the ARES platform describing how oxygen diffuses across alveolar walls into capillaries, binds to haemoglobin in red blood cells, is pumped by the heart through the circulatory system, and diffuses from muscle capillaries into muscle cells down a concentration gradient — while CO_2 travels the reverse route.
What did I learn?	Oxygen does not travel directly from the lungs to the muscles — it is carried by haemoglobin in red blood cells through the circulatory system. The heart acts as a pump that drives oxygenated blood from the lungs to every cell in the body. During exercise, the heart beats faster to increase the rate of oxygen delivery to muscle cells and the rate of carbon dioxide removal. Diffusion at both the alveolar surface and the muscle capillary surface is driven by concentration gradients: oxygen moves from high concentration to low, and carbon dioxide moves in the opposite direction.
How does this explain the phenomenon?	The complete journey of one oxygen molecule — alveolus → capillary wall (diffusion) → haemoglobin in red blood cell → pulmonary vein → left side of heart → aorta → muscle capillary → muscle cell (diffusion) — explains why both breathing rate AND heart rate rose during the step-up activity. The lungs increase oxygen supply; the heart increases delivery speed. Together, these two systems ensure muscle cells receive enough oxygen to keep releasing energy during exercise. This closes the puzzle of the heart–lung connection first raised in Lesson 1.

LESSON 6 (80 minutes): Aerobic Respiration: Unlocking Energy from Ugali Inside Your Cells	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of the lesson, the learner should be able to: (a) write and interpret the word equation and balanced chemical equation for aerobic respiration; (b) annotate a mitochondrion diagram to show where glucose and oxygen are used and where carbon dioxide, water, and ATP are produced; (c) explain conceptually how ATP represents usable energy for muscle cells; (d) connect aerobic respiration to the observed rise in breathing and pulse rate during the anchoring phenomenon.
Knowledge	Aerobic respiration is the process by which glucose is broken down in the presence of oxygen to release energy in the form of ATP, producing carbon dioxide and water as by-products. The word equation is: glucose + oxygen → carbon dioxide + water + ATP (energy). The balanced chemical equation is: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$. The mitochondrion is the primary site of aerobic respiration in eukaryotic cells. ATP (adenosine triphosphate) is the universal energy currency of the cell.
Skills	Learners will: write and balance the chemical equation for aerobic respiration; annotate a mitochondrion diagram with reactants, products, and energy output; interpret data linking oxygen consumption to ATP production; construct a conceptual energy-yield calculation linking food intake (ugali/glucose) to ATP output; update their Driving Question Board with evidence from this lesson.
Attitudes	Learners appreciate that the immediate, personal experience of breathing faster and feeling muscle burn during exercise is directly explained by events occurring at the molecular level inside every muscle cell. Learners develop curiosity about how food choices affect energy availability and value the connection between Biology and everyday Kenyan life.
Key Inquiry Question	How do animals obtain energy from food?
Purpose in Storyline	Lessons 1–5 established that: breathing rate and pulse rate both rise sharply during exercise (anchoring phenomenon); gas exchange surfaces (alveoli) bring oxygen into the blood; the diaphragm and intercostal muscles drive ventilation mechanics; and the circulatory system delivers oxygen-rich blood to working muscles. A critical question on the DQB has remained unanswered: 'What actually HAPPENS to the oxygen once it arrives at the muscle cell — and why does the cell need it so urgently?' Lesson 6 answers this directly. Learners now open the 'black box' of the muscle cell and discover aerobic respiration — the chemical process that uses the oxygen and glucose to manufacture ATP, the molecule that powers every muscle contraction. This explains why oxygen demand doubles during exercise and sets up Lesson 7, which will investigate what happens when oxygen supply is insufficient (anaerobic respiration and the burning sensation).
Safety Notes	No hazardous chemicals are used in this lesson. The physical activity (brief step-ups at the start) should be moderated for learners with known cardiovascular or respiratory conditions — those learners may act as timers and data recorders instead. Remind learners not to hyperventilate deliberately. All diagrams involving molecular structures are representational only.

B. LESSON OVERVIEW
Lesson 6 is the pivotal OBSERVE & EXPLAIN lesson in which learners move from the macroscopic evidence gathered in Lessons 1–5 (rising breathing and pulse rates, gas exchange surfaces, ventilation mechanics, circulatory delivery) down to the cellular and molecular level. The anchoring phenomenon — doubled breathing rate and spiked pulse during 2 minutes of step-ups — has so far been explained in terms of HOW oxygen gets to the muscle. This lesson answers WHY the muscle cell needs that oxygen so urgently. Learners begin with a brief recall of their own phenomenon data, then predict what they think happens inside a muscle cell during exercise. They observe the word equation and balanced chemical equation for aerobic respiration through a guided worked example framed around a Rift Valley maize farmer's day of physical labour fuelled by ugali. Using a printed mitochondrion diagram, learners annotate reactants, products, and energy output, then engage in a structured

sensemaking discussion connecting molecular events to their whole-body experience. The lesson closes with a DQB update in which learners formally mark the driving question branch 'Why does my body need more oxygen?' as 'EXPLAINED' and begin asking the next question: 'What happens when there isn't enough oxygen?'

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners briefly revisit the class phenomenon data table (resting vs. exercise breathing rate and pulse rate) displayed on the board or recalled from their notebooks. Each learner independently writes a 2–3 sentence response in their science notebook to the prompt: 'You now know that oxygen travels from the air → lungs (alveoli) → blood → muscle cell. Write your best prediction: what do you think the oxygen DOES inside the muscle cell, and why does the cell need more of it when you are exercising?' Learners share predictions with a shoulder partner for 90 seconds, then three learners share with the class. Predictions are recorded on a sticky note and posted in the 'PREDICTIONS' column of the DQB — they will be revisited at the end of the lesson.	Display the class phenomenon data table prominently. Say: 'Look at your resting breathing rate and your post-exercise breathing rate. You doubled or nearly doubled the amount of air moving through your lungs. Something inside your muscle cells is demanding that extra oxygen. Today we are going to find out exactly what that something is.' Pose the prediction prompt verbally and in writing: 'What do you think oxygen DOES inside a muscle cell during exercise?' Allow 3 minutes of independent silent writing — enforce silence, do not give hints. After writing, say: 'Share with your partner. You have 90 seconds — each person must speak.' Cold-call 3 learners using name cards (do not accept volunteers only). After each response, say: 'Interesting — hold that idea. Let's see what the evidence tells us.' Do NOT confirm or correct predictions yet. Post sticky notes on DQB.	Strategy: PREDICT-BEFORE-INSTRUCTION (Science Notebook Entry). Purpose: activating prior knowledge about energy and food, surfacing misconceptions (e.g. 'oxygen gives you energy' rather than 'oxygen enables glucose to release energy'), and creating cognitive dissonance that the Observe phase will resolve. By physically posting their predictions on the DQB, learners develop personal investment in whether their model was correct.	Observable indicator: Every learner has written at least 2 sentences in their notebook before sharing. Teacher scans the room during writing — look specifically for learners who mention 'glucose' or 'food' as the energy source (shows prior knowledge) vs. those who think 'oxygen IS the energy' (key misconception to address in Phase 3). Mentally note 1–2 misconception holders for targeted cold-calling in Phase 3.
Observe Phase 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners receive a structured worksheet containing: (A) a context paragraph about Mwangi, a maize farmer in Nakuru who eats ugali for breakfast before a long day of digging, (B) a blank space for the word equation, (C) a blank space for the balanced chemical equation, (D) a printed outline diagram of a mitochondrion with unlabelled compartments, and (E)	Distribute worksheets. Say: 'Meet Mwangi. He is a maize farmer near Nakuru. He woke up at 5 a.m., ate a large plate of ugali, and has been digging irrigation channels for three hours. His muscles are working hard — just like yours were during step-ups. The question is: how does ugali end up powering his muscles?' Write the word equation step by step — do	Strategy: GUIDED WORKED EXAMPLE WITH PROGRESSIVE ANNOTATION. Purpose: Breaking the chemical equation into three sequenced steps (word → symbolic → spatial/diagrammatic) ensures learners do not memorise symbols without understanding reactants and products. Anchoring each chemical species to Mwangi's farming scenario — and	Observable indicators: (1) During pair atom-counting, teacher circulates to check that learners correctly identify 6 C on each side, 12 H in glucose → 12 H in $6H_2O$, and 18 O total on each side. A common error is miscounting H in water — address immediately if seen in more than 3 pairs. (2) Mitochondrion diagrams: check that learners draw arrows for

📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	a conceptual ATP-yield table. The teacher delivers a guided worked example in three steps. STEP 1 (Word Equation): Teacher writes on the board — glucose + oxygen → carbon dioxide + water + energy (ATP) — and asks learners to copy and annotate: circle 'glucose' and write 'from ugali/starch digested in Mwangi's gut'; circle 'oxygen' and write 'delivered by breathing + heartbeat (Lessons 1–5)'; circle 'carbon dioxide' and write 'exhaled — explains why CO_2 rises during exercise'; circle 'ATP' and write 'powers muscle contraction — this is why Mwangi can dig all day'. STEP 2 (Balanced Chemical Equation): Teacher writes $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + ATP and guides learners to verify atom balance by counting C, H, and O atoms on each side. Learners work in pairs to count and confirm balance. STEP 3 (Mitochondrion Annotation): Learners annotate the printed mitochondrion diagram — labelling the outer membrane, inner membrane, cristae, and matrix; drawing arrows showing glucose and oxygen entering; drawing arrows showing CO_2, H_2O, and ATP leaving; writing 'site of aerobic respiration' across the diagram.	NOT write the whole equation at once. Write 'glucose' and pause: 'Where does glucose come from in Mwangi's body? Partner A, tell Partner B — you have 20 seconds.' Cold-call 2 pairs. Write '+ oxygen' and pause: 'How did that oxygen get to Mwangi's muscle cell? Give me the full journey in one sentence — I am going to cold-call three people.' Allow 15 seconds wait time. Cold-call 3 learners, requiring them to reference alveoli → blood → muscle cell (connecting to Lessons 1–5). Write the products and say: 'Mwangi breathes out more CO_2 when he digs hard. We predicted this in Lesson 2 using the limewater test. This equation tells us WHY.' For the balanced equation, say: 'Science is precise. Let's check the atoms are balanced.' Model counting C atoms (6 on each side), then ask learner pairs to count H and O independently. Give 3 minutes. Cold-call 2 pairs to report. For the mitochondrion, say: 'This organelle is often called the powerhouse of the cell. Your job in the next 8 minutes is to make that phrase mean something — annotate every arrow, every label, so that someone looking at your diagram could explain the entire process.' Circulate and prompt: 'Where exactly does the reaction happen — outer membrane or matrix?' and 'What leaves the mitochondrion — what enters?'	back to the step-up phenomenon — makes the abstract equation personally and culturally meaningful. The mitochondrion annotation forces spatial reasoning: learners must decide which molecules move IN and which move OUT, consolidating directionality of the reaction.	BOTH reactants entering AND all three products leaving. Flag any learner who draws ATP entering the mitochondrion (reversal error). (3) Quick show-of-hands: 'Thumbs up if glucose enters, thumbs down if glucose is made inside the mitochondrion.' Expected: all thumbs up — if not, pause and re-explain.
Explain Phase 📹 **VIDEO:** [Cellular respiration introduction] Source: Khan	Learners engage in a structured 'Claim–Evidence–Reasoning' (CER) written exercise in their science notebooks. The prompt is displayed on the board: 'During 2 minutes of step-ups, Achieng's	Display the CER prompt. Say: 'This is the moment we have been building towards since Lesson 1. You have all the pieces. Your job is to connect them into one coherent explanation.' Enforce 6	Strategy: CLAIM–EVIDENCE–REASONING (CER) FRAMEWORK + BECAUSE CHAIN. Purpose: The CER framework structures scientific argumentation — learners must	Observable indicators: (1) CER notebooks — teacher collects 5 random notebooks at the end of Phase 3 and checks that each CER includes: a specific claim (not vague), at least one reference to

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	breathing rate rose from 16 to 34 breaths per minute and her pulse rose from 74 to 128 bpm. Using the equation for aerobic respiration, explain WHY both rates increased.' Learners write independently for 6 minutes, then share with a partner, then three learners share with the class. After the CER discussion, the teacher facilitates a whole-class 'BECAUSE chain': a learner states one fact and the next learner must connect it with 'because…' — working from ATP demand backwards to breathing rate. Example chain: 'Achieng's muscles contracted faster → because they needed more ATP → because aerobic respiration consumes glucose and oxygen to make ATP → because glucose came from the ugali she ate → because oxygen had to be delivered faster → because her breathing rate and heart rate increased.' The teacher also addresses the key misconception identified in Phase 1: 'Oxygen IS the energy.' Teacher explicitly states: 'Oxygen is not the energy source — oxygen is the enabler. Glucose is the fuel. Oxygen allows glucose to be fully broken down to release the energy stored in its chemical bonds as ATP.'	minutes of silent independent writing. Circulate and read over shoulders — do not give answers, but prompt struggling learners with: 'Start with ATP. What needs ATP? What makes ATP? What does making ATP need?' After writing, say: 'Share with your partner — Partner B argues for Achieng's breathing rate, Partner A argues for her pulse rate. You have 2 minutes.' Cold-call 3 learners to present their CER to the class. After each, ask the class: 'Do you agree with their reasoning? Thumbs up, thumbs down, or sideways.' For the BECAUSE chain, start it yourself: 'Achieng's leg muscles needed to contract rapidly and repeatedly—' then point to a learner: 'Continue the chain with BECAUSE.' Move quickly around the room — aim for 8–10 links. When the chain reaches breathing rate, say: 'We have just traced a path from a muscle fibre all the way to a breath of air. That is what Biology looks like when it is connected.' Address the oxygen misconception explicitly and ask learners to correct any incorrect sticky-note predictions from Phase 1.	state a claim (breathing rate increased), cite evidence (the equation shows oxygen is consumed), and provide reasoning (more ATP demand → more O_2 needed → faster breathing). The BECAUSE chain builds systems thinking by making explicit the causal cascade from molecular (ATP) to cellular (respiration) to organ (lung, heart) to whole-body (breathing rate). Together, these strategies prevent learners from holding isolated facts and force integration across all six lessons.	the chemical equation or ATP, and a causal reasoning chain. (2) During the BECAUSE chain, note whether learners can spontaneously say 'aerobic respiration' or 'ATP' — if learners only say 'oxygen is needed,' the molecular level has not yet been internalised. (3) Misconception check: after addressing 'oxygen IS energy,' cold-call 2 learners who gave that prediction in Phase 1 and ask them to self-correct aloud.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES	The class gathers around the DQB. Learners re-read all sticky notes and question cards posted since Lesson 1. Working in groups of three, learners are assigned one of the following DQB question branches and must decide: (a) FULLY EXPLAINED by today's lesson, (b) PARTIALLY EXPLAINED — we have some evidence, or (c) STILL OPEN —	Bring the class to the DQB. Say: 'We have been collecting evidence for six lessons. Today is a milestone — we can now fully answer one of our biggest questions. Let's see which one.' Assign DQB branches to groups — use name cards to assign, not volunteers. Give groups 3 minutes to discuss and decide. Cold-call each group to report. When the	Strategy: DRIVING QUESTION BOARD CURATION — EXPLAIN/STILL OPEN SORT. Purpose: Explicitly sorting DQB cards into 'explained,' 'partial,' and 'open' categories externalises the structure of scientific knowledge-building. Learners experience the authentic scientific feeling of satisfaction (one question resolved) and renewed curiosity (a	Observable indicators: (1) One-sentence summary cards — teacher reads them during group work. A strong summary names aerobic respiration, ATP, and connects to breathing rate. A weak summary says only 'you need oxygen to exercise' — prompt those learners: 'Can you add the word ATP and the word mitochondrion?' (2) New question

for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	we need more lessons. Groups report back. The class votes and physically moves cards to the appropriate column. Key milestone: the question 'Why does breathing rate increase during exercise?' is now marked FULLY EXPLAINED with a green star — learners write a one-sentence summary on a new card: 'Aerobic respiration in muscle cells consumes oxygen and glucose to make ATP; more exercise = more ATP demand = more O_2 needed = faster breathing and heartbeat.' Learners also identify the NEW question the lesson has raised: 'What happens when the oxygen supply cannot keep up with the ATP demand?' — this is posted as a new red card on the DQB to anchor Lesson 7.	breathing-rate branch is nominated for FULLY EXPLAINED, ask the class: 'Does everyone agree? If you disagree, you must say WHAT is still missing.' Allow 10 seconds of wait time before accepting consensus. Once the green star is placed, say: 'Write that one-sentence summary now — in your own words, not mine.' After 2 minutes, cold-call 2 learners to read their sentence aloud. Then say: 'But notice — today's equation assumed oxygen was always available. Achieng felt a burning sensation in her legs EVEN THOUGH she was breathing hard. That burning is our next mystery. Post your question.' Receive learner-generated questions about the burning sensation and post the best-worded one as the new red card.	new, deeper question emerges). The one-sentence summary card requires learners to consolidate and compress their understanding — a powerful metacognitive act. The new red card for Lesson 7 maintains narrative momentum.	cards — the quality of learner-generated questions reveals depth of understanding. Questions that reference 'not enough oxygen' or 'what else can make ATP' show readiness for Lesson 7 (anaerobic respiration).
Model Building Phase 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner retrieves their individual cumulative model, which has been built across Lessons 1–5 and currently shows: (L1) the whole-body outline with breathing rate and pulse rate data; (L2) the gas exchange pathway from air → alveolus → blood; (L3) the ventilation mechanism (diaphragm, intercostal muscles); (L4) the alveolus cross-section with diffusion arrows; (L5) the circulatory delivery pathway to muscle. Learners now add the cellular layer: inside the muscle cell, they draw a mitochondrion, label it, and add the aerobic respiration equation (word form is acceptable, chemical form earns an extension star). They draw arrows showing: glucose (from blood, originally from ugali) entering the mitochondrion; oxygen (from blood, originally from	Distribute individual models. Say: 'Your model has been growing for six lessons. Today you are zooming in to the deepest level — inside the muscle cell itself. Add the mitochondrion and make the equation visible in your model.' Allow 10 minutes for individual model annotation — circulate constantly. Prompt with: 'Where is the glucose in your model coming from? Trace it all the way back to the ugali.' and 'Show me the CO_2 leaving the cell — where does it go next in your model?' and 'Why does your model now explain the phenomenon from Lesson 1?' After 10 minutes, nominate one learner to update the class poster model — the learner narrates their additions aloud while the class checks their own model against the class version. Close by asking three learners (cold-call, use name	Strategy: CUMULATIVE MODEL REVISION — ZOOM-IN FROM ORGAN TO CELL TO MOLECULE. Purpose: The cumulative model is the primary tool for tracking the growth of explanatory understanding across the unit. Each lesson adds a layer, creating a visible record of increasingly sophisticated mechanistic reasoning. The 'model update sentence' is a metacognitive anchor — learners must articulate WHY they made the change, not just WHAT they drew. Tracing glucose from ugali through digestion → blood → mitochondrion → ATP makes the model a complete causal chain from food intake to physical performance, directly addressing the driving question.	Observable indicators: (1) Mitochondrion is drawn INSIDE the muscle cell (not beside it or in the blood). (2) Both reactants (glucose AND oxygen) are shown entering with labelled arrows. (3) All three products (CO_2, H_2O, ATP) are shown leaving or remaining, with destinations indicated. (4) The model update sentence contains a causal connector ('because,' 'which causes,' 'so that'). (5) CO_2 pathway is continuous — leaving mitochondrion → blood → alveolus → exhaled (connecting back to Lesson 2 and 4 evidence). Teacher flags any model that shows ATP leaving the cell (ATP is not transported out — it is used within the cell) for correction at the start of Lesson 7.

	the alveolus via breathing) entering the mitochondrion; CO_2 leaving the mitochondrion → into blood → to alveolus → exhaled; H_2O leaving; ATP staying inside the cell and labelled 'powers muscle contraction → step-ups → rise in breathing rate.' Learners write one 'model update sentence' at the bottom: 'I added aerobic respiration inside the mitochondrion because this explains why oxygen is consumed and CO_2 is produced when muscles work hard, which explains the rise in breathing rate and pulse rate that I measured on Day 1.' The class model (large poster) is also updated by one nominated learner at the board while the class annotates their individual models simultaneously.	cards): 'What is ONE thing your model can now explain that it could not explain at the start of this unit?' Collect individual models to check annotations before returning them at the start of Lesson 7. Before closing, preview Lesson 7: 'Tomorrow we investigate what happens when the oxygen supply cannot keep up — the burning sensation in your legs finally gets its explanation.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. MISCONCEPTION TRACKING — Did the explicit correction of 'oxygen IS the energy' (vs. 'oxygen ENABLES glucose to release energy as ATP') land with the learners who held this misconception in Phase 1? Did those learners self-correct confidently during the BECAUSE chain, or did they still conflate oxygen with energy? What will you do differently in the Lesson 7 warm-up to reinforce this distinction?

2. EQUATION DEPTH — How many learners could write the balanced chemical equation ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + ATP) without copying from the board by the end of the lesson? Did the atom-counting exercise make the equation feel logical rather than arbitrary? If fewer than 70% of learners could reproduce it independently, consider opening Lesson 7 with a 3-minute equation recall activity.

3. CULTURAL RELEVANCE — Did the Mwangi/ugali/Nakuru context genuinely engage learners, or did it feel bolted on? Did any learners spontaneously extend the context (e.g., referencing sukuma wiki, beans, or athletes from the Rift Valley)? Learner-generated extensions are a strong signal of authentic connection.

4. CER QUALITY — From the 5 random notebooks collected after Phase 3, how many CER entries contained all three components (claim + evidence from equation + causal reasoning chain)? What was the most common gap — missing claim, missing equation evidence, or broken reasoning chain? Adjust Phase 3 scaffolding in future lessons accordingly.

5. MODEL COHERENCE — When learners updated their cumulative models in Phase 5, how many successfully traced the glucose pathway all the way from ugali

(food) through digestion → blood → mitochondrion → ATP? Gaps in this chain reveal where prior-lesson connections are not secure. Use the collected models to identify the most common missing link and address it in Lesson 7. 6. DQB MOMENTUM — Did the act of placing the green 'FULLY EXPLAINED' star on the breathing-rate question create visible excitement or satisfaction in the class? Did learners generate a genuinely interesting new question for the red card (about burning sensation / insufficient oxygen)? The quality of the new red card question is your best predictor of learner readiness and curiosity for Lesson 7 (anaerobic respiration).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed the word equation (glucose + oxygen → carbon dioxide + water + ATP) and the balanced chemical equation ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + ATP) for aerobic respiration, framed through the context of a Rift Valley maize farmer fuelled by ugali. They observed a printed mitochondrion diagram and annotated the entry of reactants and exit of products.
What did I learn?	Learners learned that aerobic respiration occurs in the mitochondria of cells and uses glucose (from digested food such as ugali) and oxygen (delivered by the breathing and circulatory systems investigated in Lessons 1–5) to produce ATP (usable energy for muscle contraction), carbon dioxide, and water. The increased demand for ATP during exercise explains why both breathing rate and pulse rate rise — the body must deliver more oxygen and glucose to the mitochondria and remove more carbon dioxide.
How does this explain the phenomenon?	The lesson explained the core mechanism behind the anchoring phenomenon: when learners performed step-ups, their muscle cells needed more ATP to power contractions, so aerobic respiration in the mitochondria increased its rate, consuming more oxygen and glucose and producing more carbon dioxide — which is precisely why breathing rate doubled and pulse rate spiked. The equation now gives a molecular explanation for every whole-body change measured on Day 1.

LESSON 7 (80 minutes): Anaerobic Respiration: Yeast Balloons, Burning Legs, and Oxygen Debt	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate anaerobic respiration in yeast and explain lactic acid build-up and oxygen debt in human muscle cells during intense exercise.
Knowledge	Learners explain the chemical equations for aerobic and anaerobic respiration, distinguish between the lactic acid pathway (animals) and the ethanol + CO_2 pathway (yeast/fungi), and define oxygen debt.
Skills	Learners set up a controlled practical (Yeast Balloon experiment), collect and compare quantitative data (balloon circumference over time), write balanced word equations, and construct explanatory written arguments linking evidence to the anchoring phenomenon.
Attitudes	Learners appreciate that science explains personal bodily experiences; they demonstrate patience and precision during the practical and respect peers' data contributions.
Key Inquiry Question	How do animals obtain energy from food?
Purpose in Storyline	In Lessons 1–6 learners established how oxygen reaches muscle cells via breathing mechanics and gas exchange surfaces. Lesson 7 now tackles the burning mystery: why do muscles ache and feel on fire during intense exercise like the step-ups — even when learners were breathing hard? By observing yeast producing CO_2 anaerobically (inflating a balloon with no oxygen), learners build the conceptual bridge to lactic acid fermentation in their own muscles, fully resolving Supporting Phenomenon 1 ('burning legs') and Supporting Phenomenon 4 ('yeast balloon'), and updating the Driving Question Board with the final piece of the cellular energy puzzle.
Safety Notes	Yeast suspension is non-hazardous but hands should be washed after handling. Glucose solution is a mild skin irritant in large quantities — wipe spills immediately. Balloons should be checked for latex allergies before distribution; non-latex alternatives must be available. Warm water baths (~37–40 °C) must be monitored; learners must not use boiling water. No open flames near balloons. Broken glassware (if flasks used) must be reported immediately and swept — not picked up by hand.

B. LESSON OVERVIEW

Learners open by revisiting the step-up phenomenon and the unresolved 'burning legs' mystery logged on the Driving Question Board. They set up the Yeast Balloon practical in small groups — mixing dried yeast, glucose solution, and warm water in a bottle and sealing a balloon over the neck — then monitor CO_2 production over 20 minutes while completing a compare-contrast analysis of aerobic vs. anaerobic respiration equations. Midway, a short 'burning legs' video clip (supporting phenomenon 1) is replayed and learners pause to explain lactic acid build-up in writing. Groups share findings, the class constructs a side-by-side equation summary, and learners compose a structured written explanation of oxygen debt. The DQB is updated to mark the burning-leg question as resolved. The lesson closes with a cumulative model revision connecting anaerobic respiration to all prior lessons in the sequence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners individually re-read the 'burning legs' question posted on	Display the DQB 'burning legs' sticky note prominently. Say:	Predict-Press-Hold (PPH): Learners commit to a written	Circulate during silent writing and read over shoulders. Look for: (1)

📹 **VIDEO:** [Biodiversity flourishes in Phanerozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	the Driving Question Board from Lesson 1 ('Why do our leg muscles burn during step-ups even though we are breathing hard?'). They write a 3-sentence prediction in their science notebooks answering: (a) What do you think is happening inside the muscle cells? (b) Do you think oxygen is involved or not? (c) Where does the 'burn' feeling come from? Learners then share predictions with a shoulder partner before two or three are cold-called to share with the class. The teacher also places an uninflated balloon over an empty bottle at the front as a curiosity hook, without explanation.	'Before we do anything today, I want you to go back to a question we have had on our board since Lesson 1. Read it carefully. In your notebook, write three sentences — what do YOU think is happening inside those muscle cells when they burn?' Allow 3 minutes silent writing (enforce no talking). After 3 minutes, say: 'Share your prediction with your partner. You have 90 seconds.' Then cold-call 3 learners using the class register (rows 2, 5, and 8): 'Tell us your prediction — what is happening in those cells?' After each response, use neutral probing: 'Interesting — what evidence from our earlier lessons supports that idea?' Wait 10 seconds after each probe for self-correction. Do NOT confirm or deny predictions. Say: 'Hold those ideas. By the end of this lesson, you will be able to judge your own prediction.' Point to the balloon on the bottle: 'And this bottle will help us figure it out.'	prediction, articulate their reasoning aloud, and are explicitly told to suspend judgement until evidence arrives. This activates prior knowledge from Lessons 1–6 (aerobic respiration, oxygen transport) and creates productive cognitive tension — especially for learners who correctly know oxygen is involved but cannot yet explain why it sometimes is not enough.	learners who mention 'lactic acid' or 'fermentation' — they have prior knowledge to build on; (2) learners who say 'muscles run out of oxygen' — partially correct, probe further; (3) learners who have no cellular-level explanation — seat them adjacent to stronger peers for the practical. Record 3 representative prediction categories on the board (no names) to revisit during Phase 3.
Observe Phase 📹 **VIDEO:** [Biodiversity flourishes in Phanerozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 set up the Yeast Balloon practical. Each group receives: 1 small plastic bottle (500 mL), 1 balloon, 1 sachet of dried yeast (~7 g), 10 g sugar (glucose or table sugar), 200 mL warm water (~37–40 °C, pre-measured in a cup), a ruler, and a marker pen. Procedure: (1) Pour warm water into the bottle. (2) Add sugar and swirl for 30 seconds. (3) Add yeast and swirl again. (4) Immediately stretch the balloon over the bottle neck. (5) Every 5 minutes for 20 minutes, measure the circumference of the balloon with a ruler (or use a piece of string and ruler) and record in a results table. Learners also record qualitative observations (smell,	Distribute materials to pre-assigned groups. Before learners begin, say: 'Before you mix anything, predict as a group: will the balloon inflate? Why or why not? Write one group prediction in the top box of your sheet. You have 60 seconds.' After 60 seconds, say: 'Begin your setup — step 1 now.' Circulate continuously. At each group, ask: 'What gas do you think is inflating the balloon?' (Wait 10 seconds.) 'Where is that gas coming from?' At the 10-minute mark, pause all groups: 'Stop. Measure your balloon circumference now and record it. Then eyes on the screen.' Play the video clip. During or immediately after, say: 'On your	Parallel Evidence Streams (PES): Learners simultaneously gather TWO lines of evidence — the yeast balloon (macroscopic, quantitative, supporting phenomenon 4) and the video of the athlete (human context, supporting phenomenon 1) — and are asked to connect both to the same underlying process. This mirrors how scientists triangulate evidence from multiple sources and prevents learners from treating yeast respiration as an isolated, irrelevant topic.	Use the observation sheets as a real-time window into understanding. Specifically check: (1) Are learners writing CO_2 as the product of yeast anaerobic respiration? (2) Do any learners write 'oxygen' as a reactant in the anaerobic equation — a key misconception to address in Phase 3? (3) When annotating the video, do learners mention lactic acid or simply say 'muscles tire'? Collect 4–5 observation sheets at the end of the phase for targeted feedback before the next lesson.

	appearance of liquid). While waiting for results, learners simultaneously complete a printed Observation Sheet: they write the word equations for (a) aerobic respiration and (b) anaerobic respiration in yeast, and (c) anaerobic respiration in animals, using a textbook or provided reference card. At the 10-minute mark, the teacher plays the 2-minute 'burning legs' video clip (supporting phenomenon 1 — e.g., footage of a Kenyan 800 m runner like David Rudisha finishing a race, legs visibly straining, or equivalent athletics footage) and learners annotate their observation sheet with what they think is happening in the runner's cells at that moment.	observation sheet, write what you think is happening inside the runner's muscle cells at this exact moment. Use what you know about respiration.' Allow 2 minutes writing. After video, say: 'Back to your balloons — measure again at 15 and 20 minutes.' In the final 5 minutes of the phase, cold-call 2 groups: 'Group 3 — what is your balloon circumference at 20 minutes? What does that tell us?' and 'Group 6 — did you notice a smell? Describe it. What might that smell tell a scientist?' Wait 10 seconds after each question before accepting answers.		
Explain Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher facilitates a whole-class construction of a side-by-side comparison table on the board. Learners copy the completed table into their notebooks. The table has three columns — Aerobic Respiration, Anaerobic Respiration (Animals/Lactic Acid), Anaerobic Respiration (Yeast/Ethanol) — and rows for: Reactants, Products, Energy Released (relative: high/low), Oxygen Required? (yes/no), Location in cell, Real-life example. Learners then work individually for 10 minutes to write a structured explanation (minimum 8 sentences) answering the prompt: 'Explain why lactic acid builds up in your leg muscles during intense step-ups and what oxygen debt means, using evidence from today's yeast balloon experiment and your knowledge of respiration.' A sentence-starter scaffold is	Begin the compare-contrast table by saying: 'Let's build this together. I will ask, you will tell me.' Point to aerobic column: 'Reactants for aerobic respiration — who can give me both reactants? Think back to Lesson 5.' Wait 10 seconds. Cold-call 1 learner. Write the answer. Repeat for each cell of the table, cold-calling a different learner each time (minimum 9 different learners across the table). For the anaerobic animal row, pause and say: 'Here is the key question. No oxygen is used — so what is being broken down? And what waste product builds up?' Wait 15 seconds. If no confident answer, prompt: 'Think about what you felt in your legs during the step-ups on Day 1.' After completing the table, say: 'Now I want you to write this in YOUR OWN WORDS. Not a list — a story. Imagine you are a muscle cell during the 800 m race. What is	Compare-Contrast Anchor Table + Structured Scientific Argument (SSA): The three-column table forces learners to hold aerobic and both anaerobic pathways in working memory simultaneously, making the distinction between lactic acid and ethanol pathways explicit and durable. The SSA writing task then requires learners to weave evidence (yeast balloon data, video, personal step-up experience) into a coherent causal chain — the hallmark of scientific sensemaking. This directly mirrors NGSS SEP 6: Constructing Explanations.	During the 10-minute writing period, circulate and use a 3-colour sticky dot system on notebooks: green dot = explanation includes lactic acid, oxygen debt, AND connects to the balloon evidence; yellow dot = mentions lactic acid but oxygen debt is missing or unclear; red dot = explanation is descriptive only, no causal mechanism. Use red-dot learners as a target group for the DQB phase. After the three readings, ask the class by show of hands: 'Does this explanation answer WHY lactic acid builds up — not just WHAT it is?' Quantify the show of hands and note it as class progress data.

	available on the board for learners who need it. Three volunteers read their explanations aloud and the class offers structured peer feedback using the format: 'I agree with… because… / I would add… because…'	happening to you?' Display the sentence starters: 'During intense exercise, muscle cells need… / When oxygen supply is insufficient… / Lactic acid accumulates because… / Oxygen debt refers to… / Evidence from the yeast balloon shows that…' Allow 10 minutes silent individual writing. After 3 volunteer readings, use Pose-Pause-Pounce: pose the feedback question ('What is the strongest scientific idea in this explanation?'), pause 10 seconds, then cold-call a peer to respond — not the writer.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Biodiversity flourishes in Phanerozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives one sticky note. They write one sentence that completes this stem: 'We now know that the burning feeling in our legs happens because…' Sticky notes are added to the DQB under the 'Burning Legs' question. The class then does a Gallery Walk of the DQB (2 minutes) to read their peers' responses. A class secretary (rotating role) physically moves the 'Burning Legs' sticky note from the 'Questions' column to the 'Explained' column — a visible, ceremonial moment marking the resolution of a mystery that has been open since Lesson 1. Learners also identify any NEW questions that today's lesson has raised and post them on the 'New Questions' column (e.g., 'How quickly does the body clear lactic acid after exercise?' or 'Can yeast be used to make food in Kenya?').	Distribute sticky notes and say: 'One sentence. What do we NOW know about the burning legs mystery? You have 2 minutes.' After posting, say: 'Take 2 minutes to walk and read at least 5 of your classmates' notes. Look for: does anyone explain it better than you did? If yes, put a small tick on their note.' After gallery walk, call the class secretary forward: 'Today we answer a question we have had since Day 1.' Have the secretary ceremonially move the sticky note. Lead a brief moment of acknowledgement: 'That is what scientists feel when evidence finally answers a question they have been investigating for weeks.' Then say: 'But has today's lesson opened any NEW questions in your mind? Think about the yeast making ethanol — is there any everyday use of that in Kenya? Post new questions now.' Allow 2 minutes for new sticky notes. Read 2–3 new questions aloud and say: 'These are great — some of these connect to what we will see in Lessons 8 and 9.'	DQB Resolution Ceremony + New Question Generation: The physical act of moving a question from 'unresolved' to 'explained' gives learners a concrete, emotionally satisfying marker of scientific progress. New question generation immediately prevents closure from becoming complacency and models the iterative nature of scientific inquiry — every answer opens new doors, connecting to real Kenyan contexts (fermentation in mandazi-making, chang'aa production, bread baking).	Read all sticky notes during the gallery walk. Sort them mentally into: (1) causal explanation (lactic acid + anaerobic + insufficient oxygen) — target; (2) surface description ('muscles get tired') — needs follow-up; (3) misconception ('oxygen is toxic to muscles') — address at the start of Lesson 8. Count: what proportion of the class (out of total posted notes) offered a causal explanation? This gives a class-level readiness indicator for the summative assessment in Lesson 10.

Model Building Phase 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative 'Body During Exercise' model diagram begun in Lesson 1 and revised in Lessons 3, 5, and 6. Today's revision task: (1) Add a new panel or annotation inside a muscle cell showing the TWO possible respiration pathways — aerobic (with oxygen, producing CO_2 + water + lots of ATP) and anaerobic (without enough oxygen, producing lactic acid + a little ATP). Use arrows to show when each pathway is active. (2) Draw or annotate to show lactic acid accumulating in the cell and entering the bloodstream. (3) Add a small key showing: Green arrow = aerobic pathway; Red arrow = anaerobic (lactic acid) pathway. (4) Write one sentence on the model: 'Oxygen debt = the extra oxygen needed after exercise to break down accumulated lactic acid.' Learners who finish early add a second panel showing the yeast pathway (ethanol + CO_2) and label it as a comparison. Models are displayed briefly at the desk for a peer to leave a written 'Notice + Wonder' post-it before being stored in the learner's portfolio.	Say: 'Take out your model from your portfolio. You are going to make it smarter today — it should now show what happens INSIDE a muscle cell when oxygen runs out.' Display a teacher exemplar model on the board (a rough sketch, deliberately imperfect, showing only one pathway) and say: 'My model shows aerobic respiration. What is MISSING from my model?' Cold-call 2 learners (wait 10 seconds each). After they identify the anaerobic pathway, say: 'Exactly. Add that pathway to YOUR model now. You have 12 minutes. Be precise — use the word equations from your comparison table.' Circulate and give live feedback: 'I can see your aerobic pathway — now show me what the cell does when oxygen is not enough. Draw the lactic acid building up.' In the final 2 minutes, say: 'Pass your model one person to the right. Write one Notice (something they did well scientifically) and one Wonder (a question their model raises) on a sticky note and leave it on their model.' Collect 5 models to review for portfolio assessment.	Cumulative Model Revision (CMR) with Dual-Pathway Annotation: Revisiting and revising the SAME model across multiple lessons builds a visible record of growing understanding and mirrors how scientific models are iteratively refined as evidence accumulates. The dual-pathway annotation (aerobic vs. anaerobic, animal vs. yeast) requires learners to integrate today's new concepts with all prior lessons — this is NGSS SEP 2 (Developing and Using Models) at its deepest level.	As you collect 5 models, use a 3-point rubric: (3) Both pathways shown with correct reactants and products, lactic acid correctly located, oxygen debt annotated; (2) Both pathways shown but products partially incorrect OR lactic acid not connected to oxygen debt; (1) Only one pathway shown or major factual errors present. Record rubric scores against learner names and use to form differentiated support groups for Lesson 8's review opener.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) Did the Yeast Balloon practical produce visible CO_2 inflation within 20 minutes? If not, what variables may have affected yeast activity (water temperature too cool, old yeast sachets, insufficient glucose)? How would you adjust the setup? (2) When learners wrote their 8-sentence explanations of lactic acid and oxygen debt, what proportion achieved a causal account vs. a descriptive account? What sentence-starter or additional scaffolding would help the descriptive writers? (3) The DQB Resolution Ceremony is a high-engagement moment — did learners respond with genuine interest, or did it feel procedural? How can you make the transfer of the sticky note more meaningful for your specific class? (4) Were there any learners who raised the connection between yeast anaerobic respiration and everyday Kenyan food production (mandazi, uji, fermented milk/mursik)? If so, note their names — they may serve as peer explainers in Lesson 8 when biotechnology applications are introduced. (5) Review the 5 collected cumulative models: do learners correctly place lactic acid INSIDE the muscle cell and show it entering the blood, or do they confuse the site of production? This misconception, if present, must be addressed at the start of Lesson 8 before moving to the circulatory system's role in clearing lactic acid.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a balloon inflate over a bottle containing yeast, glucose, and warm water over 20 minutes, recording increasing balloon circumference at 5-minute intervals. They also observed a video clip of a Kenyan distance runner experiencing visible muscle fatigue at the finish line.
What did I learn?	Learners learned that anaerobic respiration in yeast produces ethanol and carbon dioxide (inflating the balloon) and that anaerobic respiration in animal muscle cells produces lactic acid. They learned that both pathways release energy without oxygen, but release far less ATP than aerobic respiration. They defined oxygen debt as the extra oxygen required after intense exercise to break down accumulated lactic acid back to CO_2 and water.
How does this explain the phenomenon?	Learners explained in writing why lactic acid builds up in leg muscles during intense step-up exercise — because oxygen supply to muscle cells is insufficient to sustain aerobic respiration alone, forcing cells to switch to the lactic acid pathway. They connected the balloon evidence (CO_2 production without oxygen) to the broader principle of anaerobic energy release, resolving the 'burning legs' mystery that has been on the Driving Question Board since Lesson 1.

LESSON 8 (80 minutes): Respiratory Quotient (RQ): What Fuel Is Your Body Burning?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce learners to the concept of Respiratory Quotient (RQ), enable them to calculate RQ values for different food substrates, and use RQ data to infer the metabolic fuel source of an organism.
Knowledge	Learners will know that: (1) Respiratory Quotient (RQ) is the ratio of CO_2 produced to O_2 consumed during cellular respiration (RQ = CO_2 produced ÷ O_2 consumed); (2) different macronutrients yield characteristic RQ values — carbohydrates ≈ 1.0, fats ≈ 0.7, proteins ≈ 0.9; (3) RQ values reflect which substrate is being oxidised in cellular respiration; (4) a mixed or shifting RQ in athletes indicates a switch in fuel source during prolonged exercise.
Skills	Learners will be able to: (1) apply the RQ formula to calculate RQ for given gas-exchange data; (2) interpret RQ values to identify the substrate being respired; (3) analyse RQ data from athletic scenarios to draw evidence-based conclusions; (4) construct and annotate worked examples in their science notebooks.
Attitudes	Learners will: (1) appreciate that quantitative analysis deepens biological understanding beyond observation alone; (2) show curiosity about how the body selects and switches between fuel sources; (3) value precision and accuracy in biological calculations.
Key Inquiry Question	How do animals obtain energy from food?
Purpose in Storyline	In Lessons 1–7 learners established HOW gases move — from the environment into the lungs, across respiratory surfaces, and into the bloodstream. Lesson 8 shifts the lens to WHY: it quantifies exactly what the cells do with oxygen once it arrives. By calculating RQ values for ugali (maize starch), nyama choma (fat), and githeri (protein/beans), learners gain a chemical 'fingerprint' of cellular respiration and can now explain the burning sensation in their leg muscles during the step-up exercise as evidence of a fuel switch. This directly advances the driving question by connecting the increased O_2 demand observed in the phenomenon to the specific substrates being combusted inside muscle cells.
Safety Notes	No hazardous materials are used in this lesson. Standard classroom safety applies. Learners should be reminded not to run inside the classroom if any brief physical demonstration is used to revisit the phenomenon.

B. LESSON OVERVIEW

Lesson 8 is an EXPLAIN lesson in which learners are formally introduced to Respiratory Quotient (RQ) as a quantitative tool for identifying the metabolic fuel a body is burning. Beginning with a recall of the burning sensation felt in leg muscles during the anchoring step-up exercise, learners are guided to ask: 'How can we tell, chemically, what the muscles were using for fuel?' The teacher introduces the RQ formula (RQ = CO_2 produced ÷ O_2 consumed), then models calculations step-by-step using three Kenyan food substrates — maize starch from ugali (RQ = 1.0), animal fat from nyama choma (RQ ≈ 0.7), and mixed protein from githeri beans (RQ ≈ 0.9). Learners practise 4 calculation problems independently before engaging in a whole-class discussion about what an athlete's RQ during a Nairobi Marathon tells us about fuel source switching. The lesson closes with learners revising their cumulative explanatory model to annotate which substrate is being respired at rest versus during intense exercise, directly revisiting the original phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown three images on the board: a plate of ugali (maize starch), a piece of nyama choma (fat-rich meat), and a bowl of githeri (beans and maize — protein and carbohydrate). They are asked: 'When your muscles were burning during the step-ups, which of these foods do you think was being broken down inside your muscle cells — and does it matter which one? Would different foods require different amounts of oxygen?' Learners write a 2-sentence prediction in their notebooks, then share with a seat-partner before selected pairs report to the class.	Display the three food images prominently. Read the prompt aloud: 'Look at these three foods. During the step-up exercise your muscles were on fire. Which of these was burning inside your cells — and does the choice of food change how much oxygen your body needs?' Allow 10 seconds of silent thinking (WAIT TIME). Instruct: 'Write your prediction — two sentences, no wrong answer yet.' After 2 minutes, say: 'Turn and tell your partner.' After 1 minute, cold-call 3 pairs (one per food image) using a random name stick: 'Pair at the back — what did you predict about ugali?' Record 3–4 representative predictions verbatim on the board under the heading PREDICTIONS. Do NOT correct errors yet. Close with: 'Hold those predictions — by the end of this lesson you will be able to test each one with mathematics.'	Elicitation of prior knowledge using a food-context prediction prompt. Connecting the familiar sensory experience (muscle burn from the phenomenon) to abstract chemistry activates existing schema and creates cognitive dissonance that motivates the RQ concept. Recording predictions publicly creates accountability and a reference point for end-of-lesson revision.	Observable indicator: learners' written predictions reveal whether they associate oxygen consumption with food type. Listen for — and note on a clipboard — any learner who correctly anticipates that fat might need more oxygen (RQ logic), as these learners can be strategically used as peer explainers in Phase 3.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a one-page structured reading (or teacher-narrated worked example on the board, if printing is unavailable) titled 'Measuring the Body's Fuel Gauge.' The reading presents: (1) the definition of RQ and its formula; (2) the balanced chemical equation for aerobic respiration of glucose ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$), showing that 6 molecules of CO_2 are produced for every 6 molecules of O_2 consumed → RQ = 6/6 = 1.0; (3) the balanced equation for a representative fat (tripalmitin), showing that 51 CO_2 are produced for 72.5 O_2 consumed → RQ ≈ 0.70; (4) a note that protein RQ ≈ 0.9 (exact value varies with amino acid	Distribute the reading or project it on the board. Say: 'You have 8 minutes to read and annotate. Underline the formula every time you see it. Circle the numbers being divided. Write the RQ value next to the food it belongs to.' Set a visible timer for 8 minutes. Circulate — after 3 minutes pause and say: 'Show me with your fingers: how many RQ values have you calculated so far?' Use this to gauge pacing. At the 8-minute mark, bring the class to attention. Use a colour-coded mini-lecture (3 minutes maximum) to walk through each equation on the board: 'Ugali is made from maize starch — glucose. Watch what happens: six CO_2 out, six O_2 in. Six divided by	Annotated structured reading combined with teacher-modelled equation analysis (Close Reading + Worked Example). Learners move from passive reading to active annotation, then observe the teacher decompose each chemical equation step-by-step. The consistent three-food Kenyan context (ugali / nyama choma / githeri) anchors abstract molar ratios in familiar objects, reducing cognitive load and supporting transfer.	Observable indicator: during the 'show me fingers' check, any learner showing 0 fingers receives targeted support from the teacher during circulation. After the mini-lecture, ask learners to hold up their notebooks — check that each food has a correct RQ annotation (1.0, 0.7, 0.9). Learners who cannot annotate correctly are paired with a peer explainer identified in Phase 1.

	composition). Learners annotate the reading by underlining the RQ formula, circling the CO_2 and O_2 values in each equation, and writing the resulting RQ beside each food image.	six is…?' Cold-call 2 learners. 'Nyama choma — fat. The numbers are bigger, but more oxygen is needed. Seventy-two point five oxygen molecules in, fifty-one CO_2 out. What is fifty-one divided by seventy-two point five?' Allow 15 seconds WAIT TIME. Cold-call 1 learner with a calculator. Write 0.70 on the board. 'Githeri beans — protein — RQ roughly 0.9. We will use 0.9 as our working value.' Pause and ask: 'What pattern do you notice about which food has the lowest RQ?' Allow 10 seconds WAIT TIME. Cold-call 2 learners.		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners complete 4 structured calculation problems in their notebooks, progressing from scaffolded to independent: Problem 1 (scaffolded): 'A resting person consumes 250 mL of O_2 and produces 250 mL of CO_2 per minute. Calculate RQ. What substrate are they respiring?' Problem 2 (guided): 'A Kericho tea-farm worker doing light weeding consumes 600 mL O_2 and produces 420 mL CO_2 per minute. Calculate RQ. Is this person primarily burning ugali (carbohydrate) or nyama choma (fat)?' Problem 3 (independent): 'A long-distance runner from Iten training at altitude consumes 3,200 mL O_2 per minute and produces 2,240 mL CO_2 per minute. Calculate RQ and identify the dominant fuel.' Problem 4 (extension/discussion): 'At the start of the Nairobi Marathon, the same runner's RQ is measured at 0.97. By the 30 km mark it has dropped to 0.72. What does this change tell you about how the runner's fuel source has shifted? Why might this	Distribute or project the 4 problems. Say: 'Work through Problems 1 to 3 on your own — show every step: write the formula, substitute the numbers, calculate, then write a one-sentence interpretation. You have 10 minutes. Problem 4 is a discussion problem — attempt it, but we will tackle it together.' Set a 10-minute timer. Circulate actively — crouch to learner level for Problems 2 and 3. Listen for the common error of inverting the ratio ($O_2 \div CO_2$) and address immediately: 'Check your formula — which gas is on top?' After 10 minutes: 'Form a group of 3 with the people nearest you. Compare your answers. If you disagree, work through the maths together. You have 3 minutes.' At 13 minutes, call the class. Cold-call different learners for each problem (minimum 4 cold-calls, 1 per problem): 'Amina — give us Problem 2. Walk us through every step.' For Problem 4, facilitate whole-class discussion: 'What does RQ = 0.97 at the start tell us? What does RQ = 0.72 at 30 km tell	Scaffolded problem progression (I Do → We Do → You Do) with peer comparison and whole-class synthesis. Each problem is deliberately set in a Kenyan physical context (resting person, tea-farm worker, Iten runner, Nairobi Marathon) so learners practise applying RQ in situations that feel real and local. Problem 4 bridges calculation to conceptual interpretation, the highest cognitive demand of the lesson, and directly revisits the burning-muscle mystery from the phenomenon.	Observable indicators: (1) during circulation, check that learners write the formula before substituting — if not, prompt: 'Show me your formula first'; (2) during group comparison, listen for productive disagreement about Problem 3 or 4 — these are high-value teachable moments; (3) during cold-calls, assess whether learners can verbalise the interpretation ('RQ of 0.72 means fat is the dominant fuel') not just the numerical answer. Any learner who cannot interpret Problem 4 is flagged for targeted questioning during the DQB phase.

	happen?' After individual work (10 minutes), learners compare answers in groups of 3 and agree on a group answer before a whole-class debrief.	us? Why would a runner switch fuels?' Allow 15 seconds WAIT TIME before each cold-call. Explicitly connect to the phenomenon: 'Remember the burning in your legs during step-ups? That burning links to this fuel switch — keep that in mind as we build our model.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class Driving Question Board is revisited. Learners receive a sticky note (or write in a shared table on the board if sticky notes are unavailable). Each learner writes one sentence answering either: (a) 'What new evidence does RQ give us about what is happening inside our muscle cells during the step-up exercise?' OR (b) 'What new question does today's lesson raise for you about fuel, oxygen, and the body?' Learners post their notes under the appropriate column: WHAT WE NOW KNOW / WHAT WE STILL WONDER. The teacher reads selected notes aloud and annotates the DQB with a red marker to show connections to previous lessons' evidence (breathing mechanics, gas exchange, haemoglobin transport).	Direct learners' attention to the DQB. Say: 'This board has been tracking our journey since the step-up exercise in Lesson 1. Today we added a new tool — RQ. Take your sticky note and write one sentence: either something you now know, or something you still wonder.' Allow 3 minutes of silent writing. As learners post, read 4–5 notes aloud and ask the author: 'Say more about that.' Use a red marker to draw arrows between the new RQ notes and previous DQB entries about breathing rate doubling, haemoglobin carrying oxygen, and the burning sensation. Say explicitly: 'Look at this arrow — the reason your breathing rate doubled during step-ups is partly because your muscles were switching from burning carbohydrate to burning fat, which needs MORE oxygen per unit of energy. Your body was responding to a change in RQ.' Cold-call 2 learners to paraphrase this connection in their own words.	Driving Question Board curation with teacher-drawn connection arrows. This strategy makes the cumulative nature of scientific evidence visible. By physically linking RQ (today's lesson) to breathing rate and the burning sensation (the phenomenon), the teacher makes explicit the storyline thread that learners are building toward a complete explanation of the driving question.	Observable indicator: the quality of WHAT WE NOW KNOW notes reveals whether learners can connect RQ to the phenomenon rather than simply restating the formula. Target indicator for mastery: a learner writes something like 'The burning in my legs during step-ups might mean my muscles were switching to fat as fuel (lower RQ), which needs more O_2 — that is why I had to breathe faster.' Any note that only restates 'RQ = $CO_2 \div O_2$' without a phenomenon connection indicates surface-level understanding requiring follow-up.
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos	Learners open their cumulative explanatory model (begun in Lesson 1 and revised in each subsequent lesson). Today's revision task: add an RQ annotation layer to the cellular respiration section of their model. Specifically, learners must: (1)	Say: 'Open your cumulative model to the cellular respiration section. You have 8 minutes to add today's RQ layer. Three things must appear: the RQ value at rest, the RQ value during hard exercise, and the ratio of CO_2 to O_2 on your arrow.' Display the three required	Cumulative model revision with a three-element checklist and peer narration. Requiring learners to annotate RQ values directly onto their existing model of gas exchange, haemoglobin transport, and cellular respiration forces integration across all eight	Observable indicator: the model narrative must contain three components to indicate mastery — (1) a correct RQ value for at least one substrate, (2) a stated direction of change in RQ during exercise, and (3) an explicit link between the RQ change and

📖 **HTML:** [Cellular Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	draw or label the mitochondrion in the muscle cell and annotate it with 'RQ ≈ 1.0 at rest (carbohydrate fuel)' and 'RQ ≈ 0.7 during prolonged exercise (fat fuel)'; (2) draw an arrow from the mitochondrion to the blood vessel showing CO_2 leaving and O_2 entering, with the ratio labelled; (3) add a speech-bubble thought from a muscle cell: 'When I switch from glucose to fat, I need more O_2 — so the lungs and heart must work even harder.' Learners write a 3-sentence model narrative below their diagram explaining how RQ links to the breathing rate and pulse rate changes observed in the phenomenon.	elements on the board as a checklist. Circulate and use targeted prompts: 'What fuel gives RQ = 1.0?' / 'If RQ drops to 0.7, what does the cell need more of?' / 'How does that connect to what you felt during the step-ups?' At 8 minutes, cold-call 2 learners to hold up their models and narrate their 3-sentence explanation to the class. After each narration, ask the class: 'Does their model answer part of our driving question — why does the body work so much harder to get oxygen during exercise?' Allow 10 seconds WAIT TIME. Close the lesson by pointing to the driving question posted on the wall: 'We now know that the body works harder to get oxygen partly because the fuel being burned inside cells changes — and fat demands more oxygen per unit of energy than glucose. Next lesson we investigate what happens when oxygen runs out entirely.'	lessons. The speech-bubble convention (muscle cell 'thinking') humanises the biochemistry and helps learners maintain a coherent narrative rather than isolated facts. The 3-sentence narrative directly practises the scientific communication skill of evidence-based explanation.	increased breathing/heart rate from the phenomenon. Collect 6–8 notebooks at random at the end of the lesson to review model narratives before Lesson 9. Any learner whose narrative lacks the phenomenon connection receives written feedback: 'Good calculation — now tell me: how does this RQ change explain what YOU felt during the step-up exercise?'

D. TEACHER REFLECTION

After the lesson, consider: (1) Were learners able to move beyond rote calculation to interpret RQ values in context — especially Problem 4 about the Nairobi Marathon runner's fuel switch? If not, plan a 5-minute retrieval starter for Lesson 9 using a new Kenyan athlete scenario. (2) Did the DQB notes show genuine phenomenon-to-concept connections, or did learners treat RQ as an isolated formula exercise? If notes were superficial, use the first 5 minutes of Lesson 9 to cold-call 3 learners to re-narrate the link between RQ and the burning sensation from the step-ups. (3) Were any learners confused by the stoichiometry of the fat equation (RQ ≈ 0.7)? Consider preparing a simplified visual for Lesson 9 showing that fat molecules are 'oxygen-poor' and therefore need more O_2 from outside. (4) Check the collected notebooks: if more than 30% of model narratives lack a phenomenon connection, restructure the model-building phase opener of Lesson 9 to include a whole-class think-aloud narration before independent work. (5) Note which learners consistently gave correct interpretations during cold-calls — these learners can serve as peer tutors during Lesson 9's investigation into anaerobic respiration, which will directly address the burning sensation mystery.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	During the step-up exercise at the start of our unit, we observed that breathing rate and pulse rate both doubled — and some learners felt a burning sensation in their leg muscles even while breathing hard.

What did I learn?	Today we learned that the body can burn different fuels — maize starch (carbohydrate) from ugali, fat from nyama choma, and protein from githeri — and each fuel produces a different ratio of CO_2 to O_2, measured as the Respiratory Quotient (RQ). Carbohydrate gives RQ = 1.0, fat gives RQ ≈ 0.7, and protein gives RQ ≈ 0.9. During prolonged exercise, the body shifts from burning carbohydrate to burning fat, which requires MORE oxygen — explaining why the lungs and heart must work even harder.
How does this explain the phenomenon?	The burning sensation in leg muscles during the step-up exercise is connected to a shift in fuel source: as exercise continues, muscles switch toward fat (lower RQ), demanding more oxygen per unit of energy. This is why breathing rate and heart rate rise so sharply — the body is responding not just to increased energy demand, but to a change in the type of fuel being burned inside the muscle cells.

LESSON 9 (80 minutes): Practical & Data Analysis: Factors Affecting the Rate of Respiration	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners design and conduct a controlled investigation into the effect of temperature on yeast respiration rate, tabulate and graph their data, identify the optimum temperature, and explain enzyme denaturation — then connect these findings to why body temperature rises during exercise.
Knowledge	Learners will know that: (a) the rate of cellular respiration is affected by temperature; (b) enzymes controlling respiration have an optimum temperature (~37 °C in mammals); (c) above the optimum temperature, enzymes denature and respiration rate falls; (d) body temperature rises during exercise because cellular respiration releases heat as a by-product.
Skills	Learners will be able to: (a) formulate a testable hypothesis about temperature and respiration rate; (b) design a controlled investigation identifying independent, dependent, and control variables; (c) collect, tabulate, and graph experimental data accurately; (d) interpret a graph to identify an optimum and explain the denaturation region; (e) construct a scientific explanation linking evidence to the driving question.
Attitudes	Learners will: (a) appreciate precision and repeatability in scientific data collection; (b) show responsibility by following safety guidelines when handling hot water; (c) demonstrate curiosity by connecting yeast enzyme behaviour to their own muscle cells.
Key Inquiry Question	How do animals obtain energy from food?
Purpose in Storyline	In Lessons 1–8 learners established that exercise increases breathing and heart rate to supply more oxygen to muscles, and that aerobic and anaerobic respiration release energy in cells. Lesson 9 now asks: what controls HOW FAST those reactions happen? By manipulating temperature in a yeast investigation, learners discover the enzyme-driven nature of cellular respiration and gain the final piece needed to answer the DQB question — why does body temperature rise during exercise? This directly extends the anchoring phenomenon: the burning sensation in muscles and the rise in skin temperature after step-ups are now explicable through enzyme kinetics and metabolic heat release.
Safety Notes	Hot-water baths at 50 °C: learners must use tongs or heat-resistant gloves; teacher circulates continuously during this phase. Ice water at 10 °C: prolonged immersion of hands should be avoided. Yeast suspension: non-toxic, but learners should wash hands after handling. Broken glassware to be reported immediately. No learner should ingest any materials.

B. LESSON OVERVIEW

Learners open by revisiting the anchoring phenomenon — they felt body heat rise after step-ups in Lesson 1. The teacher poses the question: could temperature be affecting how fast their cells respire? In the Predict phase learners sketch a prediction graph of CO_2 production vs. temperature for yeast. In the Observe phase groups conduct a controlled experiment: equal amounts of yeast–sugar suspension are placed in water baths at 10 °C, 25 °C, 37 °C, and 50 °C; CO_2 output is measured by counting bubbles per minute (or by balloon inflation diameter) over 5 minutes. In the Explain phase groups tabulate, graph, and compare class-pooled data, identifying the optimum temperature and explaining the enzyme denaturation drop at 50 °C. In the DQB phase learners add a new card: "Respiration releases heat — is that why my body got warm during step-ups?" The lesson closes with model revision: learners annotate their cellular respiration diagram with a temperature–rate curve and the phrase "enzyme-controlled reactions."

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually recall the step-up activity from Lesson 1: their skin felt warm, muscles burned, breathing doubled. The teacher displays a blank axes graph (x-axis: temperature in °C; y-axis: rate of CO_2 production) and asks each learner to sketch their predicted curve for yeast respiration across 10 °C, 25 °C, 37 °C, and 50 °C in their notebooks. Learners annotate the sketch with a one-sentence justification. Pairs briefly compare and discuss differences in their predictions before a whole-class share-out.	Display the blank axes on the board. Say: 'Look back at your notebook entry from Lesson 1 — you recorded that your body felt warmer after step-ups. Today we are going to find out whether temperature changes how fast cells respire. Before we touch any equipment, I want you to sketch what YOU think the graph will look like for yeast at four different temperatures.' Allow 3 minutes of silent individual work. Then say: 'Compare your sketch with your partner. Do you agree on what happens at 50 °C?' Allow 2 minutes. Cold-call 3 pairs using lolly sticks. For each response ask: 'What is your reasoning — why did you draw it that way?' Apply 10-second wait time after each question before accepting answers. Record 3–4 contrasting prediction sketches on the board without evaluating them — label them 'Prediction A', 'Prediction B', etc. Say: 'We will return to these at the end of the lesson.'	Prediction Sketching with Justification — asking learners to commit a hypothesis to paper before observing activates prior knowledge of enzymes (covered in earlier Biology strands) and creates cognitive dissonance when results differ, deepening retention of the explanation.	Circulate and scan notebooks during the 3-minute sketch. Observable indicator: learner has drawn axes, plotted at least two data points or a curve shape, and written a justification sentence. Flag learners whose sketches show a straight line (suggesting they have not connected temperature to enzyme activity) for targeted questioning during the experiment.
Observe Phase 📹 **VIDEO:** Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12	Groups of four receive an equipment tray: 4 boiling tubes each containing 10 ml of 5% glucose–yeast suspension, a delivery tube fitted with a short rubber tube ending in a small balloon (or a beaker of water with the delivery tube submerged for bubble counting), a thermometer, and a stopwatch. Each group manages one temperature condition, and the class collectively covers all four conditions (10 °C ice bath, 25 °C room-temperature water bath, 37	Before distributing equipment say: 'We are scientists today — let us be precise. Every group has the same amount of yeast, the same amount of glucose, and the same time. The ONLY thing that changes between groups is temperature. What type of variable is temperature in our experiment?' Wait 10 seconds; cold-call one learner: 'That is the independent variable — well done. What is the dependent variable — what are we measuring?' Wait 10 seconds; cold-call a different learner.	Jigsaw Data Pooling — each group generates one data point; the class table aggregates all four, making the full pattern visible only through collaboration. This mirrors real scientific practice where no single laboratory generates all the evidence, and motivates learners to trust and interrogate shared data.	Observable indicators: (1) each group's results table has three trial readings and a calculated mean (not just one reading); (2) the thermometer reading is recorded beside each trial (demonstrating understanding of variable control); (3) at least one group member can state the independent and dependent variables without prompting when the teacher visits the station.

🔍 Search ARES for similar readings	°C warm-water bath, 50 °C hot-water bath prepared by the teacher). After 5 minutes of equilibration, each group counts CO_2 bubbles per minute (or measures balloon diameter in mm using a ruler) for 3 successive minutes and records all three readings. Groups enter their mean value into a shared class data table drawn on the board.	Distribute equipment. Circulate every 2 minutes. At the 50 °C station say: 'Use the tongs to lower your tube — do not put your hands in the hot water.' At each station, prompt: 'Read your thermometer now — is the temperature stable?' If a group records wildly different trial values ask: 'What might have caused that variation? Is that a problem?' After all groups enter data on the board, say: 'Look at the class table — does anything surprise you compared to your prediction sketch? Don't answer yet — we will unpack this together.'		
Explain Phase 📹 **VIDEO:** Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner plots the class-pooled data (mean CO_2 rate vs. temperature) on a pre-printed grid in their notebook, draws a best-fit curve, and labels: (a) the optimum temperature, (b) the region of increasing rate, and (c) the region of denaturation. Groups then answer three structured questions: (1) Why does the rate increase from 10 °C to 37 °C? (2) Why does the rate fall sharply between 37 °C and 50 °C? (3) How does this connect to the burning sensation in your leg muscles and the warmth you felt after step-ups in Lesson 1? Groups share answers whole-class; the teacher builds a consensus explanation on the board using the phrase 'enzyme-controlled reactions.' Learners write a final two-sentence scientific explanation in their notebooks.	Say: 'Plot the class data on your grid now — you have 5 minutes.' Circulate; if a learner draws a straight line say: 'Look carefully at what happens between 37 °C and 50 °C — should that be a straight line?' After graphing, display the three structured questions on the board. Say: 'Discuss question 3 in your group — this is the big one. How does yeast at 50 °C connect to YOUR muscles after step-ups?' Allow 3 minutes of group talk. Cold-call three different groups for question 3 — explicitly pick one group that had a very different prediction sketch and ask: 'How does your prediction compare to the actual data?' After answers, say: 'The enzymes that control respiration in your muscle cells are the same type as in yeast — protein molecules that change shape when too hot. At 37 °C — normal body temperature — they work at their best. During hard exercise, heat builds up in the muscle. Your body works hard to keep temperature from going too high.' Write on board: 'Respiration	Claim–Evidence–Reasoning (CER) Writing — learners must state a claim (respiration has an optimum temperature), cite their graph as evidence, and provide reasoning (enzyme shape and denaturation) in exactly two sentences. This structured writing consolidates scientific argumentation and directly links the lab data back to the anchoring phenomenon.	Collect the two-sentence CER explanations (written in notebooks) as an exit check. Observable indicators of proficiency: the explanation names 'enzymes', references the data (e.g., 'the graph showed the highest rate at 37 °C'), and makes a causal link to the phenomenon (e.g., 'this is why body temperature must stay near 37 °C even during exercise'). Learners who write only 'it got too hot' without mentioning enzymes are flagged for a brief follow-up conversation.

		is enzyme-controlled. Optimum temperature for human enzymes ≈ 37 °C. Above optimum → denaturation → rate falls.' Apply 12-second wait time after each board statement before asking: 'Can someone add to this or challenge it?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Enzyme reaction velocity and pH](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cellular Respiration (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one new sticky-note card for the DQB. They choose from two prompts: (A) 'Something I can now explain about the step-up phenomenon that I could not explain before,' or (B) 'A new question this lesson has raised for me.' Cards are posted and the class does a 2-minute gallery sort — grouping cards under the emerging categories: 'Temperature and Enzymes', 'Why Body Temp Rises', and 'Still Wondering.' The teacher reads aloud two 'Still Wondering' cards to the class to signal that productive curiosity is valued.	Say: 'Our driving question asks why your body works so much harder during exercise AND what happens inside cells when there is not enough oxygen. Today's experiment gave us a new angle — temperature. Write your DQB card now — you have 90 seconds.' After posting, gesture to the board: 'I see several cards about body temperature rising. Let us place those together.' Facilitate the sort with learners doing the physical grouping. Read two 'Still Wondering' cards aloud (e.g., 'If enzymes denature at 50 °C, how come athletes in Iten training in the Rift Valley do not destroy their enzymes when they get really hot?'). Say: 'That is a brilliant question — hold it. Lessons 10 and 11 will bring us closer to it.' Point to the prediction sketches still on the board: 'Look at Prediction B from the start of the lesson — whose was it? How does it compare to what we found?' Allow the author to respond.	DQB Gallery Sort — physically clustering cards by theme makes the accumulating evidence structure visible to all learners and reinforces the narrative arc of the unit. Identifying 'Still Wondering' questions validates inquiry as ongoing and previews upcoming lessons, maintaining motivation.	Observable indicator: every learner has posted a card (100% participation check). Quality indicator: the card contains a subject-specific term (enzyme, denaturation, optimum, temperature, respiration) rather than a vague statement. The teacher photographs the DQB at the end of class to track conceptual growth across the unit.
Model Building Phase 📹 **VIDEO:** [Enzyme reaction velocity and pH](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners open to their cumulative cellular respiration diagram (first drawn in Lesson 6 and revised in Lessons 7 and 8). They add two new annotations: (1) a small temperature–rate curve sketch inside the mitochondria symbol, labelled 'enzyme-controlled; optimum ~37 °C'; (2) a heat-release arrow exiting the cell,	Say: 'Open to your cumulative model. You have been building this since Lesson 6. Today you are adding the temperature dimension — the enzymes inside this mitochondria are sensitive to temperature, and the reactions release heat.' Demonstrate the two annotations on the board diagram. Say: 'Add these to YOUR model	Cumulative Model Revision — progressive annotation of the same diagram across lessons externalises conceptual growth and helps learners see that scientific models are refined iteratively as new evidence is gathered, mirroring the practice of scientists at institutions such as the Kenya Medical Research	Observable indicators: (1) the temperature–rate curve is present and correctly shaped (rising to ~37 °C, falling at 50 °C); (2) the heat-release arrow is labelled with a connection to body temperature; (3) the connecting sentence at the bottom references the anchoring phenomenon (step-ups, burning muscles, or body warmth). Models

📖 **HTML:** [Cellular Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	labelled 'metabolic heat → body temperature rises during exercise.' Learners then write one sentence connecting the model to the anchoring phenomenon at the bottom of the diagram page.	now, in your own style — you may use arrows, colour, or labels, but both pieces of information must be visible.' Allow 5 minutes. Circulate and ask three learners: 'Read me the sentence you wrote connecting this to the step-ups from Lesson 1.' After sharing, say: 'Your model now shows not just WHAT happens in cellular respiration, but WHY the rate changes and WHERE the heat comes from. That is scientific modelling — well done.' Close by pointing to the DQB driving question written on the wall: 'We have now answered: why does body temperature rise during exercise. What part of the driving question are we still working on?' Wait 12 seconds; cold-call two learners.	Institute (KEMRI).	lacking the denaturation drop in the curve indicate the learner needs reinforcement on enzyme denaturation before Lesson 10.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Practical management — Did all four temperature conditions produce clearly different results? If the 50 °C group showed similar results to 37 °C, was the temperature actually maintained? Plan to use a larger water bath container or more frequent thermometer checks next time. (2) Graph interpretation — Could learners independently identify the optimum and the denaturation region, or did they require heavy scaffolding? If more than a third needed prompting, consider pre-teaching the concept of 'optimum' using the familiar example of ugali cooking — maize flour thickens best at a specific temperature — before the graph plotting next time. (3) Phenomenon connection — Did learners' CER sentences authentically link yeast data back to their own muscles, or did they treat the two as separate topics? If the connection was weak, open Lesson 10 by re-reading three anonymised CER sentences and asking the class to strengthen the reasoning together. (4) DQB quality — Were the 'Still Wondering' cards genuinely novel, or were they repeating questions already answered? Novel questions signal healthy inquiry; repeated questions signal a misconception that needs addressing at the start of Lesson 10. (5) Equity check — Which learners did not post a DQB card or did not complete the model revision? Reach out individually before Lesson 10.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Yeast–sugar suspension in water baths at 10 °C, 25 °C, 37 °C, and 50 °C produced different rates of CO_2 bubbles per minute. The highest bubble rate was recorded at 37 °C; the rate at 50 °C was lower than at 37 °C, and the rate at 10 °C was the lowest of all.
What did I learn?	The rate of cellular respiration is controlled by enzymes. Enzymes work faster as temperature rises — up to an optimum of approximately 37 °C for yeast and human cells. Above the optimum, the enzyme's shape is permanently altered (denaturation) and the reaction rate drops sharply. Cellular respiration also releases heat as a by-product, which is why body temperature rises during exercise.
How does this explain the	After the step-ups in Lesson 1, learners felt their muscles burn and their skin become warm. We can now explain this: the enzymes

phenomenon?	driving cellular respiration in muscle cells released metabolic heat as respiration rates surged to meet energy demand. The body must maintain temperature close to 37 °C — the optimum for its enzymes — which is why sweating and increased blood flow to the skin begin during vigorous exercise in the Kenyan heat.

LESSON 10 (80 minutes): Factors Affecting Rate of Respiration (continued) & Importance of Gaseous Exchange	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explain the remaining factors that affect the rate of cellular respiration and to construct a concept map showing why gaseous exchange and respiration are essential to life.
Knowledge	Learners will be able to: (a) explain how substrate concentration, oxygen availability, and surface area affect the rate of cellular respiration; (b) describe the importance of respiration in providing energy for movement, growth, active transport, heat regulation, and reproduction.
Skills	Learners will be able to: (a) analyse data and graphical representations showing the effect of varying substrate concentration and oxygen availability on respiration rate; (b) construct and annotate a concept map linking gaseous exchange to life processes; (c) evaluate the significance of aerobic capacity in real-life Kenyan contexts including agricultural labour, sports performance, and high-altitude survival.
Attitudes	Learners will: (a) appreciate the integrated nature of gaseous exchange and cellular respiration in sustaining life; (b) value evidence-based reasoning when explaining biological phenomena; (c) show curiosity about how their own body manages energy demands during everyday physical activity.
Key Inquiry Question	How do animals obtain energy from food, and why does efficient gaseous exchange determine how well that energy can be released?
Purpose in Storyline	In Lessons 8 and 9 learners established that temperature and enzyme activity (including the role of pH) affect respiration rate. Lesson 10 completes that factor analysis by adding substrate concentration, oxygen availability, and surface area, then zooms out to ask: WHY does any of this matter? Learners synthesise all unit evidence into a concept map that explicitly links the anchoring phenomenon — doubled breathing and pulse rates after step-ups — to the ultimate purpose of gaseous exchange: keeping every cell supplied with the substrate and oxygen it needs to release energy for movement, growth, active transport, heat regulation, and reproduction.
Safety Notes	No hazardous chemicals are used in this lesson. All data analysis is paper/digital-based. If the optional brief step-up activity is used to re-anchor the phenomenon, ensure learners with known cardiac or respiratory conditions (asthma) sit out or self-moderate. Clear the aisle between desks before any movement activity.

B. LESSON OVERVIEW

Lesson 10 is the second and final EXPLAIN lesson on factors affecting respiration rate. The lesson opens by briefly reconnecting to the anchoring phenomenon data (class step-up breathing and pulse rates from Lesson 1) before moving into a structured class discussion of the three remaining factors: substrate concentration, oxygen availability, and surface area of respiratory membranes. Learners analyse two short graphical data sets — one showing glucose concentration vs. CO_2 output in yeast (a proxy model), one showing respiration rate in athletes at different altitudes in the Kenyan Rift Valley — and use CLAIM–EVIDENCE–REASONING (CER) to explain each graph. In the second half of the lesson, learners individually draft and then collaboratively refine a concept map showing the importance of respiration, anchored to five life-process nodes: movement, growth, active transport, heat regulation, and reproduction. The lesson closes with a DQB update and a cumulative model revision, explicitly annotating where the three new factors appear in the whole-unit respiratory model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Worked example: logistic model equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners re-examine the class data table from Lesson 1 (resting vs. post-exercise breathing rate and pulse rate) displayed on the board or projected on screen. They are asked to silently read two short scenario cards: – SCENARIO A: 'Achieng has eaten a large plate of ugali and githeri before the 400 m race at the Nyayo Stadium inter-schools athletics meet. Her muscles have plenty of glucose. Does having MORE glucose automatically mean her muscles release energy faster?' – SCENARIO B: 'Kamau is a maize farmer in the Rift Valley highlands at 2,400 m altitude. The air at that altitude has the same percentage of oxygen (21%) but lower air pressure, so each breath delivers fewer oxygen molecules. How do you predict his respiration rate compares to a farmer working at sea level in Mombasa doing the same task?' Learners write a 2-sentence prediction for each scenario in their notebooks, explicitly linking their prediction to the phenomenon ('This connects to our step-up data because…'). No feedback is given yet.	1. Display the original class data table and say: 'Before we continue building our explanation, I want to take you back to where this all started — your own bodies after those step-ups in Lesson 1. Look at the numbers. Keep them in mind for the next 5 minutes.' 2. Distribute or project the two scenario cards. Say: 'You have 3 minutes — silent, individual thinking. Write a prediction for EACH scenario. Use the sentence stem: I predict… because in our step-up data we saw…' 3. Set a visible 3-minute timer. 4. After 3 minutes, cold-call 2 learners on Scenario A and 2 on Scenario B (4 cold-calls total). Do NOT confirm or correct answers. Say: 'Hold those ideas — we are going to test them with evidence right now.'	PHENOMENON RE-ANCHOR + PREDICTIVE WRITING — Returning to the original data before new instruction activates prior knowledge from the full evidence-gathering sequence (Lessons 1–9) and creates cognitive dissonance that motivates the new content. Linking predictions to the phenomenon ensures learners do not treat new factors as isolated facts.	Observable indicator: Every learner has written two predictions in their notebook before cold-calling begins. Teacher scans notebooks during the 3-minute window, noting whether learners reference oxygen, glucose/substrate, or the phenomenon data — these are the conceptual entry points for targeted questioning in Phase 3.
Observe Phase 📹 **VIDEO:** Worked example: logistic model equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners work in pairs to analyse two printed (or projected) graphs: GRAPH 1 — 'Substrate Concentration vs. Rate of CO_2 Production in Yeast Given Different Concentrations of Glucose Solution (0%, 2%, 5%, 10%, 20%).' The curve rises steeply, then plateaus. A data	1. Distribute graph sheets and observation worksheets. Say: 'You have 10 minutes with your partner. Focus on DESCRIBING the patterns first — do not try to explain yet. Use the prompts on your sheet.' 2. Set a 10-minute timer. Circulate, pausing at pairs who conflate correlation with causation on	STRUCTURED DATA ANALYSIS with OBSERVATION SCAFFOLDING — Separating 'describe' from 'explain' is a deliberate NGSS Science and Engineering Practice (Analysing and Interpreting Data). It prevents premature closure and ensures learners build explanations from evidence rather than recalling	Observable indicator: Pairs can accurately describe the shape of Graph 1 (rises then plateaus) and the trend in Graph 2 (decreases with altitude) using the correct axis variables, without yet explaining the mechanism. Any pair that writes an explanation in the DESCRIBE box has conflated observation with inference —

📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	table accompanies the graph. GRAPH 2 — 'Estimated Aerobic Respiration Rate (arbitrary units) of Kenyan Long-Distance Runners Tested at Nairobi (1,795 m), Eldoret (2,085 m), and a High-Altitude Training Camp near Mt. Kenya (3,000 m) — same workload applied at each site.' Values decrease with altitude despite identical effort. For each graph, pairs complete a structured observation sheet with three prompts: – DESCRIBE: What pattern do you see in the data? (No explanation yet.) – IDENTIFY: Which variable on the x-axis represents which factor affecting respiration? – CONNECT: Does this graph support or challenge your prediction from Phase 1? Learners also read a 4-sentence text box titled 'Surface Area and Gaseous Exchange' explaining that a larger respiratory surface (e.g., the enormous combined alveolar surface area of ~70 m^2 in human lungs) allows more oxygen to diffuse per unit time, directly increasing the maximum possible rate of aerobic respiration. They annotate one key sentence.	Graph 1 (common error). Ask: 'What exactly is changing on the x-axis? What is being measured on the y-axis? Are those the same thing?' 3. At 7 minutes, give a 3-minute warning: 'You should be on the CONNECT prompt now.' 4. After 10 minutes, bring the class together. Ask pairs to share their DESCRIBE answers only (no explanations yet). Cold-call 3 pairs — one per graph section and one on the text box annotation. Say: 'We are only describing right now — save your explanations for our discussion.' 5. Briefly validate descriptions without explaining mechanisms: 'Good — you have the pattern. Now let us figure out WHY.'	memorised phrases. The Kenyan altitude context (Eldoret, Mt. Kenya) makes Graph 2 personally and culturally meaningful.	teacher notes these pairs for targeted support in Phase 3.
Explain Phase 📹 **VIDEO:** Worked example: logistic model equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES	PART A — CLASS DISCUSSION (Remaining Factors, ~18 minutes): Using the graphs from Phase 2 as evidence, the class works through each of the three remaining factors in a teacher-facilitated discussion. For each factor, learners are invited to construct a CLAIM–	PART A: 1. Write on the board: 'CLAIM | EVIDENCE | REASONING'. Say: 'We are going to build three CER explanations together — one per factor. For each one, someone gives me the CLAIM first. Then we find the EVIDENCE in your	CLAIM–EVIDENCE–REASONING (CER) FRAMEWORK for Part A — a named NGSS Practice (Constructing Explanations) that forces learners to distinguish empirical evidence from logical inference, building scientific argumentation skills. CONCEPT	Observable indicators: (1) During CER discussion, learners can identify the evidence from the correct graph without prompting. (2) Concept maps contain all five mandatory nodes, at least one Kenyan example per node, and at least one explicit link to gaseous

for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	EVIDENCE–REASONING (CER) explanation: 1. SUBSTRATE CONCENTRATION: Claim — increasing glucose concentration increases respiration rate up to a limit. Evidence — Graph 1 plateau. Reasoning — at the plateau, enzymes (including those of glycolysis and the Krebs cycle) are working at maximum capacity; adding more substrate cannot increase rate further because the enzyme active sites are already saturated. 2. OXYGEN AVAILABILITY: Claim — lower oxygen availability reduces aerobic respiration rate. Evidence — Graph 2 (Kenyan runners at altitude). Reasoning — oxygen is the final electron acceptor in the electron transport chain; without sufficient O_2, the chain slows, ATP production falls, and the cell must rely more on anaerobic respiration (explaining the burning sensation in muscles — lactic acid accumulation — that learners felt during their own step-ups in Lesson 1). 3. SURFACE AREA OF RESPIRATORY MEMBRANE: Claim — a larger respiratory surface allows a higher maximum rate of gas diffusion and therefore a higher maximum aerobic respiration rate. Evidence — text box annotation; also linked to Lesson 5 data on alveolar surface area. Reasoning — diffusion rate is proportional to surface area (Fick's Law revisited); diseases that reduce surface area (e.g.,	graphs. Then we build the REASONING together.' 2. For each factor, wait 10 seconds after posing the question before accepting answers. Cold-call at least 2 learners per factor (6 cold-calls minimum across Part A). 3. For oxygen availability, say: 'Think back to Lesson 1 — some of you felt a burning sensation in your legs during the step-ups. You were breathing hard, so oxygen was coming in. But was ALL of that oxygen reaching your muscle cells fast enough? What does Graph 2 tell us about what happens when oxygen delivery is limited?' Wait 15 seconds. 4. Explicitly connect lactic acid to the anchoring phenomenon: 'This burning feeling — this is the answer to one of our original driving questions. Write that in the margin of your graph sheet right now: BURNING = LACTIC ACID = ANAEROBIC RESPIRATION WHEN O_2 IS INSUFFICIENT.' 5. For surface area, ask: 'If a disease destroys alveoli — reducing your lung surface area from 70 m² to 30 m² — what does Fick's Law predict will happen to how much oxygen can enter your blood per breath? And therefore, what happens to your maximum aerobic respiration rate? Cold-call 2 learners.' PART B: 1. Say: 'Now I want you to answer the BIG question: why does any of this matter? Build your concept map. You have 12 minutes working alone — then 3 minutes comparing with your partner.' 2. Set timer. Circulate. Prompt	MAPPING for Part B — a metacognitive sensemaking tool that makes the relational structure of knowledge visible; requiring Kenyan examples prevents surface-level copying and demands genuine application of understanding to familiar contexts.	exchange per node. (3) Learners can verbally explain — when cold-called — why the burning sensation in their legs during step-ups is evidence of anaerobic respiration caused by insufficient oxygen delivery, thereby directly connecting Lesson 10 content to the anchoring phenomenon.

	emphysema) permanently reduce aerobic capacity. PART B — CONCEPT MAP CONSTRUCTION (~20 minutes): Learners individually draft a concept map (spider or hierarchical, their choice) centred on the node 'CELLULAR RESPIRATION PROVIDES ENERGY FOR:' with five mandatory branch nodes: MOVEMENT, GROWTH, ACTIVE TRANSPORT, HEAT REGULATION, REPRODUCTION. For each node they must add: – one specific Kenyan example (e.g., Movement → Kipchoge running a marathon; Active Transport → absorption of glucose from githeri in the small intestine; Heat Regulation → maintaining body temperature while harvesting maize in Kericho in the cold highlands) – one linking phrase showing the connection to gaseous exchange (e.g., 'which requires a continuous supply of O_2 via the alveoli') After 12 minutes of individual work, pairs compare maps and add at least one idea from their partner's map in a different colour. Groups of four then share one 'best connection' with the class.	learners who have thin Kenyan examples: 'What does a tea picker in Kericho need energy for that is specific to their work? Where does that energy come from? What gas exchange has to happen first?' 3. After pair comparison, cold-call one group of four to share their 'best connection.' Ask: 'How does that connection link back to our step-up data from Lesson 1?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Worked example: logistic model](#)	Learners return to the class Driving Question Board. The original driving question is re-read aloud: 'Why does my body work so much harder to get oxygen into my cells during exercise — and what happens inside those cells when	1. Re-read the driving question aloud and say: 'Ten lessons in — let us see how far we have come. Write your sticky note. You have 3 minutes.' 2. Set a 3-minute timer. Circulate and read notes. Identify 2–3 high-	DRIVING QUESTION BOARD PROGRESSIVE REFINEMENT — By physically connecting new cards to prior evidence cards with labelled arrows, learners make visible the cumulative, iterative nature of scientific explanation-	Observable indicator: Sticky notes explicitly name one of the three new factors AND include a connective phrase linking it to the driving question or to the step-up phenomenon data. Notes that only restate the factor without

equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	there is not enough?' Each learner writes one sticky note (or one entry in a shared digital board) responding to the prompt: 'Using what we learned today about [substrate concentration / oxygen availability / surface area — choose one], I can now explain part of our driving question because…' Learners also scan the existing DQB cards from Lessons 1–9 and draw a coloured arrow from their new card to any earlier card it connects with, labelling the arrow with a linking phrase (e.g., 'This extends Lesson 5's alveolar surface area finding because…'). The class briefly identifies which parts of the driving question are now FULLY answered, which are PARTIALLY answered, and which still need evidence (preview: Lesson 11 will address the circulatory link).	quality notes and 1 note that has a misconception to address publicly. 3. After posting, cold-call the 2–3 learners with strong notes to read them aloud. Ask the class: 'Does everyone agree this answers part of our driving question? Thumbs up / thumbs sideways / thumbs down.' 4. Address the misconception note without naming the writer: 'I noticed one note said [paraphrase misconception]. Let us think about that together — is that consistent with what Graph 2 showed us?' 5. End by pointing to the remaining unanswered portion of the driving question — the circulatory link — and say: 'Next lesson we will complete the circuit. Literally.'	building. This mirrors how real scientists revise models as new evidence accumulates, aligning with the NGSS practice of Engaging in Argument from Evidence.	connecting to the driving question indicate a learner who has memorised content but not yet integrated it — flag these learners for targeted support in Lesson 11.
Model Building Phase 📹 **VIDEO:** Worked example: logistic model equations Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative respiratory model — the annotated diagram they have been building and revising since Lesson 2, which currently shows breathing mechanics, alveolar gas exchange, and the basic cellular respiration equation with temperature and enzyme/pH effects labelled. Using a coloured pen different from all previous revision colours, learners add three new annotations to their model: 1. An arrow from the 'glucose in blood' label pointing into the mitochondrion, annotated: 'Rate increases with ↑ substrate (glucose) concentration — up to enzyme saturation limit.' 2. An arrow from the 'O_2 in blood' label pointing to the electron	1. Say: 'Take out your cumulative model. Choose a new colour — this is Lesson 10's layer. Add the three annotations I am putting on the board now.' Write the three annotation prompts on the board verbatim. 2. Set 8 minutes for individual model annotation. Circulate, checking that learners are adding all three annotations and that the altitude/lactic acid annotation explicitly references the burning sensation from the step-up phenomenon. 3. At 8 minutes, say: 'Now add your WHY IT MATTERS ring — five spokes, five Kenyan examples. 5 minutes.' 4. At 13 minutes, initiate gallery swap. Say: 'Exchange with the pair behind you. Read their model.	CUMULATIVE MODEL REVISION with COLOUR-LAYERED ANNOTATIONS — Using a new colour for each lesson's additions makes the growth of understanding literally visible; learners and teachers can trace which lesson introduced which idea. The 'WHY IT MATTERS' ring transforms the model from a mechanistic diagram into a functional explanation, directly addressing the 'so what?' dimension of the driving question. GALLERY SWAP (Peer Feedback) applies the NGSS practice of Obtaining, Evaluating, and Communicating Information in a low-stakes, collaborative format.	Observable indicator: The cumulative model contains all three new Lesson 10 annotations in the correct location (pointing into the mitochondrion or alveolar membrane as appropriate), the altitude/lactic acid annotation explicitly names the anchoring phenomenon ('burning sensation during step-ups'), and the WHY IT MATTERS ring has five populated spokes with Kenyan examples. Models missing the phenomenon reference in annotation 2 indicate incomplete integration of new content with the unit's central evidence — these learners should be prioritised for one-to-one follow-up at the start of Lesson 11.

	transport chain section of the mitochondrion, annotated: 'Rate decreases if O_2 availability falls (e.g., high altitude, Rift Valley highlands — less O_2 per breath → less ATP → anaerobic respiration → lactic acid → burning sensation.' 3. A bracket around the alveolar membrane labelled: 'Surface area ∝ diffusion rate (Fick's Law) — larger surface → more O_2 in → higher max aerobic respiration rate.' Learners also add a new outer ring to their model — a 'WHY IT MATTERS' ring — with five labelled spokes corresponding to the five importance nodes from their concept map (movement, growth, active transport, heat regulation, reproduction), each with a Kenyan example written in the margin. Pairs do a 2-minute 'gallery swap' — exchanging books with another pair, reading their model, and writing one 'warm' comment (something explained clearly) and one 'wonder' question (something they would like to see added).	Write one warm comment and one wonder question on a sticky note and attach it to their book. You have 2 minutes.' 5. After gallery swap, cold-call 2 learners to share the 'wonder question' they received. Ask the class: 'Can anyone answer that wonder question using evidence from today's lesson or any previous lesson?' Wait 10 seconds before taking responses. 6. Close by saying: 'Your model now shows not just HOW the body exchanges gases and respires — it shows WHY those processes matter for every life function. Next lesson, we complete the model by connecting the lungs to the heart and circulatory system — the final piece of our driving question puzzle.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. INTEGRATION CHECK — When learners constructed their CER explanations for oxygen availability, did they spontaneously connect the altitude data (Graph 2) to the lactic acid burning sensation from the Lesson 1 step-up activity? If fewer than half the class made this connection unprompted, plan a brief 5-minute phenomenon re-anchor at the start of Lesson 11 before introducing the circulatory system content.

2. CONCEPT MAP DEPTH — Did learners' concept maps show genuine application (Kenyan-specific examples for all five nodes) or surface-level repetition of definitions? If most maps used generic examples (e.g., 'energy for movement' without specifying a Kenyan athlete or labour context), introduce a 'specificity challenge' in Lesson 11: any claim made in discussion must include a named Kenyan person, place, or food.

3. CUMULATIVE MODEL COMPLETENESS — After the gallery swap, how many 'wonder questions' from peers pointed to gaps in the model that correspond to Lesson 11 content (circulatory system link)? If many wonder questions were about the heart and blood, use these as the opening hook for Lesson 11 — display them on the DQB and say: 'You asked these questions. Today we answer them.'

4. EQUITY CHECK — During cold-calling, were responses distributed across gender and seating zones? Note any learners who have not been cold-called in the last

two lessons and prioritise them in Lesson 11.

5. MISCONCEPTION WATCH — The most common misconception in this lesson is: 'More oxygen always means faster respiration.' Learners may not yet appreciate that substrate (glucose) availability can be the limiting factor even when oxygen is plentiful. If this appeared in DQB sticky notes or concept maps, address it explicitly at the start of Lesson 11 with a brief limiting-factor scenario using a Kenyan farming analogy (e.g., 'You have plenty of water but no seeds — the harvest is still zero.').

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners analysed two graphs: (1) glucose concentration vs. CO_2 output in yeast, showing respiration rate rises then plateaus; (2) aerobic respiration rate in Kenyan runners tested at Nairobi, Eldoret, and a high-altitude Mt. Kenya training camp, showing rate decreases with altitude despite identical workload. Learners also annotated a text box on surface area and gaseous exchange, connecting alveolar surface area (~70 m^2) to diffusion rate via Fick's Law.
What did I learn?	Three additional factors affect the rate of cellular respiration: (1) Substrate concentration — increasing glucose increases respiration rate up to the point of enzyme saturation, after which rate plateaus; (2) Oxygen availability — reduced O_2 (e.g., at high altitude in the Rift Valley highlands) slows aerobic respiration, forcing cells into anaerobic respiration and producing lactic acid — the source of the burning sensation felt during step-ups in Lesson 1; (3) Surface area of the respiratory membrane — a larger surface area (obeying Fick's Law) allows greater O_2 diffusion per unit time, raising the ceiling on aerobic respiration rate. Respiration is important for: movement (e.g., Kipchoge's marathon performance), growth, active transport (e.g., glucose absorption from githeri), heat regulation (e.g., working in cold Kericho highlands), and reproduction.
How does this explain the phenomenon?	The doubled breathing rate and pulse rate observed after step-ups in Lesson 1 reflect the body's urgent attempt to increase oxygen delivery to muscle cells so they can maintain aerobic respiration at the rate demanded by exercise. When oxygen delivery cannot keep pace — as it cannot at high altitude or during very intense exercise — cells switch to anaerobic respiration, producing lactic acid and causing the burning sensation many learners felt. The efficiency of this entire system depends on substrate availability, oxygen availability, and the surface area of the respiratory membranes — all three factors that were investigated in this lesson.

LESSON 11 (40 minutes): Model Building and Driving Question Board Consolidation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all evidence gathered across Lessons 1 to 10 into a single, fully annotated cumulative model and formally close every question posted on the Driving Question Board, demonstrating that the anchoring phenomenon — the sharp rise in breathing rate and pulse rate during step-ups — is now fully explained by integrated knowledge of respiratory surfaces, breathing mechanics, gas transport, and cellular respiration.
Knowledge	Learners can integrate and accurately represent: (a) comparative respiratory surfaces in earthworms, fish, insects, and mammals with their adaptive features; (b) the mechanics of inhalation and exhalation including diaphragm and intercostal muscle roles and pressure changes; (c) the pathway of oxygen from alveolus to muscle cell mitochondrion and carbon dioxide in reverse; (d) the balanced equations and energy outcomes of aerobic and anaerobic respiration; (e) the concept and calculation of respiratory quotient; and (f) how temperature, substrate concentration, and oxygen availability affect respiration rate.
Skills	Learners construct a multi-component annotated scientific model; use precise biological vocabulary in written explanations; apply peer-review criteria to evaluate the completeness and accuracy of a classmate's model; and write evidence-based answers to each DQB question.
Attitudes	Learners appreciate that scientific understanding is built progressively from evidence rather than from a single observation; they demonstrate intellectual honesty when peer-reviewing by offering specific, constructive feedback rather than generic praise.
Key Inquiry Question	How do animals exchange gases with their environment? How do animals obtain energy from food? Both inquiry questions are now answered in full and their connection to the lived experience of exercise is made explicit.
Purpose in Storyline	Lesson 11 is the evidence-consolidation and sense-making summit of the unit. Every prior lesson contributed one or more pieces of evidence; here learners assemble those pieces into a coherent, unified model and verify that no DQB question remains unanswered before the summative transfer task in Lesson 12.
Safety Notes	No hazardous materials are used. Ensure learners with physical disabilities are included fully in the recall of the step-up phenomenon through discussion rather than re-performance of exercise. Peer review must be conducted respectfully; remind learners that critique is directed at the model, not the person.

B. LESSON OVERVIEW

The lesson opens with a rapid whole-class recall of the anchoring phenomenon data from Lesson 1 to re-anchor learners in the original puzzle. In the Predict phase learners individually predict which components their final model must include before seeing the checklist. In the Observe phase learners examine a partially completed exemplar model and identify what is missing or inaccurate. In the Explain phase learners construct their own fully annotated cumulative model using a teacher-provided scaffold and conduct a structured peer review. The DQB phase formally closes every original question with written evidence-based answers. The Model Building phase gives learners time to revise their model in light of peer feedback and to add a final explanatory caption connecting the completed model back to the step-up phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners spend 4 minutes individually writing a list — without opening notes — of every biological process or structure they believe must appear in a complete explanation of why breathing rate and pulse rate doubled during the Lesson 1 step-up exercise. They then share with a shoulder partner and combine lists, starring items they are uncertain about. The teacher takes a quick show-of-hands tally on the board: 'Raise your hand if your list includes alveoli... lactic acid... RQ... Fick's Law.' This surfaces gaps before model construction begins.	Display the original class data table from Lesson 1 on the board or a printed copy. Say: 'Before we look at any checklist, I want to know what YOUR mind has stored across ten lessons. Look at that data — breathing rate doubled, pulse doubled, some of you felt your legs burn. What do you now know that you did not know in Lesson 1? Write every concept, structure, or equation that belongs in a full explanation. You have 3 minutes — go.' Set a visible timer. After 2 minutes say: 'One more minute — push yourself to name at least eight things.' After sharing with partners, conduct the tally and say: 'The starred items — the ones you are not sure about — are exactly where your model needs the most work today.' WAIT TIME: after each show-of-hands tally pause 5 seconds before commenting so learners can observe which items fewer classmates included.	Activating prior knowledge through unprompted free recall followed by partner comparison creates a personal gap-analysis that motivates the model-building task. Anchoring the recall to the original phenomenon data keeps the unit narrative coherent.	The teacher scans partner lists during circulation and notes which concepts (commonly RQ, Fick's Law, anaerobic pathway) are absent from most lists. These are flagged for emphasis during the Explain phase model-building scaffold.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner or pair receives a printed A4 sheet showing a partially completed exemplar cumulative model — a student-style drawing that includes a labelled alveolus and capillary, a mitochondrion with the aerobic equation partially written, and an arrow labelled 'oxygen travels here' but with no mention of haemoglobin, no anaerobic pathway, no comparative respiratory surfaces panel, no breathing mechanics labels, no RQ notation, and no reference to factors affecting rate. Learners are given 4 minutes to use a red pen to circle everything that is missing	Distribute the exemplar sheet face-down. Say: 'This is a model built by a learner who stopped studying after Lesson 5. Your job is not to admire it — your job is to find every gap and every inaccuracy using what you now know. Red pen only. You are the examiner. Go.' Circulate and prompt with: 'Does this model explain the burning legs? Does it tell me what happens when oxygen runs out? Does it show me why a fish gill works differently from a human alveolus?' After 4 minutes call two or three learners to share one gap each. Write each gap on the board as a checklist item. Say: 'Every	Critiquing an incomplete exemplar externalises the evaluation criteria before learners apply them to their own work, reducing anxiety and making quality standards concrete and co-constructed rather than imposed.	The teacher photographs or quickly notes the gaps identified by learners. If a critical component such as the anaerobic respiration equation is missed by most learners during the critique, the teacher adds it explicitly to the board checklist and makes a note to monitor that element during peer review.

	or inaccurate, referring to their own notes and the list they just made. They annotate directly on the sheet: 'MISSING: lactic acid pathway', 'MISSING: gill lamellae comparison', 'WRONG: no pressure change labels', and so on.	item on this board is something YOUR model must include. Let us now build it together.'		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a blank A3 model scaffold sheet divided into five labelled zones: Zone A — Comparative Respiratory Surfaces (earthworm skin, fish gill lamella, insect trachea, human alveolus); Zone B — Human Breathing Mechanics (inhalation and exhalation, labelled with muscle names, direction of rib movement, and pressure change); Zone C — Gas Exchange Pathway (alveolus to muscle mitochondrion including haemoglobin, blood vessels, and heart); Zone D — Respiration Equations (aerobic and anaerobic side by side with ATP yield noted, and one worked RQ calculation for a Kenyan food example of their choice — ugali, nyama choma, or githeri); Zone E — Factors Affecting Respiration Rate (temperature with enzyme denaturation noted, substrate concentration, oxygen availability with high-altitude Rift Valley example). Learners have 12 minutes to annotate all five zones. They may use their exercise books, lesson notes, and the board checklist. When complete, learners swap models with a partner and use a printed 10-point peer-review checklist to score and comment on the partner's model. Checklist items include: 'Is Fick's Law named in Zone A?', 'Is lactic acid shown in Zone D?', 'Is the Rift Valley altitude example present in	Distribute A3 scaffold sheets. Say: 'Five zones, twelve minutes. Every zone must have labels AND brief written explanations — diagrams alone are not enough. The burning-legs question, the doubled-breathing question, and the pulse question must all be answerable by someone reading your model. Begin.' Set a visible timer. At the 6-minute mark say: 'You should be finishing Zone B or starting Zone C. If you are still on Zone A, skip to Zone C and return.' At 11 minutes say: 'One minute — make sure Zone D has both equations and a food example.' After the swap, say: 'You are now a peer reviewer. Your job is to be honest and specific. Do not write good job — write exactly what is missing or what could be clearer. Your signature goes on the review, so own it.' WAIT TIME: after issuing the peer-review instruction, wait 10 seconds in silence before learners begin writing, allowing them to read the checklist fully.	The five-zone scaffold mirrors the storyline sequence of the unit, making the cumulative nature of the model visible. Requiring written explanations alongside diagrams prevents superficial labelling and demands genuine sensemaking. Peer review using an explicit checklist trains learners to evaluate scientific models against criteria rather than aesthetics.	The teacher circulates during model building and uses sticky dot stickers to mark models that are on track (green) or need a prompt (yellow). During peer review the teacher reads three or four peer comments aloud (anonymised) and asks the class: 'Is this feedback specific enough to help the learner improve?' This models the quality of feedback expected.

	Zone E?', and so on. Partners write one specific improvement suggestion in a speech bubble drawn at the bottom of the sheet.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class DQB — the physical or digital board created in Lesson 1 — is displayed. Learners read through all original questions posted. Working individually, each learner selects any three questions from the board and writes a complete evidence-based answer in their exercise book. Each answer must: name at least one lesson in which the evidence was gathered, use correct biological vocabulary, and connect back to the step-up phenomenon. The teacher then calls on volunteers to read answers aloud for two or three questions. For each question the class votes — thumbs up if fully answered, sideways if partially answered. Questions still marked sideways are addressed by the teacher with a 60-second targeted explanation before the lesson closes.	Point to the DQB and say: 'Every question on this board was asked by someone in this room in Lesson 1. That person did not know the answer then. Do they know it now? Do YOU know it now? Pick three questions — not the easiest three, the three that matter most to you — and write a full answer with evidence. Three minutes.' After writing, facilitate the read-aloud and class vote. For any question still unresolved say: 'This question connects to [specific lesson content]. The answer is [targeted explanation]. Write it down.' Close this phase by asking: 'Has anyone thought of a NEW question that our unit has opened up rather than closed?' Take two responses and acknowledge them as evidence of scientific thinking. WAIT TIME: after asking 'Pick three questions', wait 8 full seconds before learners begin writing so they genuinely scan the whole board rather than grabbing the first question they see.	Returning to the DQB created in Lesson 1 closes the narrative arc of the unit and makes the growth in learner knowledge visible and personal. The class vote creates a low-stakes public accountability structure and surfaces any persistent misconceptions before the summative lesson.	The teacher collects exercise books from five to six learners at random to read their three DQB answers as an informal check on depth of understanding. Learners whose answers reveal persistent gaps are identified for targeted support during the revision time at the start of Lesson 12.
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12	Learners retrieve their model from the peer-review exchange. They have 5 minutes to revise their model in response to the peer feedback written on it, using a different colour pen so revisions are visible. Finally, in the bottom margin of their A3 sheet, every learner writes a single caption of no more than three sentences that begins with the words: 'When I did step-ups in Lesson 1, my breathing rate doubled because...' This caption must reference at	Say: 'The peer feedback on your model is a gift. Use a different colour — blue or green — and make every improvement your reviewer suggested that you agree with. If you disagree with a suggestion, write a one-sentence defence next to it. Five minutes.' After 4 minutes say: 'Now the most important part. At the bottom of your sheet write the caption that starts: When I did step-ups in Lesson 1... — three sentences, three zones, three key terms. This	Requiring revision in a different colour makes the learning process itself visible — the model shows not just what the learner knows but how their thinking was refined through peer dialogue. The final caption anchors all abstract knowledge back to the lived, embodied experience of the anchoring phenomenon, completing the unit narrative loop.	The teacher reads captions on collected models that evening and sorts them into three groups: fully integrated (three terms, three zones, clear causal chain), partially integrated (missing one term or one zone link), and surface level (correct terms but no causal explanation). This sorting directly informs the targeted recap at the start of Lesson 12.

🔍 Search ARES for similar readings	least three zones of their model and must use the terms alveolar gas exchange, aerobic respiration, and either lactic acid or oxygen debt. Models are handed in at the end of the lesson for teacher review before Lesson 12.	is your promise to yourself that you understand.' Circulate and read captions as they are written, offering quiet individual prompts such as: 'Can you add what happens in the mitochondrion?' or 'What term did we use for the oxygen that has to be paid back?' Collect all models as learners leave and say: 'These come back to you in Lesson 12 as your revision tool for the final transfer task. Treat them as the best study notes you own.' WAIT TIME: after saying 'write the caption', pause 15 seconds to allow learners to plan before writing, rather than rushing to fill the space.		

D. TEACHER REFLECTION

After the lesson, consider: Did learners integrate all five zones with genuine explanatory connections, or did they treat zones as isolated compartments? Were peer-review comments specific and honest, or did social pressure produce only positive feedback? Did the DQB vote reveal any concept that remains poorly understood across the class — if so, which concept and how will I address it in the first 5 minutes of Lesson 12? Did the step-up caption reveal that learners can apply knowledge causally rather than just recite it? Which learners are still separating aerobic and anaerobic respiration as alternatives rather than understanding that the shift is driven by oxygen availability? Note the names of learners whose models show surface-level annotation only — they will need a direct prompt in Lesson 12 before attempting the transfer task.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed an incomplete exemplar cumulative model and identified missing components including the anaerobic lactic acid pathway, comparative respiratory surfaces panel, breathing mechanics pressure labels, haemoglobin as the oxygen carrier, and factors affecting respiration rate. Learners also observed their own model improve in response to specific peer feedback written using an explicit checklist.
What did I learn?	Learners synthesised knowledge from all ten prior lessons into a single, five-zone annotated cumulative model covering comparative respiratory surfaces, human breathing mechanics, the full gas exchange and transport pathway, aerobic and anaerobic respiration equations with RQ, and factors affecting respiration rate. Learners confirmed that every question on the Driving Question Board — all originally triggered by the Lesson 1 step-up phenomenon — can now be answered with biological evidence.
How does this explain the phenomenon?	Learners explained in a written model caption how the doubling of breathing rate and pulse rate during step-ups results from the integrated action of alveolar gas exchange, haemoglobin-mediated oxygen transport, aerobic cellular respiration in muscle mitochondria, and — when oxygen supply is insufficient — the anaerobic production of lactic acid that causes the burning sensation felt in Lesson 1. This explanation directly advances the driving question by demonstrating that the body works harder to

	supply oxygen because mitochondria consuming glucose in aerobic respiration continuously deplete local oxygen, and that when even doubled breathing cannot meet demand, anaerobic respiration fills the energy gap at the cost of lactic acid accumulation.

LESSON 12 (80 minutes): Summative Review & Transfer: Explaining the Body's Response to Exercise	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all unit learning by constructing a comprehensive scientific explanation of the anchoring phenomenon and transferring understanding to a novel scenario.
Knowledge	Learners integrate knowledge of respiratory surfaces, breathing mechanics, gas transport, cellular respiration (aerobic and anaerobic), and the role of the circulatory system to explain resting vs. post-exercise physiological changes.
Skills	Learners write a structured scientific explanation using unit vocabulary, compare their Lesson 1 model with their final model, and apply integrated understanding to a novel Kenyan context (swimming the Winam Gulf of Lake Victoria).
Attitudes	Learners appreciate the power of evidence-based scientific reasoning, demonstrate intellectual honesty in self-assessment, and celebrate the collaborative journey of answering hard questions about their own bodies.
Key Inquiry Question	How do animals exchange gases with their environment? How do animals obtain energy from food?
Purpose in Storyline	This is the capstone lesson of the unit. Learners return to the very data they collected in Lesson 1 — their own resting and post-exercise breathing and pulse rates — and now possess all the conceptual tools to explain what was once puzzling. The lesson moves from individual written explanation, through peer critique, to transfer of the full explanatory framework onto a novel scenario involving a Kenyan long-distance swimmer. The Driving Question Board is formally completed, and learners celebrate the questions they have answered across 12 lessons.
Safety Notes	No hazardous materials or physical activity in this lesson. Learners who may find revisiting exercise data uncomfortable (e.g., due to health conditions) may use the class average data rather than personal data. Emotional safety: remind learners that self-assessment is for personal growth, not ranking.

B. LESSON OVERVIEW

Lesson 12 is the summative capstone of Sub-Strand 3.3. Learners open by revisiting their personal Lesson 1 data (resting vs. post-exercise breathing and pulse rates) and their earliest explanatory model. Using a structured Final Explanation scaffold, they write a comprehensive account of the anchoring phenomenon — integrating respiratory surfaces, breathing mechanics, gas transport via blood, aerobic cellular respiration, lactic acid fermentation, and circulatory-respiratory coupling. A Socratic whole-class discussion surfaces gaps and refines explanations. Learners then tackle a novel transfer scenario: Akinyi, a competitive swimmer from Kisumu, attempts to cross the Winam Gulf of Lake Victoria. They predict and explain her physiological responses using unit vocabulary. The lesson closes with a formal DQB completion ceremony — learners physically move question cards to the 'Answered' column — and a written self-assessment against the unit's key inquiry questions. Teacher collects Final Explanation writing as summative evidence of learning.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners receive a printed copy (or view on screen) of their original	Distribute Lesson 1 data tables and initial models (kept by teacher	Retrieval of Prior Model (Metacognitive Anchoring): By	Teacher listens for whether learners accurately recall naive

📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Lesson 1 class data table showing resting and post-exercise breathing rates and pulse rates. They also receive their Lesson 1 initial model (the rough sketch they drew before any instruction). Working individually for 5 minutes, learners silently re-read their Lesson 1 model and answer in their notebooks: 'What did you think was happening in your body during exercise in Lesson 1? Write 2–3 sentences in your own words, as your Lesson 1 self would have explained it.' This activates memory of prior understanding and creates a concrete baseline for comparison.	since Lesson 1). Say: 'Do not look ahead to your later notes yet. I want you to genuinely remember what you thought in Lesson 1 — before we investigated anything.' Allow 5 minutes of silent individual writing. Circulate and observe: are learners genuinely recalling naive ideas, or importing later knowledge? After 5 minutes, cold-call 3 learners: 'Amina, read me your Lesson 1 explanation.' After each, say: 'Thank you — hold that thought. We are going to see how far your thinking has travelled.'	consciously re-inhabiting their Lesson 1 thinking, learners create a cognitive 'before' snapshot. This makes the growth in understanding visible and meaningful. The contrast between naive and scientific explanations is itself a powerful learning event.	ideas (e.g., 'I thought we breathe faster just because we are tired') rather than retrofitting current knowledge. This is NOT graded — it is a metacognitive warm-up. Observable indicator: learners write in simple, non-technical language consistent with pre-instruction thinking.
Observe Phase 📹 **VIDEO:** Elements and atoms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners spend 10 minutes on a silent Gallery Walk around the classroom. Posted around the walls are 8 'Evidence Stations' — each a brief summary card (prepared by teacher) representing one major evidence activity from the unit: (1) Lesson 1 breathing/pulse data table; (2) diagram of a respiratory surface with labelled features (thin walls, moist, large surface area, rich blood supply); (3) the class results from the breathing mechanics investigation (chest measurements at rest vs. inhale); (4) the dissection or diagram of a fish gill; (5) the chemical equation for aerobic respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + ATP; (6) the lactic acid fermentation equation and the 'burning legs' explanation from the Lesson 8 muscle fatigue investigation; (7) the diagram of the alveolus-capillary interface showing oxygen and CO_2 diffusion; (8) the circulatory system diagram showing how oxygenated blood travels from lungs to	Set up the 8 Evidence Stations before class. As learners walk, circulate quietly. Do not answer questions — redirect with: 'What does the evidence at that station tell you? Write it down.' After 10 minutes, call the class back. Ask: 'Raise your hand if you ticked more than 6 questions on the DQB as fully answered.' Then: 'Raise your hand if there is still a question on the DQB you are unsure about.' Cold-call 2 learners with remaining uncertainty: 'Korir, which question still feels incomplete to you — and which evidence station is most relevant to answering it?' Allow 10 seconds WAIT TIME before calling on a peer to help.	Evidence Audit (Synthesis Preparation): The Gallery Walk is not passive review — each 'This evidence tells me…' sentence requires learners to extract an explanatory claim from data. The DQB Audit externalises metacognitive monitoring: learners evaluate their own explanatory completeness before writing begins.	Collect notebooks briefly (or circulate during the walk) to check that 'This evidence tells me…' sentences are claim-based, not merely descriptive. Indicator of mastery: learner writes 'This evidence tells me that oxygen crosses into the blood because the alveolus wall is only one cell thick, making diffusion rapid' rather than 'This is a diagram of the alveolus.' Flag learners writing only descriptions for targeted support during Phase 3.

	muscles. At each station, learners write one sentence in their notebook: 'This evidence tells me...' They also do a DQB Audit: they review each question on the Driving Question Board and place a tick in their notebook next to questions they believe have now been fully answered.			
Explain Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cellular Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners write their Final Explanation — the centrepiece of the lesson — using a structured scaffold with five required components: (1) THE TRIGGER: What happens in the body in the first 10 seconds of exercise, and why? (mention muscle cells, ATP demand, oxygen consumption, CO_2 production); (2) THE RESPIRATORY RESPONSE: How does the breathing system respond, and why is this effective? (mention alveoli, respiratory surface features, diffusion, diaphragm and intercostal muscles, breathing rate and depth); (3) THE CIRCULATORY LINK: How does the heart and blood connect the lungs to the muscles? (mention haemoglobin, oxyhaemoglobin, capillaries, increased heart rate); (4) INSIDE THE CELL — AEROBIC AND ANAEROBIC: What happens chemically inside the muscle cell when oxygen is sufficient vs. when it is insufficient despite fast breathing? (mention aerobic respiration equation, ATP yield, anaerobic respiration/lactic acid fermentation, burning sensation in legs); (5) RETURN TO NORMAL: What happens when exercise stops? (mention oxygen debt, continued heavy breathing, lactic acid removal). Learners have 20 minutes to write. They then	Distribute the Final Explanation scaffold sheet. Read each of the 5 components aloud, pausing after each to say: 'Think for 10 seconds about which lesson gave you the evidence for this component.' [WAIT 10 seconds per component.] Then say: 'You have 20 minutes. Use your full scientific vocabulary — ATP, alveoli, diffusion, haemoglobin, lactic acid, aerobic, anaerobic. Every claim should connect back to your own data from Lesson 1.' Circulate during writing. Prioritise learners who were flagged in Phase 2 as writing only descriptive sentences — prompt them: 'You have written what happened. Now tell me WHY that happened at the molecular level.' After 20 minutes, direct peer review: 'Exchange with your partner. Use exactly the sentence stem on your sheet. You have 5 minutes.' After peer review, cold-call 2 pairs: 'Wanjiku, what one improvement did your partner suggest, and do you agree? Why?' Allow 10 seconds WAIT TIME.	Structured Scientific Explanation Writing (CER — Claim, Evidence, Reasoning): The five-component scaffold enforces the Claim-Evidence-Reasoning structure across multiple biological systems simultaneously, requiring learners to integrate, not merely recall. The peer review introduces collaborative epistemic critique — a core Science and Engineering Practice (Engaging in Argument from Evidence). Requiring learners to respond to peer feedback ('do you agree? why?') deepens metacognitive engagement.	Teacher collects Final Explanation writing at the end of the lesson as the primary summative artefact. Mastery indicators: (a) All 5 components addressed; (b) Correct use of ≥8 unit vocabulary terms; (c) Aerobic and anaerobic respiration both explained with equations or word descriptions; (d) Explicit link between circulatory and respiratory systems; (e) 'Oxygen debt' or equivalent concept used to explain post-exercise recovery. Partial mastery: 3–4 components present, equations missing or incorrect. Beginning: 1–2 components, vocabulary sparse. Use this rubric for targeted re-teaching in follow-up lessons.

	exchange with a partner for a 5-minute Peer Review using the sentence stem: 'You explained [X] clearly. I think you could strengthen [Y] by adding [Z].'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cellular Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class gathers around the DQB (physical board or projected digital version). The teacher reads each question card aloud one by one. For each question, learners vote by show of hands: 'Fully answered,' 'Partially answered,' or 'Still open.' For each 'Fully answered' card, a volunteer learner physically moves the card to the 'Answered' column and reads a one-sentence summary answer aloud to the class. For any 'Partially answered' or 'Still open' questions, the class briefly discusses which part remains uncertain and whether it connects to future Biology topics (e.g., detailed enzyme biochemistry of respiration — Grade 11; detailed cardiac physiology — Grade 11). Three key questions are explicitly resolved with whole-class constructed answers: (Q1) 'Why does breathing rate double during exercise?' (Q2) 'Why do legs burn even when you are breathing hard?' (Q3) 'Where exactly does the oxygen go once it enters the lungs?'	Facilitate the ceremony with energy and genuine celebration: 'In Lesson 1, this question was a mystery. Who can now answer it in one sentence?' For Q1, cold-call a learner: 'Otieno, give us the full answer — from muscle cell to diaphragm.' [WAIT 10 seconds.] For Q2, cold-call a different learner: 'Chebet, explain the burning sensation — use the word anaerobic.' [WAIT 10 seconds.] For Q3, cold-call a third learner: 'Mwangi, trace the oxygen molecule from the alveolus to the mitochondrion.' [WAIT 15 seconds — this is the most complex path.] After all cards are moved, say: 'Every question on this board came from YOUR observations of YOUR own body. You answered them with evidence. That is what scientists do.' Pause for acknowledgement of the class's collective intellectual work.	DQB Completion Ceremony (Epistemic Closure & Metacognitive Celebration): The physical act of moving question cards to 'Answered' externalises conceptual progress. Requiring a one-sentence oral summary per card consolidates declarative knowledge. Explicitly naming 'partially answered' questions models scientific humility — science always opens new questions — and motivates continued inquiry in Grade 11.	Observable indicator: Can learners articulate a precise, evidence-linked answer for each of the three key questions when cold-called? Mastery: answer includes mechanism (not just observation) — e.g., not 'we breathe faster' but 'CO_2 builds up in the blood, lowering pH, which signals the medulla oblongata to increase breathing rate.' Flag learners who give observation-only answers for follow-up written feedback.
Model Building Phase 📹 **VIDEO:** [Elements and atoms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	PART A — Model Comparison (10 min): Learners place their Lesson 1 initial model and their most recent (Lesson 11) model side by side. Using a two-column comparison sheet, they annotate: LEFT COLUMN — 'What my Lesson 1 model could NOT explain (and why)'; RIGHT COLUMN — 'What my final model CAN explain (and the evidence	Introduce Part A: 'Place both models in front of you. Your Lesson 1 model was your best thinking then — honour it, but interrogate it. What was it missing?' Allow 10 minutes. Circulate and prompt: 'What evidence from Lessons 3, 6, or 9 filled that gap?' Introduce the Akinyi scenario by reading it aloud with expression: 'Akinyi grew up	Model Comparison (Visible Learning Growth) + Transfer Task (Application to Novel Scenario): The side-by-side model comparison makes cognitive growth tangible and visible — a powerful metacognitive tool. The Akinyi transfer task is deliberately set in a Kenyan context (Lake Victoria, Kisumu, a named Kenyan girl) to honour the Kenya Context	Transfer Task indicators: (1) Q1 answer must include: increased breathing rate and depth, increased heart rate, more oxygen delivered to muscles via haemoglobin, aerobic respiration occurring in muscle cells; (2) Q2 answer must include: anaerobic respiration / lactic acid fermentation, equation ($C_6H_{12}O_6 \rightarrow 2C_3H_6O_3$ + 2ATP or word

📖 **HTML:** [Cellular Respiration (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	that made it possible).' They write at least 3 entries per column. PART B — Transfer Task (15 min): Learners read the following novel scenario independently: 'Akinyi is a 17-year-old competitive swimmer from Kisumu training to swim across the Winam Gulf of Lake Victoria — a distance of approximately 40 km. During her training swim, she maintains a steady pace for the first 30 km, breathing rhythmically with her face in the water and turning to inhale every 3 strokes. In the final 10 km, she pushes her pace hard, and her coach notices she is struggling to keep a steady breathing rhythm. By the time she reaches the shore, her arm and shoulder muscles feel a severe burning sensation, and she is breathing very heavily even though she has stopped swimming.' Learners answer three questions: (1) Describe and explain Akinyi's respiratory and circulatory responses during the first 30 km of steady swimming. (2) What is happening in Akinyi's muscle cells during the final 10 km sprint? Name the process, write the equation, and explain why the burning sensation occurs. (3) Why does Akinyi continue to breathe heavily AFTER she stops swimming? What is being repaid? PART C — Self-Assessment (5 min): Learners complete a written self-assessment checklist against the two KICD Key Inquiry Questions, rating themselves: 'I can explain this confidently with evidence / I can explain this but need more practice / I am still unsure about this.'	near Lake Victoria. She has trained for two years for this swim. Her body is about to face every challenge we have studied. Your job is to be her sports physiologist.' Allow 15 minutes for independent written responses. After writing, cold-call one learner for each of the 3 questions — choose learners who have not yet been cold-called this lesson. After each response, ask the class: 'Does anyone want to add or refine that answer?' [WAIT 10 seconds.] Close with Part C self-assessment: 'Be honest — this is for you, not for a grade. Circle the response that is truly where you are today.' Collect self-assessments for teacher planning of any revision sessions needed.	Rules and to test whether learners can de-contextualise their learning from the classroom phenomenon and re-contextualise it in a new scenario. This is the hallmark of deep understanding vs. surface memorisation. The self-assessment against KICD Key Inquiry Questions closes the loop on the curriculum objectives.	equivalent), lactic acid accumulation causing burning; (3) Q3 answer must include: oxygen debt concept — oxygen needed to oxidise lactic acid and replenish ATP/CP stores, hence continued heavy breathing. Self-assessment data informs teacher's planning for any follow-up revision lessons before the end of term.

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions in your professional journal: 1. EXPLANATION QUALITY: When you read the Final Explanation writing, were learners able to integrate all five components coherently, or did certain components consistently appear weak (e.g., the circulatory link, or the anaerobic pathway)? What instructional moves in earlier lessons might have strengthened those components? 2. TRANSFER SUCCESS: How did learners perform on the Akinyi transfer task compared to the original phenomenon explanation? Were they able to map the same mechanisms onto a new context (swimming vs. step-ups), or did they revert to surface-level descriptions? If transfer was weak, what does this suggest about the depth of conceptual understanding vs. rote recall of the phenomenon? 3. DQB CEREMONY: Which questions on the DQB were the hardest for learners to answer confidently in one sentence during the ceremony? Were there patterns — e.g., questions about the molecular level (ATP, lactic acid) being harder than questions about the organ level (breathing rate, heart rate)? This has implications for how to sequence micro-to-macro vs. macro-to-micro in future units. 4. MODEL COMPARISON: Did learners' Lesson 1 vs. final model comparisons reveal genuine conceptual change, or did some learners' final models still contain misconceptions (e.g., confusing breathing with cellular respiration, or believing oxygen is 'burned' directly in the lungs)? How will you address persistent misconceptions before the end-of-strand assessment? 5. KENYA CONTEXT: Did the Akinyi/Winam Gulf scenario feel authentic and engaging to learners from the Lake Victoria region vs. learners from inland counties (Nairobi, Kericho, Rift Valley)? Consider how you might localise transfer tasks further for your specific school community in future iterations. 6. SELF-ASSESSMENT HONESTY: Did learners' self-assessments align with your own observations of their understanding? Where there was a mismatch — learners rating themselves highly but showing weak performance, or vice versa — what does this suggest about calibration of self-efficacy, and how might you build metacognitive accuracy across the year?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners revisited their own Lesson 1 resting and post-exercise breathing rate and pulse rate data, and compared their initial naive model of exercise physiology with their final integrated model built across 12 lessons.
What did I learn?	Learners integrated the complete explanatory chain: muscle ATP demand → aerobic cellular respiration → CO_2 build-up → increased breathing rate and depth → alveolar gas exchange → haemoglobin transport → aerobic/anaerobic pathways in muscle cells → lactic acid accumulation → oxygen debt. They applied this framework to the novel scenario of Akinyi swimming the Winam Gulf of Lake Victoria.
How does this explain the phenomenon?	Learners can now fully explain the anchoring phenomenon: why breathing rate doubles during exercise (increased CO_2 and decreased O_2 in blood stimulate faster, deeper breathing to meet muscle cell demand), why the heart beats faster simultaneously (to pump oxygenated blood more rapidly to working muscles), and why legs burn even during hard breathing (when oxygen delivery cannot meet demand, muscle cells switch to anaerobic respiration producing lactic acid, which is acidic and causes the burning sensation). They can also explain why heavy breathing continues after exercise stops — to repay the oxygen debt by oxidising accumulated lactic acid.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.